

GE Healthcare

LightSpeed* Dicom Conformance Statement

for DICOM v3.0 (ID/Net v3.0)

* Refer to section 1 for a list of products in the LightSpeed family to which this Conformance Statement applies.



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- Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

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REVISION HISTORY

REV.	DATE	REASON FOR CHANGE
1	1/31/2006	Appendix A.1.2 – Added 6 tags in the CT Image Module for helical (spiral) scanning (0018, 9305 – 9307) and (0018, 9309 – 9311). Appendix A – Removed MR Specific IOD Definition. Appendix B.1.6 – Added private tag (0043, 1064) Recon Filter. Appendix B.1.7 – Added private tag (0045, 103B) Noise Reduction Image Filter Desc

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1 SECTION 1 – INTRODUCTION

Note: Throughout this entire document the term “LightSpeed” refers to the following LightSpeed products:

LightSpeed VCT
LightSpeed Pro³²
LightSpeed VCT Select
BrightSpeed Edge
BrightSpeed Excel
BrightSpeed Elite
LightSpeed QX/I
LightSpeed Plus
LightSpeed Ultra
LightSpeed 16
HiSpeed QX/I
LightSpeed Pro 16
LightSpeed RT

This document applies to the following software releases for each product:

LightSpeed VCT	06MW03.x
LightSpeed Pro ³²	06MW03.x
LightSpeed VCT Select	06MW03.x
BrightSpeed Edge	06MW03.x
BrightSpeed Excel	06MW03.x
BrightSpeed Elite	06MW03.x
LightSpeed QX/I	06MW03.x
LightSpeed Plus	06MW03.x
LightSpeed Ultra	06MW03.x
LightSpeed 16	06MW03.x
HiSpeed QX/I	06MW03.x
LightSpeed Pro 16	06MW03.x
LightSpeed RT	06MW03.x

1.1 Overview

Section 1, Introduction, provides general information about the content and scope of this document.

Section 2, *Network Conformance Statement*, is the DICOM v3.0 Conformance Statement related to this product. Conformance Statements define the subset of options selected from those offered by the DICOM v3.0 standard.

Section 3, *Media Storage Conformance Statement*, is the DICOM v3.0 Conformance Statement related to Media Storage Application Profile.

Section 4, *Print SCU Conformance Statement*, is the DICOM v3.0 Conformance Statement related to DICOM Print SCU Application. This product uses DICOM Print SCU to print the images to DICOM Compliant Printers.

Section 5, *DICOM Storage Commitment SCU*, is the DICOM v3.0 Conformance Statement related to DICOM Storage Commitment SCU Application. This product uses DICOM Storage Commitment Push Model to store the images using remote DICOM entity, which is Storage Commitment SCP.

Section 6, HIS/RIS (Modality Worklist) The ModalityWorklist option for LightSpeed allows a user to query for and display DICOM modality worklist information. ModalityWorklist is providing the DICOM C-FIND service as a service class user (SCU).

Section 7, Performed Procedure Step. The PPS option for LightSpeed, allows a Modality Performed Procedure Step to be communicated to the Hospital/Radiology information system. The PPS feature is providing the DICOM Modality Performed Procedure Step service as a service class user (SCU).

Section 8, Gray Scale Presentation State. The Virtual Exam Split Option for LightSpeed provides the DICOM Modality GSPS service as a service class user (SCU).

Appendix A specifies the CT/MR IOD information object.

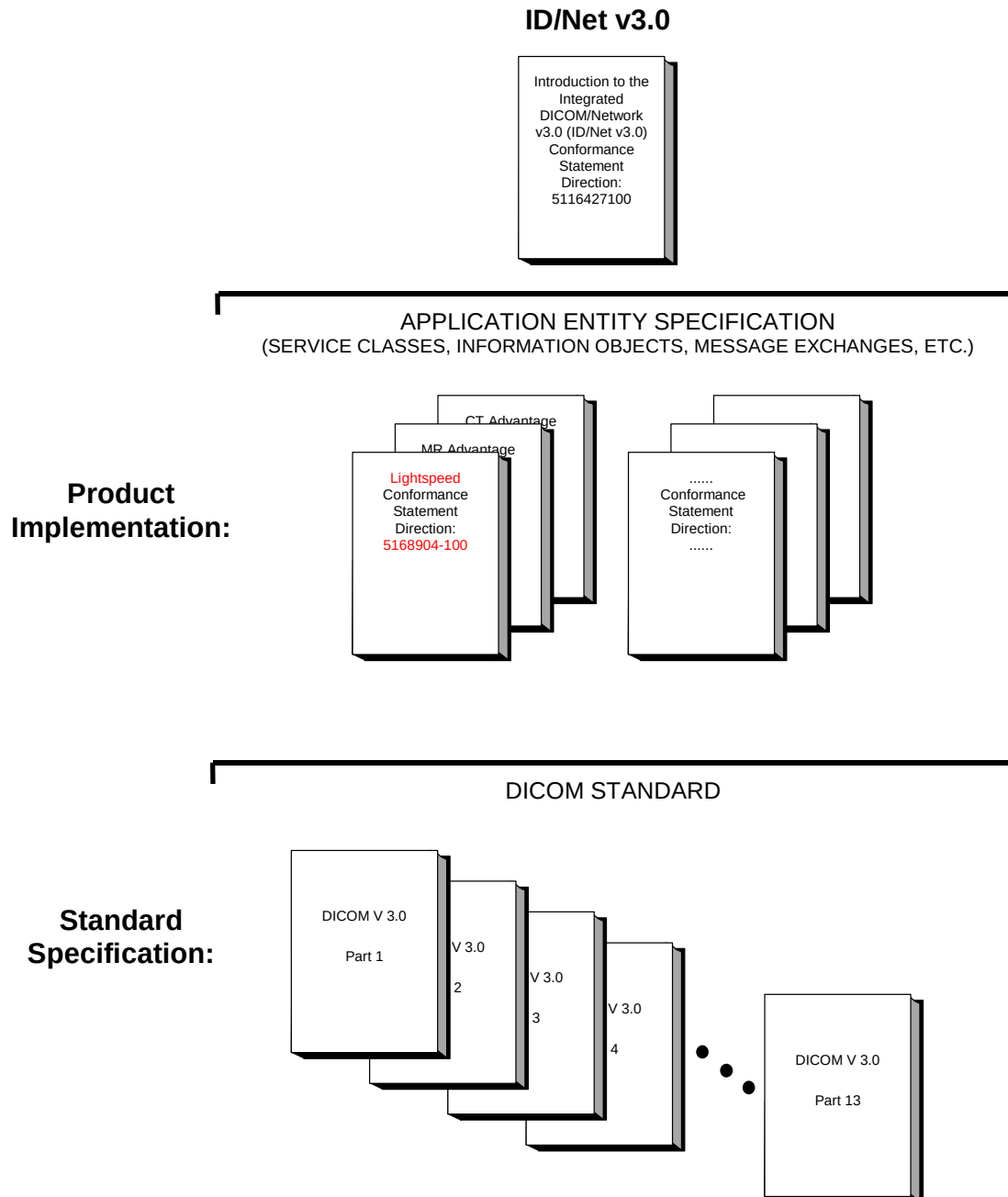
Appendix B specifies the private data element definition for CT/MR IOD.

Appendix C specifies the DICOMDIR directory information.

1.2 Overall Conformance Statement Documentation Structure

The documentation structure of the ID/Net v3.0 Conformance Statements and their relationship with the DICOM v3.0 Conformance Statements is shown in Illustration 1-1.

Illustration 1-1



The Documentation structure given in Illustration 1-1 shows the overall documentation structure for all of the GE ID/Net v3.0 Conformance Statements.

This document specifies the DICOM v3.0 implementation supported by the LightSpeed. It is entitled:

LightSpeed Conformance Statement for DICOM v3.0 (ID/Net v3.0), Direction 5114780-100

This Conformance Statement documents the DICOM v3.0 Conformance Statement and Technical Specification required to interoperate with the GE ID/Net v3.0 network interface. Introductory information, which is applicable to all GE ID/Net v3.0 Conformance Statements, is described in the document:

Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0) Conformance Statements, Direction 2118780

This introduction familiarizes the reader with DICOM terminology and general concepts. It should be read prior to reading individual products' ID/Net v3.0 Conformance Statements.

The ID/Net v3.0 Conformance Statement, contained in this document, also specifies the Lower Layer communications that it supports (e.g. TCP/IP). However, the Technical Specifications are defined in the DICOM v3.0 Part 8 standard.

For more information including Network Architecture and basic DICOM concepts, please refer to the *Introduction*.

For the convenience of developers, there is a "collector" Direction available. By ordering the collector, the Introduction described above and all of the currently published ID/Net v3.0 Product Conformance Statements will be received. The collector Direction is:

ID/Net v3.0 Conformance Statements, Direction 2117016

For more information regarding DICOM v3.0, copies of the Standard may be obtained by written request by contacting:

*ACR-NEMA Representative
NEMA
1300 N. 17th Street, Suite 1847
Rosslyn, VA 22209 USA*

1.3 Intended Audience

The reader of this document is concerned with software design and/or system integration issues. It is assumed that the reader of this document is familiar with the DICOM v3.0 standards and with the terminology and concepts that are used in those standards.

If readers are unfamiliar with DICOM v3.0 terminology they should first refer to the document listed below, then read the DICOM v3.0 Standard itself, prior to reading this Conformance Statement document.

*Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0)
Conformance Statements
Direction 2118780*

1.4 Scope and Field of Application

It is the intent of this document, in conjunction with the *Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0) Conformance Statements Direction 2118780*, to provide an unambiguous specification for GE ID/Net v3.0 implementations. This specification, called a Conformance Statement, includes a DICOM v3.0 Conformance Statement and is necessary to insure proper processing and interpretation of GE medical image data exchanged using DICOM v3.0. The GE ID/Net v3.0 Conformance Statements are available to the public.

The reader of this conformance statement should be aware that different GE devices are capable of using different Information Object Definitions. For example, a GE CT scanner may send images using the CT Information Object, MR Information Object, Secondary Capture Object, etc.

Included in this Conformance Statement are Module Definitions that define all data elements used by the GE ID/Net v3.0 implementation. If the user encounters unspecified private data elements while parsing a GE Data Set, the user is well advised to ignore those data elements (per the DICOM v3.0 standard). Unspecified private data element information is subject to change without notice. If, however, the device is acting as a “full fidelity storage device”, it should retain and retransmit all of the private data elements that are sent by GE devices.

1.5 Important Remarks

The use of these Conformance Statements, in conjunction with the DICOM v3.0 Standards, is intended to facilitate communication with GE imaging equipment. However, **by itself, it is not sufficient to insure that inter-operation will be successful.** The **user (or user’s agent)** needs to proceed with caution and address at least four issues:

- **Integration**

The integration of any device into an overall system of interconnected devices goes beyond the scope of standards (DICOM v3.0), and of this introduction and associated Conformance Statements when interoperability with non-GE equipment is desired. The responsibility to analyze the applications requirements and to design a solution that integrates GE imaging equipment with non-GE systems is the **user’s** responsibility and should not be underestimated. The **user** is strongly advised to ensure that such integration analysis is correctly performed.

- **Validation**

Testing the complete range of possible interactions between any GE device and non-GE devices, before the connection is declared operational, should not be overlooked. Therefore, the user should ensure that any non-GE provider accepts full responsibility for all validation required for their connection with GE devices. This includes the accuracy of the image data once it has crossed the interface between the GE imaging equipment and the non-GE device and the stability of the image data for the intended applications.

Such a validation is required before any clinical use (diagnosis and/or treatment) is performed. It applies when images acquired on GE imaging equipment are processed/displayed on a non-GE device, as well as when images acquired on non-GE equipment is processed/displayed on a GE console or workstation.

- **Future Evolution**

GE understands that the DICOM Standard will evolve to meet the user's growing requirements. GE is actively involved in the development of the DICOM v3.0 Standard. DICOM v3.0 will incorporate new features and technologies and GE may follow the evolution of the Standard. ID/Net v3.0 is based on DICOM v3.0 as specified in each ID/Net DICOM Conformance Statement. Evolution of the Standard may require changes to devices that have implemented DICOM v3.0. In addition, GE reserves the right to discontinue or make changes to the support of communications features (on its products) reflected on by these ID/Net DICOM Conformance Statements. The user should ensure that any non-GE provider, which connects with GE devices, also plans future evolution of the DICOM standard. Failure to do so will likely result in the loss of function and/or connectivity as the DICOM Standard changes and GE products are enhanced to support these changes.

To be kept informed of the evolution of the implementation described in this document, the user should register on the GE Internet server, accessible via anonymous ftp, by entering his/her e-mail address (GE Internet Server Address: ftp.med.ge.com: 192.88.230.11).

- **Interaction**

It is the sole responsibility of the non-GE provider to ensure that communications with the interfaced equipment does not cause degradation of GE imaging equipment performance and/or function.

1.6 References

A list of references that are applicable to all ID/Net v3.0 Conformance Statements is included in the *Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0) Conformance Statements Direction 2118780*.

1.7 Definitions

A set of definitions applicable to all ID/Net v3.0 Conformance Statements is included in the *Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0) Conformance Statements Direction 2118780*.

1.8 Symbols and Abbreviations

A list of symbols and abbreviations that are applicable to all ID/Net v3.0 Conformance Statements is included in the *Introduction to the Integrated DICOM/Network v3.0 (ID/Net v3.0) Conformance Statements Direction 2118780*.

2 SECTION 2 – NETWORK CONFORMANCE STATEMENT

2.1 Introduction

This Conformance Statement (CS) specifies the LightSpeed compliance to DICOM v3.0. It details the DICOM Service Classes and roles that are supported by this product in its version 1.0.

The LightSpeed product uses DICOM services to import images for possible further analysis and/or processing. It also uses DICOM services to export images to other DICOM-compliant machines.

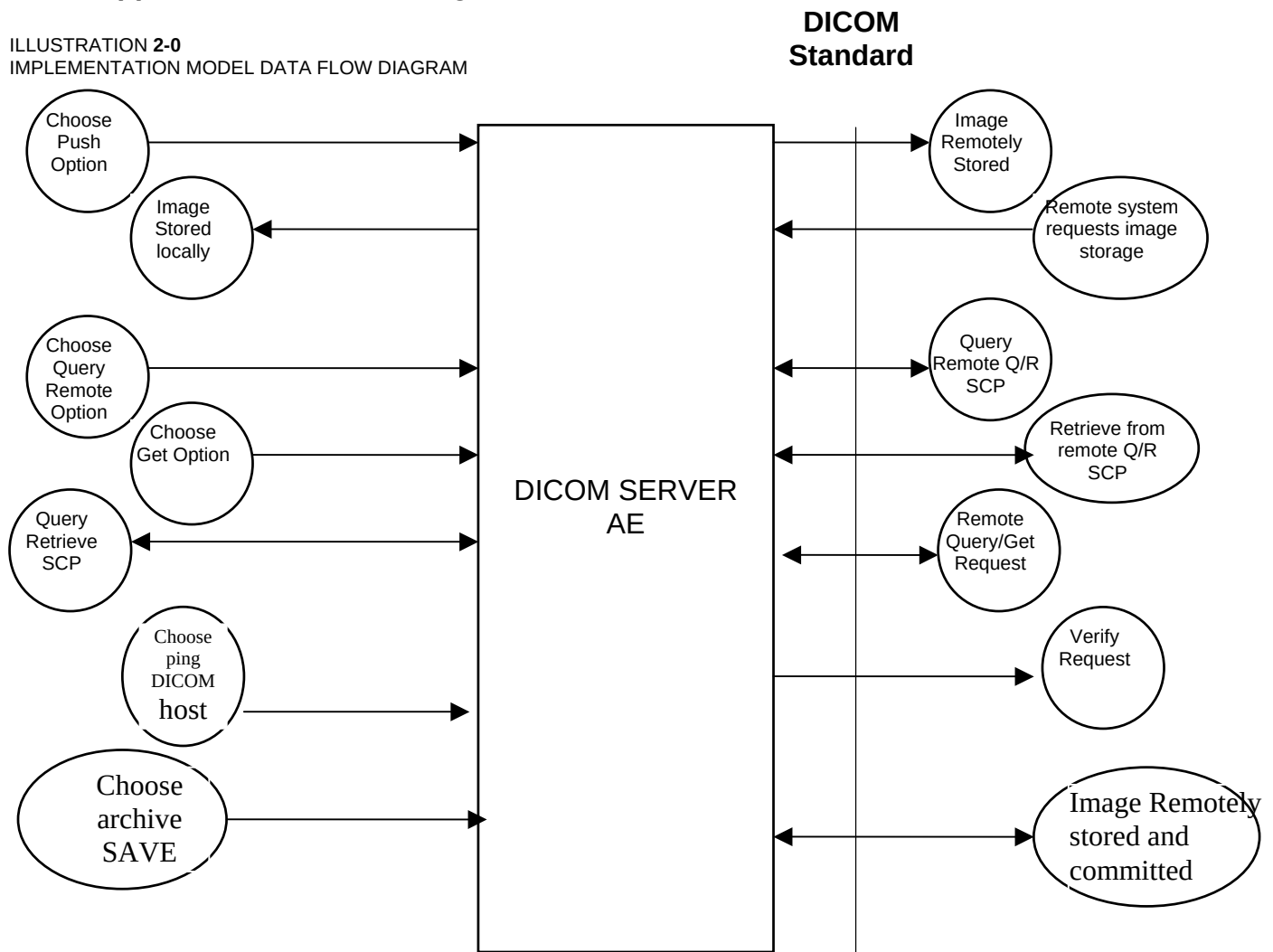
Note the format of this section follows the format of the DICOM Standard Part 2 (conformance) Annex A hence the paragraph numbering scheme. Please refer to that part of the standard while reading this section.

2.2 Implementation Model

All DICOM functionality on the LightSpeed product is handled by the DICOM Server Application Entity (AE). The DICOM Server AE is commanded to perform DICOM services through the buttons and menu selections on the main user interface panel. The DICOM Server AE is also listening to a pre-defined port for incoming connections.

2.2.1 Application Data Flow Diagram

ILLUSTRATION 2-0
IMPLEMENTATION MODEL DATA FLOW DIAGRAM



There are five Real-World Activities that will cause the DICOM Server Application Entity (DICOM Server AE) to initiate a DICOM association to a remote DICOM Application Entity.

(1) The *Choose "Push" Option* Real-World activity consists of an operator selecting one or more study, series or image in the local database manager and choosing either "Push Examination", "Push Series" or "Push Image from the "Network" pull-down menu on the local database manager to send the image(s) to a selected destination.

(2) Real-World Activity, *Query Remote*, causes the DICOM Server AE to initiate an association to the Remote DICOM AE and request the list of all studies. Once the DICOM Server AE receives the list of studies, it will select the first study (as determined through the local database manager list sort criterion) and request the list of series for that study. After receiving the list of series the DICOM Server AE will ask for the list of images for the first series in the list. The operator can then select any study in the study list to retrieve the list of series and images.

(3) Real-World Activity, *Choose “Get” Option*, will be available once the *Query Remote* activity is performed. The operator can now select one or more study (series or image) and ask the DICOM Server AE to retrieve the selected image(s) from the Remote DICOM AE by choosing “Get Examination”, “Get Series”, or “Get Images”.

(4) Real-World Activity, *Choose “Ping DICOM host” Option*, consists of an operator selecting “Ping DICOM host” from the “Network” pull down menu. This will cause the DICOM Server AE to initiate a “DICOM Verification Request” to the remote AE, to verify the remote system activeness.

(5) The *Choose “storage commitment” Option* Real-World activity consists of an operator configuring the remote node as an “*archive node*.” Select “choose media” option under “Archive” pull down menu. Select “Remote Node” in that menu. Select one or more study, series or image in the local database manager and choose either “*Save Exam*” or “*Save Series*” from the “Archive” pull down menu. This will cause the DICOM server AE to send the image(s) to the selected destination and wait for the storage commitment from the remote node. The remote node should be a storage commitment provider.

There is no Real-World activity required for the DICOM Server AE to respond to an incoming DICOM store, query or retrieve. The DICOM Server AE is always prepared to respond to a DICOM Store, Query, or Retrieve by any remote DICOM AE.

The DICOM Server AE will perform the Real-World activity Image Installation after the remote AE sends an image to the LightSpeed product.

Once a Query request is received, the DICOM Server AE will search the local database for all entries that match the keys requested by the Remote DICOM AE and send back the list of matches. The DICOM Server AE will also respond to an incoming retrieval request from a Remote AE by sending the image(s) to the Destination AE.

2.2.2 Functional Definition of AE's

DICOM Server Application Entity initiates the following operations:

- *Ping DICOM Host (Verification)*: The DICOM Server initiates an association and sends a C-ECHO-RQ message to the remote DICOM AE; the remote DICOM Server will send back a C-ECHO-RSP message with a status of “success”.
- *Push*: Initiate an association to a Remote AE to send image(s). If the Remote AE accepts the presentation context applicable to the image(s) being sent, the DICOM Server AE will send the image(s) by invoking C-STORE-RQ operation for each image on the same association.
- *Query*: Initiate an association with a Remote AE to query for images on the remote host. A Study-Root Study-Level C-FIND-RQ request will be sent to the Remote AE once an association has been established. After all responses are received, DICOM Server AE will issue a Series-Level C-FIND-RQ request to get the series for a study in the list. An Image-Level C-FIND-RQ will be issued for the first series in the series list.

- *Get*: Send a C-MOVE-RQ request to a Remote AE after successful association establishment. The DICOM Server AE's Storage SCP will receive the images over a separate association.

The DICOM Server AE waits for association requests from Remote AEs that wish to perform the following operations:

- *Verification*: If a C-ECHO-RQ message is received, the DICOM Server AE will send back a C-ECHO-RSP message with a status of "success".
- *Image Storage*: If a C-STORE-RQ message is received, the DICOM Server AE will receive the image and try to update the local database. If the image is stored successfully on storage media and the database updated a status of "success" will be returned in a C-STORE-RSP message.
- *Query*: If a C-FIND-RQ message is received the DICOM Server AE will search the database for the requested attributes and send back a C-FIND-RSP message containing a match and a status of "pending". After all matching records have been sent, a status of "success" will be returned in a C-FIND-RSP message. The Remote AE can terminate the query by sending a C-CANCEL-FIND-RQ message.
- *Retrieve*: If a C-MOVE-RQ message is received the DICOM Server AE will lookup its list of configured Remote AEs for the Destination AE. If the Destination AE is configured, the DICOM Server AE will open a new association to the Destination AE and use C-STORE-RQ to send the image(s). The DICOM Server AE will send a C-MOVE-RSP message with a status of "pending" after every five images are sent. When all images are sent or if DICOM Server AE receives a C-CANCEL-MOVE-RQ a final C-STORE-RSP will be sent back with an appropriate status.

2.2.3 Sequencing of Real-World Activities

Real-World Activity *Query Remote* must be performed before *Choose Get Option* can be performed.

2.3 AE Specifications

2.3.1 DICOM Server AE Specification

This Application Entity provides Standard Conformance to the following DICOM v3.0 SOP Classes as an SCU:

SOP Class Name (SCU)	SOP Class UID
Verification (Echo)	1.2.840.10008.1.1
CT Image Information Storage	1.2.840.10008.5.1.4.1.1.2
MR Image Information Storage	1.2.840.10008.5.1.4.1.1.4
Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7
Study Root Query/Retrieve – FIND	1.2.840.10008.5.1.4.1.2.2.1
Study Root Query/Retrieve – MOVE	1.2.840.10008.5.1.4.1.2.2.2
Storage Commitment Push Model	1.2.840.10008.1.20.1

This Application Entity provides Standard Conformance to the following DICOM v3.0 SOP classes as an SCP:

SOP Class Name (SCP)	SOP Class UID
Verification (Echo)	1.2.840.10008.1.1
CT Information Storage	1.2.840.10008.5.1.4.1.1.2
MR Information Storage	1.2.840.10008.5.1.4.1.1.4
Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7
Study Root Query/Retrieve – FIND	1.2.840.10008.5.1.4.1.2.2.1
Study Root Query/Retrieve – MOVE	1.2.840.10008.5.1.4.1.2.2.2

2.3.1.1 Association Establishment Policy

2.3.1.1.1 General

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
---------------------------------	------------------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU for association initiated by the DICOM Server AE is:

Maximum Length PDU	51 Kbytes
---------------------------	------------------

SOP class Extended Negotiation is not supported.

Maximum PDU Length of Zero (0) is not supported. The association request will be rejected when a request with “Maximum PDU length of zero” is received.

The maximum number of Presentation Context Items that is supported is 60. Note that the same Abstract Syntax may be offered multiple times with different Transfer Syntax.

The user information items sent by this product are:

- Maximum PDU Length and,
- Implementation UID

2.3.1.1.2 Number of Associations

The DICOM Server AE (SCU) will initiate only one DICOM association at a time to perform an image store to a remote host or retrieve image(s) from a Remote AE.

The DICOM Server AE (SCP) can have a maximum of four DICOM associations open simultaneously to receive and store image store or respond to an echo.

2.3.1.1.3 Asynchronous Nature

Asynchronous mode is not supported. All operations will be performed synchronously.

2.3.1.1.4 Implementation Identifying Information

The Implementation UID allows unique identification of a set of products that share the same implementation.

The Implementation UID for this ID/Net v3.0 Implementation is:

LightSpeed Implementation UID	1.2.840.113619.6.55
-------------------------------	---------------------

2.3.1.2 Association Initiation by Real-World Activity

This AE attempts to initiate a new association due to a “ Ping DICOM Host” or “Push” operation initiated by the user. A new association is also initiated when the user performs a “Query Remote” operation or issues a retrieve operation by performing a “Get” operation at Study/Series/Image level.

2.3.1.2.1 Push Image(s) to Remote AE

2.3.1.2.1.1 Associated Real-World Activity

The operator must first select a destination by choosing “Select Remote Host” from the “Network” pull-down menu on the local database manager and then choose a hostname.

The “Push” operation will cause the DICOM server AE to initiate an Association when the operator selects one or more study, series, or images in the local database manager and then chooses either “Push Examination”, “Push Series”, or “Push Image” from the “Network” pull-down menu on the local database manager.

Note: If multiple study, series, or images are chosen to be pushed, one association will be established for each of the studies, series, or images.

2.3.1.2.1.2 Proposed Presentation Contexts

The following table shows the proposed presentation contexts for the DICOM Server AE after Real-World Activity “Push” Operation has been performed.

Table 2.2.1.2.1.2-1 Proposed Presentation Contexts for DICOM Server AE and Real-World activities Push, Query, Pull (Get Images) and Verification (Dicom Ping) SCUs

Presentation Context Table – Proposal					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
		Implicit VR Big Endian (GE Private)	1.2.840.113619.5.2		
		Explicit VR Little Endian	1.2.840.10008.1.2.1		
		Explicit VR Big Endian	1.2.840.10008.1.2.2		
MR Image Storage	1.2.840.10008.5.1.4.1.1.4	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
		Implicit VR Big Endian (GE Private)	1.2.840.113619.5.2		
		Explicit VR Little Endian	1.2.840.10008.1.2.1		
		Explicit VR Big Endian	1.2.840.10008.1.2.2		

Presentation Context Table – Proposal					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7	Implicit VR Little Endian Implicit VR Big Endian (GE Private) Explicit VR Little Endian Explicit VR Big Endian	1.2.840.10008.1.2 1.2.840.113619.5.2 1.2.840.10008.1.2.1 1.2.840.10008.1.2.2	SCU	None
Study Root Query/Retrieve FIND	1.2.840.10008.5.1.4.1.2.2.1	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
Study Root Query/Retrieve MOVE	1.2.840.10008.5.1.4.1.2.2.2	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
Verification SOP Class	1.2.840.10008.1.1	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None

Note: Other Abstract Syntax and Transfer Syntax names may be offered when establishing an association. These are private syntax that may be ignored. See section 2.4.

2.3.1.2.1.2.1 SOP Specific Conformance Statement C_STORE SCU

This implementation can perform multiple C-STORE operations over a single association.

Upon receiving a C-STORE confirmation containing a successful status, this implementation will perform the next C-STORE operation. The association will be maintained if possible.

Upon receiving a C-STORE confirmation containing a Refused status, this implementation will terminate the association.

Upon receiving a C-STORE confirmation containing any status that is not Success or Refused, this implementation will consider the current request to be a failure but will continue to attempt to send the remaining images in the request on the same association.

Each C-STORE operation supports an “Association Timer”. This timer starts when the association request is sent or received and stops when the association is established. The time-out is 900 seconds.

Each C-STORE operation also supports an “Operation Inactivity Timer”. This time-out starts once the first C-STORE request has been issued (on association) or received and is reset each time a C-STORE response has been received or when subsequent C-STORES are sent. This time-out is 900 seconds.

Each C-STORE operation also supports a “Session Timer”. This timer starts when the association is established and stops when the association is ended. This time-out is 60 minutes.

If any of the three timers mentioned above expires, the connection is closed and the operation in progress is considered failed.

Note: The time-outs are configurable.

When DICOM Server AE initiates an association to issue a C-STORE, the following will occur:

1. If the image is stored locally on the LightSpeed in DICOM format ("Dicm" appears in the "Fmt" column at the exam level of the local database manager), the image will be transmitted by the DICOM Server AE with the same elements as was originally received or created locally (for the standard elements only).

2.3.1.2.2 Query Remote AE

2.3.1.2.2.1 Associated Real-World Activity

The operator must first select a destination by choosing "*Select Remote Host*" from "*Network*" pull-down menu on the local database manager and then choose a hostname. To do custom queries select "**Yes**" for "**Custom search**" option.

Note: Custom queries can be done on the following fields "**Last Name contains**", "**Exam Number**", "**Patient Id**", "**Accession Number**".

To initiate a Query select "**Receive**" from "**Network**" pull-down menu.

Note: If "**Custom Search**" option is set then a small GUI will show up. Operator will be able to enter custom query fields. Entering values in "Last Name contains" field will initiate wild card query for patient name.

The "*Query*" operation will cause the DICOM Server AE to initiate an association to the selected Remote AE when the "*Query Remote Host*" entry is selected from the "*Network*" pull-down menu. Once a list of Study/Series/Image is retrieved, the operator can invoke the "*Get*" operation by choosing "*Get Exam*" or "*Get Series*" or "*Get Image*" from the "*Network*" pull-down menu.

2.3.1.2.2.2 Proposed Presentation Contexts

When the Real-World activity "*Query*" or "*Get*" is initiated all presentation contexts shown in table 2.2.1.2.1.2-1 are proposed during association establishment, but only the Query/Retrieve-FIND related contexts are applicable to this activity.

2.3.1.2.2.2.1 SOP Specific Conformance Statement for C-FIND SCU

After the *Query* operation is initiated, the DICOM Server AE will perform a study-root C-FIND-RQ request at each of the three levels (Study, Series, and Image) in succession. The Initial Study-Level request will ask for all studies in the Remote database.

The C-FIND SCU will not perform any extended negotiation and so will only perform hierarchical query. Relational Queries are not supported.

Each C-FIND SCU operation supports an "Association Timer", "Operation Inactivity Timer" and "Session Timer" with time out values of 900 seconds, 900 seconds and 60 minutes respectively.

If a “Cancel” or “Refused” status is returned from the Remote AE the association is closed and the operation terminated.

The DICOM Server AE will parse each matching C-FIND-RSP reply and ignore the entries it fails to parse. Tables 2.2.1.2.2.2.1-1 - 2.2.1.2.2.2.1-3 shows the various fields that are requested at the Study, Series, and Image levels of the C-FIND request.

Query results are filtered based on the Modality field. Only CT/MR Screen Save images are supported.

Table 2.2.1.2.2.2.1-1: Requested Study Level Keys

Description	Type	Tag	Value
Study date	R	0008,0020	Zero length
Study time	R	0008,0030	Zero length
Patient's name	R	0010,0010	Zero length
Study id	R	0020,0010	Zero length
Study Instance UID	U	0020,000D	Zero length
Study description	O	0008,1030	Zero length
Private Creator Identification	P1	0009,00xx	GEMS_IDEN_01
Suite Id	P2	0009,xx02	Zero Length

Table 2.2.1.2.2.2.1-2: Requested Series Level Keys

Description	Type	Tag	Value
Modality	R	0008,0060	Zero length
Series number	R	0020,0011	Zero length
Series Instance UID	U	0020,000E	Series UID
Series description	O	0008,103E	Zero length
Manufacturer	O	0008,0070	Zero length
Images in series	O	0020,1002	Zero length

Table 2.2.1.2.2.2.1-3: Requested Image Level Keys

Description	Type	Tag	Value
Image number	R	0020,0013	Zero length
Image Instance UID	U	0008,0018	Image UID
Image type	O	0008,0008	Zero length
Rows	O	0028,0010	Zero length
Columns	O	0028,0011	Zero length
Image position	O	0020,0032	Zero length
Image orientation	O	0020,0037	Zero length
Slice thickness	O	0018,0050	Zero length

1 Private data element

Description	Type	Tag	Value
Slice spacing	O	0018,0088	Zero length
Gantry tilt	O	0018,1120	Zero length
Convolution kernel	O	0018,1210	Zero length
Reconstruction diameter	O	0018,1100	Zero length
Data collection diameter	O	0018,0090	Zero length
Flip angle	O	0018,1314	Zero length
Echo number	O	0018,0086	Zero length
Echo time	O	0018,0081	Zero length
Inversion time	O	0018,0082	Zero length
Repetition time	O	0018,0080	Zero length
Private Creator Identification	P	0019,00xx	GEMS_ACQU_01
Dfov Rect	P	0019,001E	Zero Length
Dfov Rect	P	0019,xx1E	Zero Length
Midscan Time	P	0019,xx24	Zero Length
Azimuth	P	0019,xx26	Zero Length
Number of Echo	P	0019,xx7E	Zero Length
Private Creator Identification	P	0021,00xx	GEMS_REL_01
Scout Anref	P	0021,xx4A	Zero Length
Private Creator Identification	P	0027,00xx	GEMS_IMAG_01
Location RAS	P	0027,xx40	Zero Length
Location	P	0027,xx41	Zero Length
Center R Coordinate	P	0027,xx42	Zero Length
Center A Coordinate	P	0027,xx43	Zero Length
Table Start Location	P	0027,xx50	Zero Length
Table End Location	P	0027,xx51	Zero Length
RAS Letter for Side of Image	P	0027,xx52	Zero Length
RAS Letter for Anterior/Posterior	P	0027,xx53	Zero Length
RAS Letter for Scout Start Location	P	0027,xx54	Zero Length
RAS Letter for Scout End Location	P	0027,xx55	Zero Length
Image Dimension X	P	0027,xx60	Zero Length
Image Dimension Y	P	0027,xx61	Zero Length

Note: Refer to section 2.3.1.2.2.1 for Custom Search/Query option.

2.3.1.2.3 Get Image(s) from Remote AE

2.3.1.2.3.1 Associated Real-World Activity

The operator must first select a destination by choosing “*Select Remote Host*” from “*Network*” pull-down menu on the local database manager and then choose a hostname. The operator then has to perform the Real-World activity “*Query*” to get a list of Study/Series/Image. Once the list of Study/Series/Image is retrieved, the operator can invoke the “*Get*” operation by choosing “*Get Exam*” or “*Get Series*” or “*Get Image*” from the “*Network*” pull-down menu.

2.3.1.2.3.2 Proposed Presentation Contexts

When the Real-World activity “Get” is initiated all presentation contexts shown in table 2.2.1.2.1.2-1 are proposed during association establishment, but only the Query/Retrieve-MOVE related contexts are applicable to this activity.

2.3.1.2.3.2.1 SOP Specific Conformance Statement for C-MOVE SCU

When the operator starts a *Get* operation at any level (Study, Series, Image) the DICOM Server AE will initiate a C-MOVE-RQ request to the Remote AE with the DICOM Server AE as the Destination AE. The Storage SCP will handle the incoming images as described in section 2.2.1.3.1.

Each C-MOVE SCU operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 300 seconds and 60 minutes respectively.

The DICOM Server AE will send a C-CANCEL-MOVE-RQ to the Remote AE if the operator “*Pauses*” or “*Clears*” the job from the local database manager Network queue.

2.3.1.2.4 Real-World Activity: Verify (DICOM Ping)

2.3.1.2.4.1 Associated Real-World Activity

The operator must first select a destination by choosing “*Select Remote Host*” from “*Network*” pull-down menu on the local database manager and then choose a hostname.

The operator must then select the “*Ping DICOM host*” form “*Network*” pull-down menu.

The DICOM server will initiate an association with the remote DICOM AE in order to verify communication at the application level. The success or failure of the verification process is displayed to the user.

2.3.1.2.4.2 Proposed Presentation Context Table

Refer to the Table 2.2.1.2.1.2-1 for the Proposed Presentation Contexts for DICOM Server AE and Real-World activity Verification

2.3.1.2.4.2.1 SOP Specific DICOM Conformance C_ECHO SCU

The LightSpeed DICOM Server AE provides standard conformance to the DICOM Verification Service Class.

Each ECHO operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 30 seconds and 60 minutes respectively

2.3.1.2.5 Real-World Activity: Storage Commitment

2.3.1.2.5.1 Associated Real-World Activity

The operator must first select a destination by choosing “*Select Remote Host*” from “*Network*” pull-down menu on the local database manager and then choose a hostname. Select the “*Update*” option, if the hostname is already present. Set the “*Archive Node*” option to “Yes”. If the remote host is not present add the remote host in the local database with “*Archive Node*” option set to “Yes”.

Note: The remote node should be a Storage Commitment SCP.

The operator must then select “Archive” pull-down menu on the local database manager. Select “*Select Media*” option menu. Select “*Remote Node*” in the menu.

Note: Only if you configure the remote node as a “*archive node*” will you see the remote node option in the archive menu.

The operator then selects one or more study, series, or images in the local database manager and then chooses “Save Examination”, “Save Series”, or “Save Image” from the “Archive” pull-down menu on the local database manager.

Note: Also refer to section 5.3.1.3

2.3.1.2.5.2 Proposed Presentation Context Table

Refer to section 5.3.1.4

2.3.1.2.5.3 SOP Specific Conformance Statement Storage Commitment SCU

Refer to section 5.3.1.4

2.3.1.3 Association Acceptance Policy

The DICOM Server AE places limitations on who may connect to it.

If the Remote AE needs to “Push Images” or “Query/Retrieve Images”, to the local system then it has to be configured in the Local system to do the same.

To configure a remote AE choose “Select Remote Host” in the “network” pull down menu. Select the Remote Host and choose “Update” option, if the Remote AE is already present. If the Remote AE is not present choose “Add” option to add the Remote AE.

Set “Push Images” option to “Yes”, if the Remote AE needs to push images to the Local system

Set “Query/Retrieve” option to “Yes”, if the Remote AE needs to “Query/Retrieve” the local system.

When the DICOM Server AE accepts an association for image storage, it will receive any images transmitted on that association and store the images on disk.

It will also respond to queries from Remote AEs by sending matching entries. Any Remote AE can request and receive a list of images on the local database. The Remote AE must be configured in the local database manager's list of Remote AE for it to be able to retrieve images from DICOM Server AE.

Any remote AE can open an association to the DICOM Server AE for the purpose of verification.

2.3.1.3.1 Receive Image(s)

This AE is indefinitely listening for associations. No operator action is required to receive an image.

2.3.1.3.1.1 Associated Real-World Activity

The Real-World Activity associated with the Receive Image(s) operation is the storage of the image on the disk drive of the LightSpeed.

2.3.1.3.1.2 Presentation Context Table

Table 2.2.1.3.1.2-1: Accepted Presentation Contexts for DICOM Server AE and Real-World Activity Receive Image(s)

Presentation Context Table - Accepted					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Verification	1.2.840.10008.1.1	Implicit VR Little Endian	1.2.840.10008.1.2	SCP	None
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	Implicit VR Little Endian	1.2.840.10008.1.2	SCP	None
		Implicit VR Big Endian	1.2.840.113619.5.2		
		(GE Private)			
		Explicit VR Little Endian	1.2.840.10008.1.2.1		
MR Image Storage	1.2.840.10008.5.1.4.1.1.4	Explicit VR Big Endian	1.2.840.10008.1.2.2	SCP	None
		Implicit VR Little Endian	1.2.840.10008.1.2		
		Implicit VR Big Endian	1.2.840.113619.5.2		
		(GE Private)			
Secondary Capture Image Storage	1.2.840.10008.5.1.4.1.1.7	Explicit VR Little Endian	1.2.840.10008.1.2.1	SCP	None
		Explicit VR Big Endian	1.2.840.10008.1.2.2		
		Implicit VR Little Endian	1.2.840.10008.1.2		
		Implicit VR Big Endian	1.2.840.113619.5.2		
		(GE Private)			
		Explicit VR Little Endian	1.2.840.10008.1.2.1		
		Explicit VR Big Endian	1.2.840.10008.1.2.2		
		Implicit VR Little Endian	1.2.840.10008.1.2		

Note: The SCP does not have a default acceptance policy if more than one acceptable transfer syntax is proposed by the SCU. It is the responsibility of the SCU to make a selection from more than one transfer sentences accepted.

2.3.1.3.1.2.1 SOP Specific Conformance to C_STORE SCP

The DICOM Server AE conforms to the SOP's of the Storage Service Class at level 1 (base). Private elements will be discarded from the image when receiving images containing non-GE private data elements. All of the **standard** type elements (1,1c,2,2c,3) will be retained.

Each C-STORE SCP operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 900 seconds and 60 minutes respectively.

Image Reception

If the DICOM Server AE returns one of the following status codes, then the C-STORE operation was unsuccessful and no image will be installed:

- 0110 (Processing Failure) Indicates that an internal system call has failed while processing an image.
- A711 (Out of Resources) Indicates that probably there was not enough disk space to store the image. The user should attempt recovery by removing some images from the LightSpeed system.
- A712 (Out of Resources) Indicates that there was not enough resource (such as memory) to store the image.
- A800 (SOP Class not supported)

In the event of a successful C-STORE operation, the image has successfully been written to disk. The image will then be accessed in the same manner as any other image by the applications on the LightSpeed system.

Images may be deleted when instructed to do so by the user. Thus the duration of the storage of the image is determined by the users of the LightSpeed system.

Image Installation

If the image installation is unsuccessful, a message will appear in the Message Log informing the user of the failure and the image will be removed.

If the image installation process finds that an element is not encoded according to the DICOM standard, it will fail to install the image and the file will be removed.

Image Installation of non-GE Created MR or CT Images

Images received from non-GE products are installed as appropriate image object without any of their private data elements. Also if some critical fields (mandatory) are missing, then the image will not be installed.

2.3.1.3.2 Verification Request from Remote AE

This AE is indefinitely listening for associations. No operator action is required to respond to a *verification* message.

2.3.1.3.2.1 Associated Real-World Activity

The Real-World Activity associated with the verification request is to send a C-ECHO response message with a status of “success” to the requesting AE.

2.3.1.3.2.2 Presentation Context Table

Table 2.2.1.3.2.2-1: Acceptable Presentation Contexts for DICOM Server AE and Real-World Activity Verification Request

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Verification	1.2.840.10008.1.1	Implicit VR Little Endian	1.2.840.10008.1.2	SCP	None

2.3.1.3.2.2.1 SOP Specific Conformance to C_ECHO SCP

The DICOM Server AE provides standard conformance to the DICOM Verification Service Class.

Each ECHO operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 30 seconds and 60 minutes respectively.

2.3.1.3.3 Query Request from Remote AE

This AE is indefinitely listening for associations. No operator action is required to respond to a *query* request.

2.3.1.3.3.1 Associated Real-World Activity

The Real-World Activity associated with the query request is to search the local database for entries that match the request and send a C-FIND response message with a status of “pending” for each matching entry.

2.3.1.3.3.2 Presentation Context Table

Table 2.3.1.3.3.2-1: Acceptable Presentation Contexts for DICOM Server AE and Real-World Activity Query Request

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Study Root Query/Retrieve FIND	1.2.840.10008.5.1.4.1.2.2.1	Implicit VR Little Endian	1.2.840.10008.1.2	SCP	None

2.3.1.3.3.2.1 SOP Specific Conformance to C-FIND SCP

Each C-FIND SCP operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 900 seconds and 60 minutes respectively.

All Required (R) and Unique (U) study, series, and image level keys for the Study-Root Query/Retrieve information model are supported. Some optional (O) keys are also supported as described in the following tables.

Table 2.3.1.3.3.2.1-1 Supported study level keys

Description	Type	Tag	Usage
Study date	R	0008,0020	Matched
Study time	R	0008,0030	Matched
Accession number	R	0008,0050	Matched
Patient's name	R	0010,0010	Matched3
Patient id	R	0010,0020	Matched
Study id	R	0020,0010	Matched
Study Instance UID	U	0020,000D	Matched
Study description	O	0008,1030	Returned
Suite Id	P	0009,0002	Returned

Table 2.3.1.3.3.2.1-2 Supported series level keys

Description	Type	Tag	Usage
Modality	R	0008,0060	Matched
Series number	R	0020,0011	Matched
Series Instance UID	U	0020,000E	Matched
Series description	O	0008,103E	Returned
Manufacturer	O	0008,0070	Returned
Images in series	O	0020,1002	Returned

Table 2.3.1.3.3.2.1-3 Supported image level keys

Description	Type	Tag	Usage
Image number	R	0020,0013	Matched
Image Instance UID	U	0008,0018	Matched
Image type	O	0008,0008	Returned
Rows	O	0028,0010	Returned
Columns	O	0028,0011	Returned
Image position	O	0020,0032	Returned
Image orientation	O	0020,0037	Returned
Slice thickness	O	0018,0050	Returned
Slice spacing	O	0018,0088	Returned
Gantry tilt	O	0018,1120	Returned
Convolution kernel	O	0018,1210	Returned
Reconstruction diameter	O	0018,1100	Returned
Data collection diameter	O	0018,0090	Returned
Flip angle	O	0018,1314	Returned
Echo number	O	0018,0086	Returned
Echo time	O	0018,0081	Returned
Inversion time	O	0018,0082	Returned

Description	Type	Tag	Usage
Repetition time	O	0018,0080	Returned
Trigger time	O	0018,1060	Returned
Dfov Rect	P	0019,001E	Returned
Midscan Time	P	0019,0024	Returned
Azimuth	P	0019,0026	Returned
Number of Echo	P	0019,007E	Returned
Scout Anref	P	0021,004A	Returned
Location RAS	P	0027,0040	Returned
Location	P	0027,0041	Returned
Center R Coordinate	P	0027,0042	Returned
Center A Coordinate	P	0027,0043	Returned
Table Start Location	P	0027,0050	Returned
Table End Location	P	0027,0051	Returned
RAS Letter for Side of Image	P	0027,0052	Returned
RAS Letter for Anterior/Posterior	P	0027,0053	Returned
RAS Letter for Scout Start Location	P	0027,0054	Returned
RAS Letter for Scout End Location	P	0027,0055	Returned
Image Dimension X	P	0027,0060	Returned
Image Dimension Y	P	0027,0061	Returned

Note: In the above tables the type field has the following meaning:

R = Required **U** = Unique **O** = Optional **P** = Private

Only keys with Usage type *Matched* will be matched against values in the database.

Values in keys of type Returned will be ignored and will be filled in with data from the database.

If an optional key is requested that does not appear in any of the tables above, that key will be ignored and no corresponding element will be returned.

If the database does not have a value corresponding to any requested optional key a zero-length element will be returned.

Sequence matching is not supported.

Range matching is supported for attributes of type date and time.

Only hierarchical query is supported.

2.3.1.3.4 Retrieve Request From Remote AE

This AE is indefinitely listening for associations. No operator action is required to respond to a *retrieve* request.

2.3.1.3.4.1 Associated Real-World Activity

The Real-World Activity associated with the Retrieve Request is to send all images corresponding to the C-MOVE request to the destination AE through a separate association.

2.3.1.3.4.2 Presentation Context Table

Table 2.3.1.3.4.2-1: Acceptable Presentation Contexts for DICOM Server AE and Real-World Activity Retrieve Request.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Study Root Query/Retrieve MOVE	1.2.840.10008.5.1.4.1.2.2.2	Implicit VR Little Endian	1.2.840.10008.1.2	SCP	None

2.3.1.3.4.2.1 SOP Specific Conformance to C-MOVE SCP

The DICOM Server AE provides standard conformance to the baseline Study-root C-MOVE Service Class SCP.

Each C-MOVE SCP operation supports an “Association Timer”, “Operation Inactivity Timer” and “Session Timer” with time out values of 900 seconds, 300 seconds and 60 minutes respectively. These time-outs are configurable in dcs.cfg as *bi_assoc_tio*, *bi_store_tio* and *bi_session_tio* respectively

All images requested in a C-MOVE-RQ will be sent over a single association. A C-MOVE-RSP with a “pending” status will be returned to the requester every five images.

The C-MOVE SCP will invoke C-STORE requests for the following SOP classes:

SOP Class Name	SOP Class UID
CT Image Information Storage	1.2.840.10008.5.1.4.1.1.2
MR Image Information Storage	1.2.840.10008.5.1.4.1.1.4
Secondary Capture image storage	1.2.840.10008.5.1.4.1.1.7
Standalone Overlay storage	1.2.840.10008.5.1.4.1.1.8

In addition to the C-MOVE response status values defined in DICOM V3.0 part 4 the following status values will be returned:

- C000 Indicates that an error occurred while retrieving records from the local database.
- C001 Indicates all other processing error.
- C011 If the Destination AE returns a “*Storage Full*” condition this status will be returned. This status will only be sent if the Destination AE returns a status of A711 and is only applicable if the Destination AE is another GE Medical Systems product.

2.3.1.3.4.3 Presentation Context Acceptance Criteria

No criterion.

2.3.1.3.4.4 Transfer Syntax Selection Policy

All protocols detailed above in the tables are supported; no priority selection is given.

2.4 Communication Profiles

2.4.1 Supported Communication Stacks (parts 8,9)

DICOM Upper Layer (Part 8) is supported using TCP/IP.

2.4.2 TCP/IP Stack

The TCP/IP stack is inherited from a UNIX Operating System.

2.4.2.1 Physical Media Support

Ethernet v2.0, IEEE 802.3. Auto senses 10/100 Base T Ethernet connection.

2.4.3 Point-to-Point Stack

A 50-pin ACR-NEMA connection is not applicable to this product.

2.5 Extensions / Specializations / Privatizations

2.5.1 Specialized Information Object Definition

Following is a list of additional private attributes defined for a standard **CT IOD**.

(0008,0008) Value 3: CT Image IOD specific specializations
AXIAL
LOCALIZER
SEGMENTED
REFORMATTED
PROCESSED
COMBINED

Value 4: CT Image implementation specific
MIN IP
MIP
AVERAGE
VOLREN
INTEGRAL
HD MIP
RAYSUM

SURFACE
MINMAX

(0018,0022) Scan Options: Defined Terms
AXIAL MODE
DYNAMIC MODE
SCOUT MODE
AXIAL MODE
STATIC XRON MODE
STATIC XROFF MODE
TUBE HEAT MODE
DAS MODE
TUBE CAL MODE
BIOPSY MODE
CINE
HELICAL
ROTGENCAL MODE

2.5.2 Private Data Elements

Refer to **Appendix B** for a complete listing of private data elements used with this implementation.

2.6 Configuration

2.6.1 AE Title/Presentation Address Mapping

The LightSpeed system allows the user to “add”, “Remove”, or “Update the mapping of remote AE Titles to IP Addresses and Ports. These options can be selected from the “Remote Host Selection” menu displayed by choosing “Select Remote Host” from the “Network” pull-down menu from the local database manager.

2.6.2 Configurable Parameters

The following fields are configurable for the DICOM Server AE:

- Local AE Title (the machine hostname)
- Local IP Address
- Local IP Netmask
- Max PDU length
- Time-outs, which are set for all hosts, are configurable in dcs.cfg:
 - *Association time-out - *bi_assoc_tio*
 - *Session timeout - *bi_session_tio*
 - *Create timeout - *bi_ncreate_tio*

Note: All configurations should be performed by a GE Field Service Engineer.

Note: The local port on which the LightSpeed system receives DICOM incoming TCP connections is port 4006.

2.7 Support of Extended Character Sets

In addition to the DICOM default character set, *LightSpeed* supports the ISO IR 100 Latin alphabet #1 supplementary set for the purpose of interchange.

3 MEDIA STORAGE CONFORMANCE STATEMENT

3.1 Introduction

This conformance statement specifies the LightSpeed conformance to the DICOM V3.0 Media Interchange. It details the DICOM Media Storage Application Profiles and roles that are supported by this product in its version 1.0.

Note the format of this section follows the format of the DICOM Standard Part 2 (conformance) Annex A, hence the paragraph numbering scheme. Please refer to that part of the standard while reading this section.

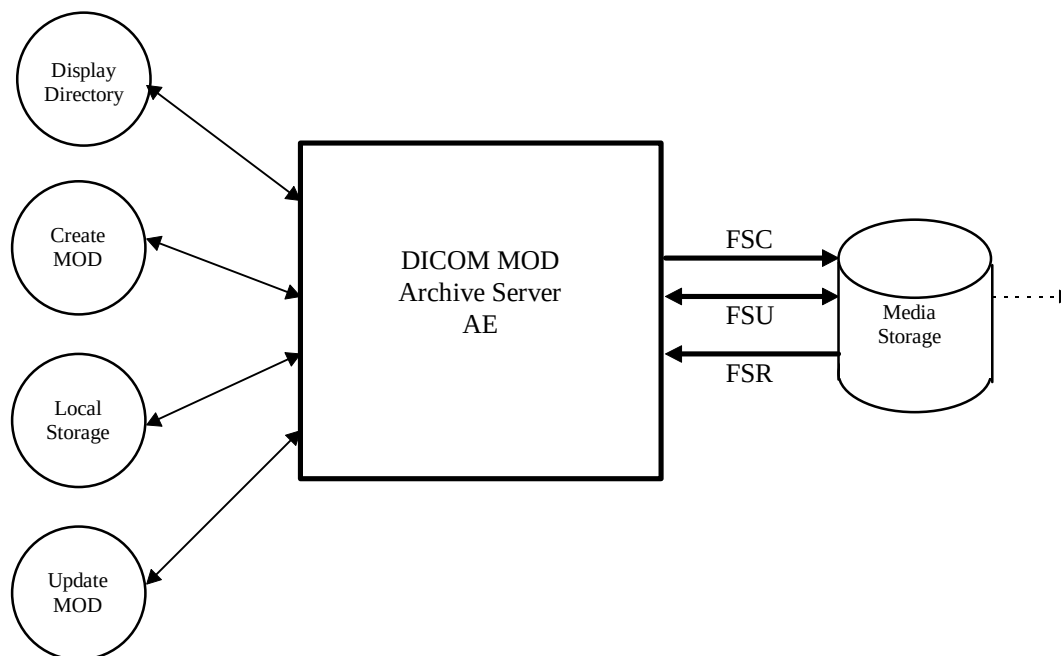
3.2 Implementation Model: MOD Archive Server

All DICOM functionality on the LightSpeed product is handled by the DICOM Server Application Entity (AE). The DICOM Server AE is commanded to perform DICOM services through the buttons and menu selections on the main user interface panel.

3.2.1 Application Data Flow Diagram

The MOD Archive Server creates and/or updates the 5.25-inch (130mm) MOD media with various DICOM SOP instances. It can process CT, MR, Secondary Capture and Overlay IOD's.

Illustration 3-1: Specific AE Application Model



The MOD Archive Server AE has a local storage that may contain various SOP instances. These may have been obtained by original creation, network (DICOM or proprietary) or by

removable media using other application entities. These instances are external to this conformance claim and the origin of SOP instances is outside the scope of this claim.

The MOD Archive Server AE can initialize Media by acting as an FSC to create a new DICOM File-set on a 2.3GB MOD media or a 1.2GB MOD media. It initializes the DICOM File-set and writes the specified SOP instances onto the MOD. The SOP instances written will be limited to instances that match the criteria of one of the Application Profiles that is supported. When updating media, a pre-existing File-set will be updated with the selected SOP instances that match one of the supported Application Profiles.

3.2.2 Functional Definitions of AE's

This Server has only one Application Entity: the MOD Archive Application.

The MOD Archive Application can perform these functions:

- It can initialize (create DOS filesystem) a piece of media, writing a new label and DICOM File-set onto the media. ("LABEL")
- It can update a piece of media by adding new SOP instances to an already existing DICOM File-set from local storage. ("SAVE")
- It can display a directory listing of the File-set on a piece of media. (QUERY)
- It can copy SOP instance from the MOD onto local storage. ("RESTORE")

3.2.3 Sequencing of Real World Activities

The updating function can only be performed on a piece of media that has already had a DICOM File- set created.

The display function can only be performed on a piece of media that has already has a DICOM File-set created. With no SOP instances having been added, the directory will be displayed empty.

The copy function can only be performed with a piece of media that has been updated and has DICOM SOP instances in the File-set.

There are no other sequencing requirements.

3.2.4 File Meta Information for Implementation Class and Version

The File Meta-Information for this implementation is:

- File Meta-Information Version 1
- LightSpeed Implementation UID..... 1.2.840.113619.6.55
- Implementation Version Name 1.0

3.3 AE Specifications

3.3.1 MOD Archive Specification

The MOD Archive provides standard conformance to DICOM Interchange Option of the Media Storage Service Class. The Application Profiles and roles are listed in Table 3.1-1.

Table 3.1-1: Application Profile, Activities and Roles for MOD Update

Application Profiles Supported	Real World Activity	Role	SC Option
STD-CTMR-MOD23	Create MOD	FSC	Interchange
	Update MOD	FSU	Interchange
	Display Directory	FSR	Interchange
	Copy To Local Storage	FSR	Interchange
STD-CTMR-MOD12	Create MOD	FSC	Interchange
	Update MOD	FSU	Interchange
	Display Directory	FSR	Interchange
	Copy To Local Storage	FSR	Interchange
PRI-CTMR-MOD12	Update MOD	FSU	Interchange
	Display Directory	FSR	Interchange
	Copy To Local Storage	FSR	Interchange

The MOD Archive Application will query the user before initializing media when a File-set is found on the media and an initialize operation has been requested.

Depending on the physical media type being 2.3GB or 1.2GB media, the system will apply the correct characteristics as an FSC to meet the appropriate Application Profile.

As an FSU role, the system will produce directory sequences for each SOP instance that is added to the File-set consistent with the STD-CTMR-MOD {12,23} profile and it will make the File-set directory so that an early installed base system that only understands the PRI-CTMR-MOD12 profile will not be able to use the media. An upgrade to the installed base system will be required to learn how to be a FSU or FSR of the STD-CTMR-MOD {12,23} profile directory sequences.

3.3.1.1 File Meta Information for the Application Entity

Following are the values set in the File Meta Information for this AE:

- Source Application Entity Title Not used by the MOD Archive Application
- Private Information Creator UID Not used by the MOD Archive Application
- Private Information..... Not used by the MOD Archive Application

3.3.1.2 Real World Activities

3.3.1.2.1 Real World Activity: Create MOD Request

The MOD Archive Application acts as an FSC using the Interchange option when requested to initialize ("LABEL") the media.

The AE will use the appropriate Application Profile depending on the physical media type that is in the optical disk drive (be it 2.3GB or 1.2GB media).

The MOD Archive Application will label the media and take the user provided list of SOP instances eliminating any SOP instances on that list that does not correspond to one of the Application Profiles in Table 3.1-1. These SOP instances are written to the media and a corresponding DICOMDIR sequence record is created. The determination of the potentially applicable Application Profile is dependent on the type of media and the associated software on which the AE has been invoked.

3.3.1.2.1.1 Application Profiles for the RWA: Create MOD

For the list of Application Profiles that invoke this AE for the Create MOD RWA, see Table 3.1-1.

For extensions and specialization's see section 3.4.

3.3.1.2.2 Real World Activity: Display Directory

The MOD Archive Application acts as an FSR using the Interchange option when requested to provide a directory listing.

When an MOD update Application is requested to provide a directory listing it will read the File-set and display the DICOMDIR directory entries for those SOP instances in the File-set that correspond to the user selected Application Profile.

3.3.1.2.2.1 Application Profiles for the RWA: MOD Directory Listing

For the list of Application Profiles that invoke this AE for the MOD directory listing RWA, see Table 3.1-1.

There are no extensions or specializations.

3.3.1.2.2.2 Options:

None applicable.

3.3.1.2.3 Real World Activity: Copy to Local Storage

The MOD Archive Application acts as an FSR when copying from the MOD to local storage.

The MOD Archive Application will copy any SOP Instance selected from an MOD Directory list from the MOD to the local storage upon request. The MOD Directory listing

Real-World Application will filter out the SOP Instances that do not match the Application Profile.

3.3.1.2.3.1 Application Profiles for the RWA: Copy to Local Storage

For the list of Application Profiles that invoke this AE for the MOD Copy to Local Storage RWA, see Table 3.1-1. For extensions and specialization's see section 3.4.

3.3.1.2.3.2 Options:

None applicable.

3.3.1.2.4 Real World Activity: Update MOD

The MOD Archive Application acts as an FSU using the Interchange option when requested to update an MOD.

The MOD Archive Application will take the selected list of SOP instances and eliminate any SOP instance that does not correspond to permissible SOP instances listed in the Application Profiles Table 3.1-1. The remaining SOP instances are written to the media that is found in the MOD Archive Application disk drive. The determination of the potentially applicable Application Profile is dependent on the type of media and associated software on which the AE has been invoked.

3.3.1.2.4.1 Application Profiles for the RWA: Update MOD Request

For the list of Application Profiles that invoke this AE for the MOD Copy to Local Storage RWA, see Table 3.1-1. For extensions and specialization's see section 3.4.

3.3.1.2.4.2 Options:

None applicable.

3.4 Standard, Augmented and Private Application Profiles

3.4.1 Class and Profile Identification

The Class of Application Profiles defined here is for Computed Tomography and Magnetic Resonance Imaging Clinical applications.

The identifier for this class is CTMR-MOD.

The specific Application Profile in this class is shown in Table 3.3.1-1.

Table 3.3.1-1: CTMR MOD Profiles

Application Profile	Identifier	Description
CT/MR Studies on 2.3GB MOD media	STD-CTMR-MOD23	Handles single 12 or 16 bit grayscale compressed / uncompressed images.

Application Profile	Identifier	Description
CT/MR Studies on 1.2GB MOD media	STD-CTMR-MOD12	Handles single 12 or 16 bit grayscale compressed / uncompressed images.
CT/MR Studies on 1.2 GB MOD media	PRI-CTMR-MOD12	Handles single 12 or 16 bit grayscale compressed / uncompressed images.

3.4.2 Clinical Contexts

This Application Profile facilitates the interchange of primary CT and MR images as well as processed CT and MR images as Secondary Capture and Standalone Overlay Images with certain defined attributes. CT, MR, SC and OV images may coexist within the same File-set.

Typical interchanges would be between acquisition devices, archives and workstations within and between institutions.

3.4.2.1 Roles and Service Class Options

This Application Profile Class uses the Media Storage Service Class defined in PS3.4 with the Interchange Option.

The Application Entity shall support one or more roles of the File-set Creator, File-set Reader and File-set Updater defined in PS3.10.

3.4.2.1.1 File Set Creator

The Application Entity acting as a File-set Creator generates a File Set under the PRI-CTMR-MOD12 Application Profile Class. Typical entities using this role would include CT or MR equipment and archive systems that generate a patient record to transfer to another institution. File Set Creator shall be able to generate the Basic Directory SOP Class in the DICOMDIR File with all types of Directory Records related to the SOP Classes stored in the File-set.

FSC for the MOD profile has no concept of finalizing the disc at the completion of most recent write session. This is a requirement of CD-R and does not apply to the MOD profiles.

3.4.2.1.2 File Set Reader

The role of File Set Reader is used by the Application Entities that receive a transferred File Set. Typical entities using this role would include display workstations and archive systems that receive a patient record transferred from another institution. File Set Readers shall be able to read all the SOP Classes defined for the specific Application Profile for which a Conformance Statement is made using all the defined Transfer Syntax.

3.4.2.1.3 File Set Updater

The role of File Set Updater is used by Application Entities that receive a transferred File Set and update it by the addition of information. Typical entities using this role would

include analytic workstations that for instance may add to the File-set an information object containing a processed (e.g., edge-enhanced) image. Stations that update patient information objects would also use this role. File-set Updaters do not have to read the images. File-set Updaters shall be able to generate one or more of the SOP Instances defined for the specific Application Profile for which a conformance statement is made and to read and update the DICOMDIR file.

FSU shall offer the ability to allow multi-session writes (additional information may be subsequently added to the volume).

3.4.3 Standard Application Profiles

The MOD Archive Server supports the standard Application Profile: CTMR-MOD as an FSC, FSR or FSU.

3.4.3.1 STD-CTMR-MOD12 Class Profile

3.4.3.1.1 SOP Classes and Transfer Syntax's

This class of Application Profiles is based on the Media Storage Service Class with the Interchange Option. (See PS3.4).

SOP Classes and corresponding Transfer Syntax's supported by the STD-CTMR-MOD12 Application Profiles is also specified in the Table 3.4.3.1.1-1.

Table 3.4.3.1.1-1: IOD's and Transfer Syntax's for CTMR-MOD Application Profiles

Information Object Definition	SOP Class UID	Transfer Syntax and UID	FSC Req.	FSR Req.	FSU Req.
Basic Directory	1.2.840.10008.1.3.10	Explicit VR Little Endian Uncompressed 1.2.840.10008.1.2.1	M	M	M
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	Explicit VR Big Endian Uncompressed 1.2.840.10008.1.2.2	O	M	O
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	JPEG Lossless Process Selection Value 14 1.2.840.10008.1.2.4.70	O	M	O
MR Image Storage	1.2.840.10008.5.1.4.1.1.4	Explicit VR Big Endian Uncompressed 1.2.840.10008.1.2.2	O	M	O
MR Image Storage	1.2.840.10008.5.1.4.1.1.4	JPEG Lossless Process Selection Value 14 1.2.840.10008.1.2.4.70	O	M	O
SC Image Storage	1.2.840.10008.5.1.4.1.1.7	Explicit VR Big Endian Uncompressed 1.2.840.10008.1.2.2	O	M	O

Information Object Definition	SOP Class UID	Transfer Syntax and UID	FSC Req.	FSR Req.	FSU Req.
SC Image Storage	1.2.840.10008.5.1.4.1.1.7	JPEG Lossless Process Selection Value 14 1.2.840.10008.1.2.4.70	O	M	O
Standalone Overlay	1.2.840.10008.5.1.4.1.1.8	Explicit VR Big Endian Uncompressed 1.2.840.10008.1.2.2	O	O	O
Standalone Overlay	1.2.840.10008.5.1.4.1.1.8	JPEG Lossless Process Selection Value 14 1.2.840.10008.1.2.4.70	O	O	O

M = Mandatory **O** = Optional

Note: The MOD Archive Application chooses the JPEG Lossless compression (selection value 1) as the default transfer syntax for storing images on the media. But the transfer syntax is selectable on a per media basis at install time using a method defined by the Application.

3.4.3.1.2 Physical Media and Media Formats

The CT/MR Application Profiles in the STD-CTMR-MOD12 require the 130mm 1.2GB 512 Byte per Sector Magneto-Optical Rewriteable physical media with the PC File System as defined in PS3.12.

3.4.3.1.3 Logical Format

The STD-CTMR-MOD12 Application profile media format conforms to the Logical Format specification in PS3.12. The following table 3.3.3.1.3-1 defines the specific values used for this profile.

Table 3.3.3.1.3-1: Boot Sector for STD-CTMR-MOD12 conformant media

Bytes(s)	Big Endian	As Stored	Description
00 - 02		0xEB3C90	Jump instruction to loader (NOPs)
03 - 10	0x4d53444F53342E30		"MSDOS4.0" The formatting DOS (vendor specific)
11 - 12	0x0200	0x0002	512 bytes/sector
13	32	0x20	sectors/cluster
14-15	0x0001	0x0100	1 sector in boot record
16	0x02	0x02	2 File Allocation Tables (FAT)
17 - 18	0x0200	0x0002	512 root directory entries
19 - 20	0x0000	0x0000	Flag for more than 65536 sector/disk. Use offset 32 value
21	0xF8	0xF8	Flag for disk type; F0H if not otherwise specified
22 - 23	143	0x8F00	143 sectors/FAT
24 - 25	31	0x1F00	31 sectors/track
26 - 27	1	0x0100	1 side (head) per disk
28 - 31	00000000	0x0...0	0 reserved or hidden sectors
32 - 35	1163306	0x2AC01100	1163306 Total sector/disk. Varies from disk to disk
36 - 37	0x0080	0x8000	Physical Drive number = 80
38	0x29	0x29	Extended boot record signature = 41
39 - 42	<i>Date Stamp</i>		Volume serial number
43 - 53	0x47454D535F4449434F4D20		"GEMS-DICOM"
54 - 61	0x0...0	0x0...0	The file system label
62 - 509	-		Don't care. Any contents acceptable
510 - 511	0xAA55	0x55AA	Signature flag

3.4.3.1.4 Directory Information in DICOMDIR

Conformant Application Entities shall include in the DICOMDIR File a Basic Directory IOD containing Directory Records at the Patient and subsidiary levels appropriate to the SOP Classes in the File-set. All DICOM files in the File-set incorporating SOP Instances defined for the specific Application Profile shall be referenced by the Directory Records.

All DICOM SOP Instance files in the File-set are stored in the MSDOS filesystem subdirectory \GEMS and are referenced from the DICOMDIR directory records.

Refer to Appendix C for a complete listing of all the optional modules and optional attributes used in the DICOMDIR definition. Please note that the mandatory attributes in each of the directory records as mentioned in Part 3 Addendum (Basic

Directory Information Object Definition) are not listed in the appendix, but are supported by the implementation.

Note: The modality attribute (0008,0060) in the DICOMDIR Series records should be CT/MR for Secondary Capture and Standalone Overlay SOP Class Images. Otherwise the LightSpeed DICOM media display browser will not list-up the series for contents rendering the retrieval to local storage impossible.

3.4.3.1.4.1 Additional Keys

None.

3.4.3.1.5 Other Parameters

This section defines other parameters common to all specific Application Profiles in the PRI-CTMR-MOD12 class that need to be specified in order to ensure interoperable information interchange.

3.4.3.1.6 Image Attribute Values

The attributes listed in Table 3.3.3.1.5.1-1 used within the CT/MR/SC/Standalone Overlay Image files, shall take the values specified.

Table 3.3.3.1.5.1-1: CTMR-MOD Profiles Required Image Attribute Values for CT and MR, SC and OV Images

Attribute	Tag	Value
Modality	(0008,0060)	CT/MR
Photometric Interpretation	(0028,0004)	MONOCHROME2
Bits Allocated	(0028,0100)	16
Bits Stored	(0028,0101)	16

Overlay data if present shall be encoded in Overlay Data (60XX,3000).

3.4.3.1.6.1 Attribute Value Precedence

None.

3.4.3.2 STD-CTMR-MOD23 Class Profile

3.4.3.2.1 SOP Classes and Transfer Syntax's

This class of Application Profiles is based on the Media Storage Service Class with the Interchange Option. (See PS3.4).

SOP Classes and corresponding Transfer Syntax's supported by the STD-CTMR-MOD23 Application Profiles is also specified in the Table 3.3.3.1.1-1.

3.4.3.2.2 Physical Media and Media Formats

The CT/MR Application Profiles in the STD-CTMR-MOD23 require the 130mm 2.3GB 512 Byte per Sector Magneto-Optical Rewriteable physical media with the PC File System as defined in PS3.12.

3.4.3.2.3 Logical Format

The STD-CTMR-MOD23 Application profile media format conforms to the Logical Format specification in PS3.12. The following table 3.3.3.2.3-1 defines the specific values used for this profile.

Table 3.3.3.2.3-1: Boot Sector for STD-CTMR-MOD23 conformant media

Bytes(s)	Big Endian	As Stored	Description
00 – 02		0xEB3C90	Jump instruction to loader (NOPs)
03 – 10	0x4d53444F53342E30		“MSDOS4.0” The formatting DOS (vendor specific)
11 – 12	0x0200	0x0002	512 bytes/sector
13	64	0x40	sectors/cluster
14-15	0x0001	0x0100	1 sector in boot record
16	0x02	0x02	2 File Allocation Tables (FAT)
17 – 18	0x0200	0x0002	512 root directory entries
19 – 20	0x0000	0x0000	Flag for more than 65536 sector/disk. Use offset 32 value
21	0xF8	0xF8	Flag for disk type; F0H if not otherwise specified
22 – 23	138	0x8A00	138 sectors/FAT
24 – 25	62	0x3E00	62 sectors/track
26 – 27	1	0x0100	1 side (head) per disk
28 – 31	00000000	0x0...0	0 reserved or hidden sectors
32 – 35	2244896	0x20412200	2244896 Total sector/disk. Varies from disk to disk
36 – 37	0x0080	0x8000	Physical Drive number = 80
38	0x29	0x29	Extended boot record signature = 41
39 – 42	<i>Date Stamp</i>		Volume serial number
43 – 53	0x47454D535F4449434F4D20		“GEMS-DICOM”
54 – 61	0x0...0	0x0...0	The file system label
62 – 509	-		Don't care. Any contents acceptable
510 – 511	0xAA55	0x55AA	Signature flag

3.4.3.2.4 Directory Information in DICOMDIR

Conformant Application Entities shall include in the DICOMDIR File a Basic Directory IOD containing Directory Records at the Patient and subsidiary levels appropriate to the SOP

Classes in the File-set. All DICOM files in the File-set incorporating SOP Instances defined for the specific Application Profile shall be referenced by the Directory Records.

All DICOM SOP Instance files in the File-set are stored in the MSDOS filesystem subdirectory \GEMS and are referenced from the DICOMDIR directory records.

Refer to *Appendix C* for a complete listing of all the optional modules and optional attributes used in the DICOMDIR definition. Please note that the mandatory attributes in each of the directory records as mentioned in Part 3 Addendum (Basic Directory Information Object Definition) are not listed in the appendix, but are supported by the implementation.

Note: The modality attribute (0008,0060) in the DICOMDIR Series records should be CT/MR for Secondary Capture and Standalone Overlay SOP Class Images. Otherwise the LightSpeed DICOM media display browser will not list-up the series for contents rendering the retrieval to local storage impossible.

3.4.3.2.4.1 Additional Keys

None.

3.4.3.2.5 Other Parameters

This section defines other parameters common to all specific Application Profiles in the PRI-CTMR-MOD12 class that need to be specified in order to ensure interoperable information interchange.

3.4.3.2.5.1 Image Attribute Values

The attributes listed in Table 3.3.3.1.5.1-1 used within the CT/MR/SC/Standalone Overlay Image files, shall take the values specified.

Overlay data if present shall be encoded in Overlay Data (60XX,3000).

3.4.3.2.5.2 Attribute Value Precedence

None.

3.4.4 Private Application Profiles

The MOD Archive Server supports a private Application Profile: PRI-CTMR-MOD12 as an FSR or FSU.

3.4.4.1 PRI-CTMR-MOD12 Class Profile

3.4.4.1.1 SOP Classes and Transfer Syntax

This class of Application Profiles is based on the Media Storage Service Class with the Interchange Option. (See PS3.4).

SOP Classes and corresponding Transfer Syntax supported by the PRI-CTMR-MOD12 Application Profiles is specified in the Table 3.3.3.1.1-1.

Note: The MOD Archive Application chooses the JPEG Lossless compression (selection value 1) as the default transfer syntax for storing images on the media. But the transfer syntax is selectable on a per media basis at install time using a method defined by the Application.

The SOP instance files in the Media File-set in the PRI-CTMR-MOD12 profile uses a non-standard JPEG compression process resulting in interoperability problems when decoded using readily available industry decompression routines. See section 3.4.2 for details of the encoding problem unique to the PRI-CTMR-MOD12 profile.

3.4.4.1.2 Physical Media and Media Formats

The CT/MR Application Profiles in the PRI-CTMR-MOD12 require the 130mm 1.2GB 512 Byte per Sector Magneto-Optical Rewritable physical media with the PC File System as defined in PS3.12.

3.4.4.1.3 Logical Format

The PRI-CTMR-MOD12 Application profile media format conforms to the Logical Format specification in PS3.12. The table 3.3.3.1.3-1 defines the specific values used for this profile.

3.4.4.1.4 Directory Information in DICOMDIR

Conformant Application Entities shall include in the DICOMDIR File a Basic Directory IOD containing Directory Records at the Patient and subsidiary levels appropriate to the SOP Classes in the File-set. All DICOM files in the File-set incorporating SOP Instances defined for the specific Application Profile shall be referenced by the Directory Records.

All DICOM SOP Instance files in the File-set are stored in the MSDOS filesystem subdirectory \GEMS and are referenced from the DICOMDIR directory records.

Refer to *Appendix C* for a complete listing of all the optional modules and optional attributes used in the DICOMDIR definition. Please note that the mandatory attributes in each of the directory records as mentioned in Part 3 Addendum (Basic Directory Information Object Definition) are not listed in the appendix, but are supported by the implementation.

Directory records found in the PRI-CTMR-MOD12 profile do not contain the Attribute "Frame of Reference" (0020,0052). This Attribute was specified as Required after the first release of this Application Profile.

Note: The modality attribute (0008,0060) in the DICOMDIR Series records should be CT/MR for Secondary Capture and Standalone Overlay SOP Class Images. Otherwise the LightSpeed DICOM media display browser will not list-up the series for contents rendering the retrieval to local storage impossible.

3.4.4.1.4.1 Additional Keys

None.

3.4.4.1.5 Other Parameters

This section defines other parameters common to all specific Application Profiles in the PRI-CTMR-MOD12 class that need to be specified in order to ensure interoperable information interchange.

See 3.4.2 for JPEG encoding problem in the PRI-CTMR-MOD12 profile.

3.4.4.1.5.1 Image Attribute Values

The attributes listed in Table 3.3.3.1.5.1-1 used within the CT/MR/SC/Standalone Overlay Image files, shall take the values specified.

Overlay data if present shall be encoded in Overlay Data (60XX,3000).

3.4.4.1.5.2 Attribute Value Precedence

None.

3.5 Extensions, Specializations and Privatization of SOP Classes and Transfer Syntax

3.5.1 Extensions, Specialization's and Privatization's of SOP Classes

The CT/MR SOP Class Images have definitions extended for Defined Terms and include GE specific Private Data elements. The following sections describe the details for these SOP classes.

3.5.1.1 SOP Specific Conformance Statement for CT SOP Class

Refer to **section 2.4** for standard extensions and **Appendix B** for private data elements.

3.5.1.2 SOP Specific Conformance Statement for MR SOP Class

Refer to **section 2.4** for standard extensions and **Appendix B** for private data elements.

3.5.2 Private Transfer Syntax Specification

The SOP instance files in the Media File-set in the PRI-CTMR-MOD12 profile uses a non-standard JPEG compression process resulting in interoperability problems when decoded using readily available industry standard JPEG decompression routines.

GEMS "PRI-CTMR-MOD12" DICOM MOD "JPEG" Encoding Information

Note: *Some earlier versions of that were intended to write JPEG Lossless Process Selection Value 14 1.2.840.10008.1.2.4.70 actually contained errors in byte ordering of encapsulation of fragment item tags, selection of entropy coding*

tables, and calculation of difference values from predictors. These errors cause the compressed images to be unreadable by non-GEMS software. All future GEMS LightSpeed software will read (but not write) archive media written with these errors.

The early release product software had a JPEG encode routine that is called the BAD_ENCODER for the following explanation.

In the BAD_ENCODER produced 'SOS' (0xFFDA) header, the entropy coding table selector codes are 0x11. They should be 0x00. This mistake is the key that specifies the BAD_ENCODER processing problems

The BAD_ENCODER computation of the pixel value difference to be entropy encoded is computed as (Predictor - value) when it should be calculated as (value - Predictor)⁴. The result is that the decompression via industry standard packages result in a negative of the original image pixel value.

The BAD_ENCODER predictor value used at the beginning of each line uses the last value of the previous line in the image. The JPEG standard says the predictor for the first line element of each line should be the first element of the line immediately above the current line. The first line, first element predictor should be the unsigned value that is half the full-scale range for the "sample precision". The first line, first element predictor is correctly determined in the BAD_ENCODER.

Explanation

This section is provided to explain what was done during Image encoding so that people who wish and need to decode the images can modify the available JPEG codec software. These packages are available from the WEB and were originally developed at Stanford and Cornell as well as others in the industry.

What was done in Generation 1 DICOM MOD encoding by GEMS:

In the JPEG codec output, the JPEG 'SOS' header defines the Huffman table selector codes to find the appropriate Huffman table (FF DA 00 08 01 00 11 01 00 00 ...). The GEMS DICOM Access Kit coded these as 0x11. They should have been 0x00. Zero was the number assigned in the "DHT" header. This error is a primary key that is used in the GEMS Generation 2 system to determine that the JPEG data was encoded using the GEMS Generation 1 coding technique.

In Generation 1 images, the predictor value used at the beginning of each line is the last pixel value from the previous line in the image (Assumes Left to Right, Top to Bottom scanning of the pixels).

The Generation 2 and industry compliant predictor for the first line element of each line should be the first element of the line above the current line. For the first line, first element

⁴ No specific direct confirmation of the (value - Predictor) policy was found, but the graphic on page 185 of the book "JPEG still image data compression standard" by Pennebaker shows the predictor entering the adder as a negative element and the value entering as positive element. Like wise, both Cornell and Stanford packages perform the computation as (value - Predictor).

predictor should be the unsigned value that is half the full-scale range for the "sample precision" (GEMS uses $(1 \ll 15) = 32768$).

The GEMS Generation 1 JPEG computation of the difference to be Huffman encoded was computed as (Predictor - value) when it should have been calculated as (value - Predictor) . The result is that the decompression via industry standard packages results in a negative of the original image and a breakdown of the pixel basis when the last pixel on a line is different from the first pixel on the next line.

It was recently (Jul-98) learned that the JPEG software distributed by the Independent JPEG Group does not include an important feature found in the GEMS, the Stanford, and the Cornell JPEG packages with respect to handling 16 bit deep images. In the case where the "huffDECODE" would return a category 16 value indicating that 16 bits should be acquired from the bit stream, the correct action is to add 32768 to the current value and to NOT get the number from the bit stream. Without this test [if (s == 16) dx += 32768; else ...], the image will exhibit a spatial offset resulting from the incorrect synchronization of the input bit stream.

This version and all future GEMS LightSpeed software will read (but not write) the Installed Base Generation 1 media. In addition, the creation of Generation 2 media as has been initiated and the new format media will be Industry compatible JPEG and will NOT use the Multi-Fragment segmentation. The ability to handle Multi-Segment data sets is also being deployed to handle the small number of Multi-Fragment images that have reached media.

3.6 Configuration

The MOD Archive Application may have either a 1.2GB or a 2.3GB drive installed.

Table 3.6-1: Supported Profiles for various Drive Configurations

Disk Drive Installed	Profiles Supported
1.2GB Optical Disk Drive	PRI-CTMR-MOD12
	STD-CTMR-MOD12
2.3GB-Optical Disk Drive	PRI-CTMR-MOD12
	STD-CTMR-MOD12
	STD-CTMR-MOD23

3.7 Support of Extended Character Sets

The MOD Archive Application will support copy of SOP instances containing the ISO IR 100 (Latin alphabet No. 1, supplementary set) and DICOM default character sets as defined in PS3.5.

4 PRINT SCU CONFORMANCE

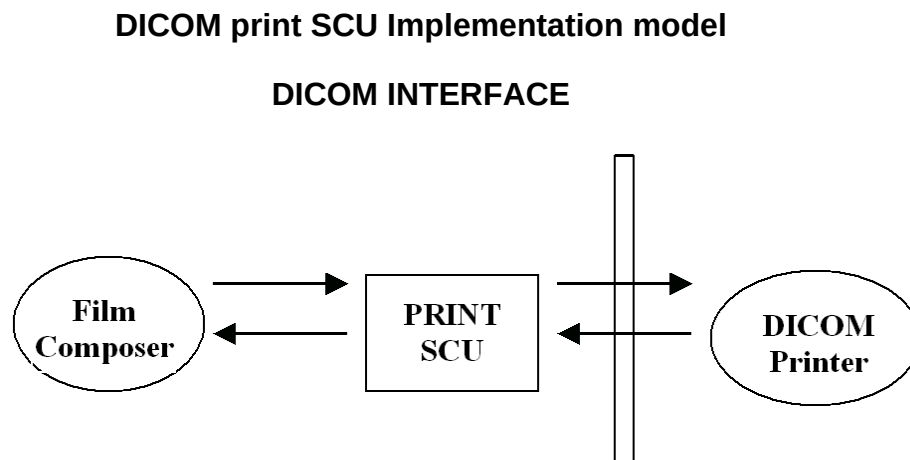
4.1 Introduction

This conformance statement specifies the LightSpeed conformance to the DICOM Print SCU Application Profile. The LightSpeed product uses DICOM Print SCU to print images on DICOM Compliant Printers.

4.2 Implementation Model: PRINT SCU

The DICOM Print SCU is a DICOM print filter that provides the capability to print images to DICOM printers. The DICOM Print filter acts as an SCU of the DICOM print management SOP class.

4.2.1 Application Data Flow Diagram



The Film Composer is the User interface and this is used to initiate the local real world activity. The user issues the print request using Film Composer. Film composer allows printer selection and it composes the pre-formatted film file. This film file is interpreted by Print SCU and it sends the appropriate messages to DICOM print SCP running on a DICOM printer.

4.2.2 Functional Definition of AE's

DICOM Print SCU Establishes the Association to the requested printer for printing the composed film.

4.2.3 Sequencing of Real-World Activities

- The DICOM printer is installed through a camera installation process. The DICOM printer is selected from the Film Composer Interface for Manual Filming. Autofilming is enabled using the camera setup via the camera installation process.

- Images may be acquired for printing in the following manner:
 - Images to be printed may be manually dropped from Viewing applications into the manual film composer slots.
 - Images to be printed may be automatically dropped from Viewing applications into the manual film composer slots.
 - Images to be printed may be manually dropped from Viewing applications into the auto film composer slots.
 - Image series to be printed may be automatically dropped from the PrintSeries application into the manual film composer slots.
 - Images to be printed may be automatically dropped from the Scanning Application into the auto film composer slots.
- Based upon the filming mode (and the attributes) used to acquire images into the film composer, films will be automatically printed when the film session is full, or the user must press the Print Button to print the images.
- The Print SCU retrieves the Print SCP status by using the N-GET service of Printer SOP Class. If the printer returns a FAILURE status the print session will be terminated. The printer status is not read.
- The film session is created using the N-CREATE service. The print session will be terminated if the printer fails to create the film session.
- The film box is created using the N-CREATE service. The print session will be terminated if the printer fails to create the film box.
- The images are placed on the film box by using the N-SET. If the printer fails to place the image on the film box, the print session will be terminated.
- The film will be printed using the N-ACTION. Only film box printing is supported. If the printer fails to print the film, the print session will be terminated.
- After the successful N-ACTION, the film box will be deleted using the N-DELETE. Any N-EVENT-REPORTS given to the PRINT SCU will be received but the data is ignored. After the film box is deleted, the association will be closed.
- Upon the successful completion of the above sequencing and the association is closed, the user will be notified of the successful print session.

4.3 AE Specifications

4.3.1 DICOM Print SCU AE Specification

Print SCU provides Standard Conformance to the following DICOM 3.0 SOP classes as an SCU:

SOP CLASS Name	SOP CLASS UID
Basic grayscale print management meta SOP class	1.2.840.10008.5.1.1.9
Print Job SOP class	1.2.840.10008.5.1.1.14

4.3.1.1 Association Establishment Policy

4.3.1.1.1 General

The Film Composer provides options to indicate the printer AE title and Printer host name. These parameters are used by the Print SCU to establish the association with the desired printer.

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
---------------------------------	------------------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU for association initiated by the DICOM Print SCU AE is:

Maximum Length PDU	16 Kbytes
---------------------------	------------------

4.3.1.1.2 Number of Associations

The Print SCU will initiate only one association with printer. Multiple associations are not supported.

4.3.1.1.3 Asynchronous Nature

The print SCU does not support asynchronous operations. All operations will be performed synchronously.

4.3.1.1.4 Implementation identifying information

The Implementation UID allows unique identification of a set of products that share the same implementation.

The Implementation UID for this ID/Net v3.0 Implementation is:

LightSpeed Implementation UID	1.2.840.113619.6.55
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4.3.1.2 Association Establishment Policy

Print SCU initiates the association with the Printer.

4.3.1.2.1 Real World Activity

4.3.1.2.1.1 Associated Real-World Activity – “Print”

The Film Composer allows the user to select printers and it also allows the user to drag and drop the images (from the viewer application) into the film. It also allows the user to manipulate some print parameters like film format and number of copies to print. When the user presses the "Print" Button, the Film composer communicates this request to Print SCU, which then tries to establish the association with the requested printer and transmits the images for printing.

4.3.1.2.1.2 Proposed Presentation Contexts

The Proposed Presentation Context Table for the Print SCU is as shown in following Table.

Presentation Context Table – Proposed					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name	UID		
Basic Grayscale Print Management Meta SOP Class	1.2.840.10008.5.1.1.9	DICOM Implicit VR Little Endian Transfer Syntax	1.2.840.10008.1.2	SCU	None
Verification SOP Class	1.2.840.10008.1.1	DICOM Implicit VR Little Endian Transfer Syntax	1.2.840.10008.1.2	SCU	None
Printer SOP Class	1.2.840.10008.5.1.1.16	DICOM Implicit VR Little Endian Transfer Syntax	1.2.840.10008.1.2	SCU	None
Print Job SOP Class	1.2.840.10008.5.1.1.14	DICOM Implicit VR Little Endian Transfer Syntax	1.2.840.10008.1.2	SCU	None

Note: Certain Presentation Contexts may be negotiated that are not used during the association. See the following sections for the DICOM Print services performed by the LightSpeed.

4.3.1.2.2 SOP Specific Conformance Statement

The Print SCU supports the following mandatory SOP classes, which are defined under the Basic Grayscale Print Management Meta SOP Class.

The Print SOP Classes supported by Print SCU:

NAME	UID
Basic Film Session SOP Class	1.2.840.10008.5.1.1.1
Basic Film Box SOP Class	1.2.840.10008.5.1.1.2
Basic Grayscale Image Box SOP Class	1.2.840.10008.5.1.1.4
Printer SOP Class	1.2.840.10008.5.11.16

The Print SCU does not support any optional SOP Classes.

Note: The asterisk (*) preceding some of the SOP attributes in the following sections indicates optional SCP attributes. The SCU is aware of these optional attributes and will not fail on the receipt of a response without the optional attributes; the limitation being that this information is not available for the User. Some SCP's will terminate the association with a failure upon the receipt of an optional attribute it does not support.

4.3.1.2.2.1 Basic Film Session SOP Class

The Print SCU supports the following DIMSE Service Elements for the Basic Film Session SOP Class.

N-CREATE – Requests the Print SCP to create an instance of Basic Film Session.

The following Attribute values are supported by the N-CREATE:

Attribute	DICOM Tag	Valid Range	Default Value
* Number of Copies	(2000, 0010)	1-99	Set by user
* Print Priority	(2000, 0020)	HIGH / MED / LOW	Set in Configuration file (Default value is HIGH)
* Medium Type	(2000, 0030)	CLEAR FILM BLUE FILM PAPER	Set in Configuration File
* Film Destination	(2000, 0040)	MAGAZINE PROCESSOR	Set in Configuration File

Note: * denotes that the attribute is optional for the SCU. However, we do provide values for all of these optional attributes and if the SCP does not support the requested value it may choose to either return a failure status or ignore the value provided and use its default value.

If Failure status is returned during N-CREATE operation of Film session the following action will be taken by Print SCU:

0x213 "Resource Limitation" message will be sent.

All other status "Failure" message will be sent.

In all the cases the print session will be terminated.

4.3.1.2.2.2 Basic Film box SOP Class

The Print SCU supports the following DIMSE Service Elements for the Basic Film Box SOP Class.

- N-CREATE – Requests the Print SCP to create an instance of Film Box.
- N-ACTION – Requests the Print SCP to print the Film Box onto Printer.
- N-DELETE – Requests the Print SCP to delete the Film Box Instance.

The Following Attribute values are supported:

Attribute	DICOM Tag	Valid Range	Default Value
Image Display Format	(2010, 0010)	STANDARD/C,R Printer Dependent	Set in User Interface
Reference Film Session Sequence	(2010, 0500)		
*Film Orientation	(2010, 0040)	PORTRAIT	Set in Configuration File
Film Size ID	(2010, 0050)	Sent zero length	Sent zero length
*Magnification type	(2010, 0060)	BILINEAR CUBIC REPLICATE NONE	Set in Configuration File
*Max Density	(2010, 0130)	0-4095	Set in Configuration File
Configuration Information	(2010, 0150)	Printer Dependent	Set in Configuration File
*Smoothing type	(2010, 0080)	Printer Dependent	Set in Configuration File
*Border density	(2010, 0100)	BLACK WHITE	Set in Configuration File
*Empty image density	(2010, 0110)	BLACK WHITE	Set in Configuration File
*Min density	(2010, 0120)	0-4094	Set in Configuration File
*Trim	(2010, 0140)	NO	NO

Note: Attributes “sent zero length” use the camera default values.

Note: * denotes that the attribute is optional for the SCU. However, we do provide values for all of these optional attributes and if the SCP does not support the requested value it may choose to either return a failure status or ignore the value provided and use its default value.

If Failure status is returned during N-CREATE operation of Film box following action will be taken by Print SCU.

0x213: “Resource Limitation” message will be sent.

0x106: “Unsupported Film Format” message will be sent.

All other status: "Failure" message will be sent

In all the cases the print session will be terminated.

If Failure status is returned during N-ACTION operation of Film box following action will be taken by Print SCU.

0xC602: "Unable to Create Print Job" message will be sent.

All other status: "Failure" message will be sent.

In all the cases the print session will be terminated.

If Failure status is returned during N-DELETE operation of Film box following action will be taken by Print SCU.

All the return status: "Failure" message will be sent and the print session will be terminated.

4.3.1.2.2.3 Basic Grayscale Image Box SOP Class

The Print SCU supports the following DIMSE Service Elements for Grayscale Image Box SOP Class.

N-SET – Requests the Printer to set the image box attributes.

The Following Attribute values are supported:

Attribute	DICOM Tag	Valid Range	Default Value
Image Position	(2020, 0010)	Based on Image Display Format	No Default Value
Pre-formatted Grayscale Image Sequence	(2020, 0110)		
Samples per pixel	(0028, 0002)	1	1
Photometric Interpretation	(0028, 0004)	MONOCHROME2	MONOCHROME2
Rows	(0028, 0010)	Image Dependent	Image Dependent
Columns	(0028, 0011)	Image Dependent	Image Dependent
Pixel Aspect Ratio	(0028, 0034)	1/1	1/1
Bits Allocated	(0028, 0100)	8	8
Bits Stored	(0028, 0101)	8	8
High Bit	(0028, 0102)	7	7
Pixel Representation	(0028, 0103)	0 (unsigned integer)	0 (unsigned integer)
*Polarity	(2020, 0020)	NORMAL	NORMAL

Note: * denotes that the attribute is optional for the SCU. However, we do provide values for all of these optional attributes and if the SCP does not support the requested value it may choose to either return a failure status or ignore the value provided and use its default value.

If Failure status is returned during N-SET operation of Image Box following action will be taken by Print SCU.

0xC605: "Resources temporarily not available" message will be sent.

All other status: "Failure" message will be sent.

In all the cases the print session will be terminated.

4.3.1.2.2.4 Printer SOP Class

N-GET DIMSE service is supported for the Printer SOP Class. If an N-EVENT-REPORT DIMSE service is received when the association is active, Print SCU handles the relevant states but the data received is ignored.

Print SCU issues the request to retrieve the following attributes:

Optional Attribute	DICOM Tag	Default Value
Printer Status	(2110, 0010)	Printer shall return Value
Printer Status Info	(2110, 0020)	Printer shall return Value
Printer Name	(2110, 0030)	Printer shall return Value
Manufacturer	(0008, 0070)	Printer shall return Value
Manufacturer Model Name	(0008, 1090)	Printer shall return Value
Device Serial No.	(1800, 1000)	Printer shall return Value
Software Versions	(1800, 1020)	Printer shall return Value
Date Last Calibrated	(1800, 1200)	Printer shall return Value
Time Last Calibrated	(1800, 1201)	Printer shall return Value

The Print SCU issues the N-GET service to retrieve the printer status. The status is processed as follows:

- If Printer status (2110, 0010) is NORMAL, the film is printed.
- If Printer status (2110, 0010) is FAILURE, the print job is terminated. The Printer Status Info (2110, 0020) attribute is not processed.
- If Printer status (2110, 0010) is WARNING, one of three things will happen:
 - a) If the Printer Status Info (2110, 0020) is "SUPPLY LOW" the status is displayed to the user and the print job continues.
 - b) If the Printer Status Info (2110, 0020) is "RECEIVER FULL" or "SUPPLY EMPTY" or "FILM JAM" the status is displayed to the user and the print job is aborted.
For all other Printer Status Info (2110, 0020) values, the status is ignored and the print job continues.

4.3.1.2.2.5 Print Job SOP Class

Print SCU looks for following attributes in N-EVENT REPORT data received from Print SCP. If Print SCU does not receive N-EVENT_REPORT it requests the Print SCP to retrieve the following set of attributes using N-GET.

Attribute Name	DICOM Tag	Default Value
Execution Status	(2100, 0020)	Value returned by Print SCP
Execution Status info	(2100, 0030)	Value returned by Print SCP
Print Priority	(2000, 0020)	Value returned by Print SCP
Creation Date	(2100, 0040)	Value returned by Print SCP
Creation Time	(2100, 0050)	Value returned by Print SCP
Printer Name	(2110, 0030)	Value returned by Print SCP
Originator	(2100, 0070)	Value returned by Print SCP

4.3.1.3 Association Acceptance Policy

The Print SCU does not accept associations.

4.4 Communication Profiles

4.4.1 Supported Communication Stacks (parts 8,9)

DICOM Upper Layer (Part 8) is supported using TCP/IP.

4.4.1.1 TCP/IP Stack

The TCP/IP stack is inherited from the UNIX Operating System.

4.4.1.1.1 API

Not Applicable

4.4.1.1.2 Physical Media Support

Ethernet v2.0, IEEE 802.3.

4.5 Standard Extended / Specialized / Private SOPs

None.

4.6 Configuration

4.6.1 AE Title/Presentation Address Mapping

The Local AE title is "hostname_DCP", where "hostname" is the unique hostname within the network.

The LightSpeed system allows the user to “add”, “Remove”, or “Update” the mapping of remote DICOM Printer AE Titles to IP Addresses and Ports. These options can be selected from the Service Desktop camera installation utility.

4.6.2 Configurable Parameters

The following fields are configurable for the DICOM Print SCU AE:

Note: All configurations should be performed by a GE Field Service Engineer.

- Local IP Address
- Local IP Netmask
- Max PDU length
- The following DICOM print parameters are configurable. The valid ranges are shown in earlier sections of this conformance statement.
 - Medium Type (2000, 0030)
 - Film Destination (2000, 0040)
 - Magnification Type (2010, 0060)
 - Min Density (2010, 0120)
 - Max Density (2010, 0130)
 - Empty Image Density (2010, 0110)
 - Border Density (2010, 0100)
 - Configuration Information (2010, 0150)
 - Smoothing Type (2010, 0080)
- The following network timers are supported and are configurable. When these timers expire the association is terminated. The default values are:
 - Association Time out = 600 seconds
 - Session Time out = 3000 seconds (total time for open association)
 - N-SET Time out = 300 seconds
 - N-ACTION Time out = 300 seconds
 - N-CREATE Time out = 300 seconds
 - N-DELETE Time out = 300 seconds
 - N-GET Time out = 100 seconds
- Time-outs, which are set for all hosts, are configurable in dprint.cfg:
 - *Association time-out - *bi_assoc_tio*
 - *Session timeout - *bi_session_tio*
 - *Create timeout - *bi_ncreate_tio*

Note: All configurations are performed by a GE Field Service Engineer.

4.7 Support of Extended Character Set

The Print SCU does not support any Extended Character Set.

5 DICOM Storage Commitment SCU Conformance

5.1 Introduction

This section of the DICOM conformance statement specifies the Storage Commitment Push Model SOP class as user, the optional attributes and service elements supported.

5.2 Implementation Model

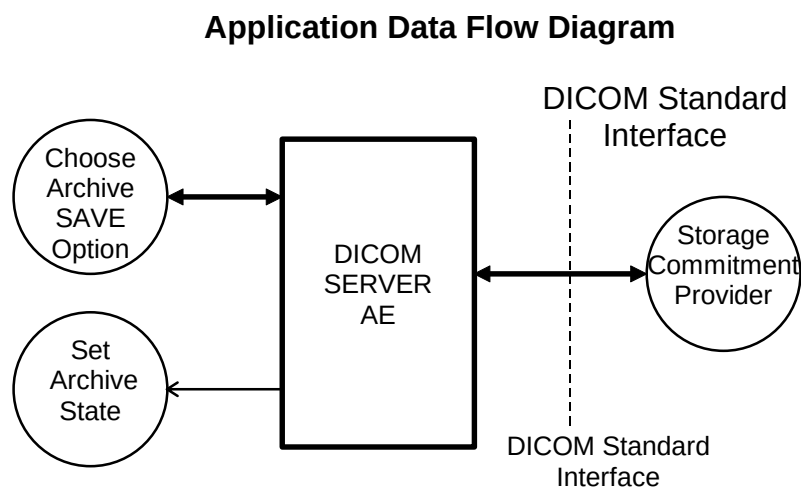
5.2.1 Application data flow diagram

Illustration 5.1.1 shows the data flow diagram of Storage Commitment SCU. There is only one Real-World activity that will cause the DICOM Server Application Entity (DICOM Server AE) to initiate an association to a remote DICOM entity that is a Service Class Provider (SCP) of the Storage Commitment SOP class.

The *Choose Archive save Option* Real-World activity consists of an operator choosing a remote DICOM AE that supports Storage Commitment as provider as the archive device and selecting one or more exam or series from the local database and choosing either *Save Exam* or *Save Series* from the archive menu. The images to be committed are sent to the remote provider entity first. The Commitment request for the transferred image instances is sent after the complete image transfer. The Commitment response can come on same association or on a different association.

There is no Real-World activity required for the DICOM Server AE to respond to an incoming Storage Commitment response from the remote DICOM AE.

Illustration 5.2.1



5.2.2 Functional definitions

DICOM Server AE initiates the following operations:

- Negotiates and establishes association with remote Storage Commitment Provider
- Sends the selected images to the remote DICOM AE.
- Closes the association.
- If all the images are transferred without failures the following steps will be executed. If there are any failures the job will be marked as failed and the Storage Commitment request will not be sent.
- Establishes a new association for sending the commitment request. Here note that a storage Commitment request does not imply that the acquisition of images is complete.
- Receives the response on same association or on a different association.
- Updates the archive flag information for successful instances. The archive flag is shared with the local archive and the two cannot be distinguished.

5.2.3 Sequencing of real-world activities

The user has to select the remote commitment provider and the exams/series to be pushed before clicking Save Exam/Series button on the Archive menu.

5.3 AE Specifications

5.3.1 DICOM Storage Commitment SCU specifications

Provides Standard Conformance to the following DICOM 3.0 SOP class as an SCU:

SOP class name	SOP class UID
Storage Commitment Push model SOP class	1.2.840.10008.1.20.1

5.3.1.1 Association establishment policies

5.3.1.1.1 General

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
--------------------------	-----------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU for association initiated by the DICOM Storage Commitment SCU AE is:

Maximum Length PDU	10 Kbytes
---------------------------	------------------

5.3.1.1.2 Number of associations

A single association is used to send the commitment request. This waits for the response from commitment provider on the same association for a configurable amount of time. It will also accept an association for receiving the response from commitment provider at any time.

5.3.1.1.3 Asynchronous nature

Asynchronous window negotiation is not supported.

5.3.1.1.4 Implementation identifying information

The Implementation UID for this GEMS Implementation is:

LightSpeed Implementation UID	1.2.840.113619.6.55
--------------------------------------	----------------------------

5.3.1.2 Association initiation policy

An association is initiated by the Storage SCU to the provider to send the images to be committed. A separate association is established to send the commitment request.

5.3.1.3 Real World Activity

5.3.1.3.1 Associated Real World Activity - "Save exam/series"

The user selects the exam/series to be committed. All the images currently in the selected exam/series will be sent to the selected remote archive node (which is also the Storage commitment SCP) using DICOM C-STORE operations. Once all the images are transferred the commitment request will be sent on a separate association.

5.3.1.3.2 Associated Real World Activity - "Auto Archive exam/series"

The Auto archive API's are used to archive the exams/series onto local archive media or the remote archive node (which shall be a Storage Commitment SCP) without manual interface. If the default device selected for Auto Archive is a remote Storage Commitment SCP then all the images currently in the specified exam/series will be sent to the selected Storage commitment SCP using C-STORE operations. On successful transfer of all the images the Storage Commitment request will be sent.

5.3.1.4 Proposed Presentation Contexts

The Proposed Presentation Context table for the DICOM Storage Commitment SCU is as shown in following Table.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Extended
Name	UID	Name	UID		Negotiation
Storage Commitment Push Model SOP Class	1.2.840.10008.1.20.1	DICOM Implicit VR Little Endian Transfer Syntax	1.2.840.10008.1.2	SCU	None

The Storage Commitment SCU can send the commitment request for following DICOM 3.0 SOP classes.

NAME	UID
CT Image Storage SOP Class	1.2.840.10008.5.1.4.1.
MR Image storage SOP Class	1.2.840.10008.5.1.4.1.
Secondary Capture Storage SOP Class	1.2.840.10008.5.1.4.1.
Standalone Overlay SOP Class	1.2.840.10008.5.1.4.1.

5.3.1.5 Request Processing

The images in the selected exam/series are sent to the remote commitment provider entity using DICOM C-STORE operations. If there are any failures in the image transfers the Storage commitment request will not be sent. The corresponding job will be marked as failed and user will be notified of the status of job.

If all the images are successfully transferred then the commitment request will be sent on a different association with the list of sop instances.

If the N-Action request fails, the job will be marked as failed otherwise following sequence of actions will take place.

The SCU waits for N-Action-Rsp from provider on the same association for a configurable amount of time. If it does not receive N-Action-Rsp during this time it will close the association. It changes the Job state to "Wait" indicating the job is waiting for the response from commitment provider. The job will be marked as failed if the response is not received by stop job time. Stop job time is the maximum duration the job can wait for responses.

A New transaction uid will be created for each retry by user. The old transaction uid is not applicable for these requests.

The Following DIMSE service Elements are supported for the Storage Commitment request and response processing.

N-ACTION – Requests the remote Storage Commitment SCP to commit to storing the image instances.

N-EVENT-REPORT – The response sent by the remote Storage Commitment SCP.

The following attributes are sent as part of the **DATA Set** for the **N-Action request**.

Attribute	Tag	Value
Transaction UID	(0008,1195)	Transaction UID
Referenced SOP Sequence	(0008,1199)	
SOP Class UID	(0008,1150)	
SOP Instance UID	(0008,1155)	

- Referenced Study Component sequence attribute is not sent.
- Storage Media File-Set ID and Storage Media File-Set UID attributes are not supported.

5.3.1.6 Response Processing

Once the N-EVENT REPORT response is received, the following actions will be taken depending on the status of response.

5.3.1.6.1 Commit response with SUCCESS status

The Archive flag information in the browser for all the successful instances will be updated. The status will be changed to “Y”.

The job queue entry will be removed

Note: The following attributes are expected as part of **DATA Set** for **N-Event-Request** from SCP

Attribute	Tag	Value
Transaction UID	(0008,1195)	Value received from SCP
Referenced SOP Sequence	(0008,1199)	Value received from SCP
SOP Class UID	(0008,1150)	Value received from SCP
SOP Instance UID	(0008,1155)	Value received from SCP

N-Event-Response will be sent on the same association itself. No DATA Set will be sent along with the response.

5.3.1.6.2 Commit response with FAILURE status

The following attributes are expected as part of **DATA Set** for **N-Event-Request** from SCP:

Attribute	Tag	Value
Transaction UID	(0008,1195)	Value received from SCP
Failed SOP Sequence	(0008,1199)	Value received from SCP
SOP Class UID	(0008,1150)	Value received from SCP
SOP Instance UID	(0008,1155)	Value received from SCP
Failure Reason	(0008,1197)	Value received from SCP

In case of complete/partial failure the user will be notified about the status and the job entry will be paused. There is no attempt made to retry automatically the failed sop instances. However the user can manually retry the failed jobs. Such requests will be treated as new requests. This will go through the whole sequence of operations once again.

The failure reason is ignored.

Failed SOP instances will have their archive flag information unaltered.

Note: The archive status flag in the browser is a shared flag with local archive. When the status is “Y”, it means that the images are archived but doesn’t specify whether on local archive device or remote archive device. It is left to the user’s discretion whether the local sop instances (with their archive flag set to “Y”) are to be deleted.

N-Event-Response will be sent on the same association itself. No DATA Set will be sent along with the response.

5.3.1.7 Association Acceptance Policy

A single association will be accepted at any time to receive Storage Commitment responses.

5.4 Communication Profiles

5.4.1 Supported Communication Stacks (parts 8,9)

DICOM Upper Layer (Part 8) is supported using TCP/IP.

5.4.2 TCP/IP Stack

The TCP/IP stack is inherited from a UNIX Operating System.

5.4.2.1 Physical Media Support

Ethernet v2.0, IEEE 802.3.

5.4.3 Point-to-Point Stack

A 50-pin ACR-NEMA connection is not applicable to this product.

5.5 EXTENSIONS/SPECIALIZATIONS/PRIVATIZATIONS

None.

5.6 CONFIGURATION

Ethernet v2.0, IEEE 802.3.

5.6.1 AE Title/Presentation Address Mapping

The system allows the user to “add”, “Remove”, or “Update the mapping of remote AE Titles to IP Addresses and Ports. These options can be selected from the “Remote Host Selection” menu displayed by choosing “Select Remote Host” from the “Network” pull-down menu from the local database manager. A remote DICOM host can be designated as a “remote archive device” by setting the “Remote archive node” radio button on the remote host configuration panel. Once a DICOM host is designated as a “remote archive device” the host will be available in the “Archive” device selection menu and can be chosen for any subsequent archive operation.

5.6.2 Configurable Parameters

The following fields are configurable for the DICOM Server AE:

- Local AE Title (the machine hostname)
- Local IP Address
- Local IP Netmask
- Max PDU length
- Time-outs, which are set for all hosts, are configurable in dcs.cfg:

Note: * denotes any SOP class (time in sec)
 *Association time-out - *bi_assoc_tio*
 * Inactivity time-out - *bi_store_tio*
 *Session time-out - *bi_session_tio*

Note: All configurations should be performed by a GE Field Service Engineer.

5.7 Support of Extended Character Sets

This does not support any Extended Character set.

6 Modality Work List Information Model Definition

6.1 Introduction

This section of the DICOM Conformance Statement specifies the compliance to DICOM conformance requirements for the relevant **Networking** features on this GEMS product. Note that the format of this section strictly follows the format defined in DICOM Standard PS 3.2 (Conformance). Please refer to that part of the standard while reading this section.

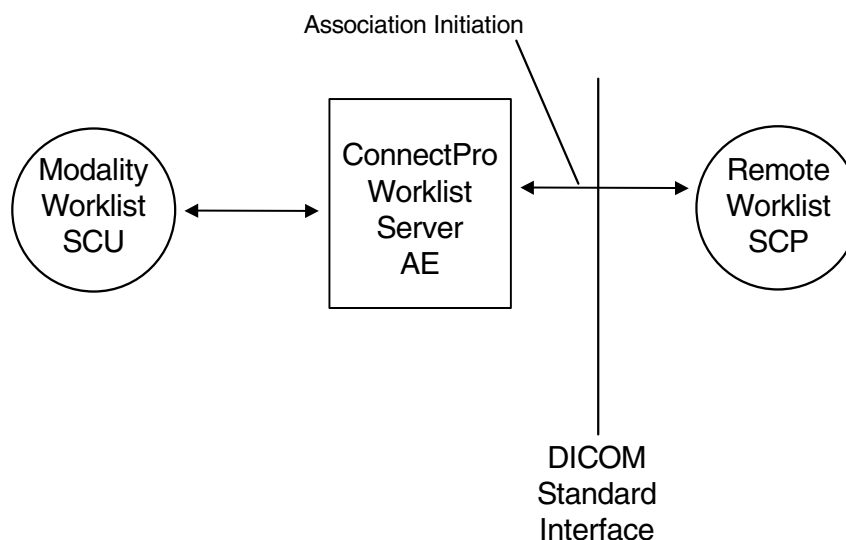
The ModalityWorklist option for LightSpeed allows a user to query for and display DICOM modality worklist information from a remote hospital or radiology department information system computer. For example, a user may wish to query for all procedures scheduled to be performed on the scanner. In this situation, ModalityWorklist is providing the DICOM C-FIND service as a service class user (SCU).

6.2 Implementation Model

All DICOM functionality provided by the ModalityWorklist feature is logically provided by the Worklist Server DICOM AE. The Worklist Server DICOM AE is commanded to perform DICOM modality worklist query services through the use of the LightSpeed user interface.

6.2.1 Application Data Flow Diagram

The Basic and Specific Application models for this device are shown in the following illustration:



6.2.2 Functional definitions

The ModalityWorklist Worklist Server AE is implemented as an application process on the scanner host computer. It runs as a daemon serving requests from the user interface to obtain modality worklists, query remote AE's and return the results to the user interface.

The ModalityWorklist Worklist Server AE initiates the following functions:

- *Query*: Initiates a DICOM association in order to query a remote AE. If the remote AE accepts a presentation context applicable to modality worklist, the Worklist Server AE will issue a modality worklist query request via the C-FIND service.

6.2.3 Sequencing of Real-World Activities

1. The user or the system initiates a modality worklist query (as a modality worklist SCU) to the modality worklist SCP with a given set of query parameters.
2. The modality worklist SCP returns responses that match the query parameters.
3. Items from the returned worklist responses are presented to the user.
4. A subset of the returned worklist responses is included in acquired DICOM images related to the responses.

6.3 AE Specifications

6.3.1 Worklist Server AE Specification

This Application Entity provides Standard Conformance to the following DICOM v3.0 SOP Classes as an **SCU**:

SOP Class Name	SOP Class UID
Modality Worklist Information Model - FIND	1.2.840.10008.5.1.4.31

6.3.1.1 Association Establishment Policies

6.3.1.1.1 General

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
--------------------------	-----------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU for an association initiated by the DICOM Worklist Server is:

Maximum Length PDU	50 Kbytes
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The SOP Class Extended Negotiation is not supported.

The maximum number of Presentation Context Items that will be proposed is 1.

The user information Items sent by this product are:

- Maximum PDU Length
- Implementation UID

6.3.1.1.2 Number of Associations

The Worklist Server AE (SCU) will initiate only one DICOM association at a time to perform a modality worklist query of a single remote AE.

6.3.1.1.3 Asynchronous Nature

Asynchronous mode is not supported. All operations are performed synchronously.

6.3.1.1.4 Implementation Identifying Information

The Implementation UID for this DICOM v3.0 Implementation is:

ModalityWorklist for LightSpeed implementation UID	1.2.840.113619.6.50
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6.3.1.2 Association Initiation Policy

The Worklist Server AE initiates a new association due to an update operation being initiated from the LightSpeed user interface.

6.3.1.2.1 Real-World Activity: Worklist Query

6.3.1.2.1.1 Associated Real-World Activity

The operator of the system initiates a query for a modality worklist by either opening the Schedule screen or by opening the Schedule screen and pressing the Update button. The choice of which of these two behaviors occurs is user configurable. The Worklist Server will then initiate an association with the remote AE in order to query for the worklist

A user can configure a number of parameters that directly control the worklist query request. The user can request worklist items that are intended for the scanner the user is working at, all items that apply to the modality of the scanner the user is working at or all worklist items available. These selections and their affects on worklist query parameters are given below:

- | | |
|-----------------------|---|
| <u>This Scanner:</u> | <ul style="list-style-type: none">• Modality, (0008,0060) - set to CT• Scheduled Station AE Title, (0040,0001) - set to local AE title |
| <u>This Modality:</u> | <ul style="list-style-type: none">• Modality, (0008,0060) - set to CT• Scheduled Station AE Title, (0040,0001) - zero-length (universal matching) |
| <u>All Scanners</u> | <ul style="list-style-type: none">• Modality, (0008,0060) - zero-length (universal matching)• Scheduled Station AE Title, (0040,0001) - zero-length (universal matching) |

The scheduled dates of procedures of interest can be specified for query by selecting a specific date range. The date ranges available are Today, Days Before Today, Days After Today and All Days. These selections and their affects on worklist query parameters are given below:

Today: Scheduled Procedure Step Start Date (0040,0002) - set to YYYYMMDD, where this date is the current date.

Days Before Today and Days After Today: Scheduled Procedure Step Start Date (0040,0002) - set to YYYYMMDD-YYYYMMDD, where this date range represents the specified number of days before today and/or after today. Note that number of days both before and after can be specified in the same query and that each always includes today.

All Days: Scheduled Procedure Step Start Date (0040,0002) - zero-length (universal matching)

6.3.1.2.1.2 Proposed Presentation Context Table

The following table shows the proposed presentation contexts for the Worklist Server AE after real-world activity "Worklist Query" has been initiated:

Presentation Context Table - Proposed					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Modality Worklist Information Model – FIND	1.2.840.10008.5.1.4.31	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None

6.3.1.2.1.2.1 SOP Specific DICOM Conformance Statement for the Worklist SOP Class

If the remote AE does not support the proposed Presentation Context, an appropriate error is logged and the operator is notified.

This implementation can receive multiple C-FIND results over a single association. Only one association is opened at a time.

Each C-FIND response received from the remote AE is parsed to verify the length/type of the items in the response (see section 2.3.1.2.1.2.2 for more information). Upon detecting any error in the response data, the Worklist Server AE will issue a C-FIND-CANCEL and, upon receipt of a C-FIND-RSP (or if an applicable timer expires), will abort the association. All previously received worklist items are retained.

On receipt of any error from the remote AE, the Worklist Server will issue a C-FIND-CANCEL and, upon receipt of a C-FIND-RSP (or if an applicable timer expires), will abort the association. All previously received worklist items are retained. Warnings received from the remote AE are ignored.

Each C-FIND operation supports a configurable “Association Timer.” This timer starts when the association request is sent or received and stops when the association is established. The default time-out value is 30 seconds.

Each C-FIND operation supports a configurable “Session Timer.” This timer starts when an association is established and stops when the association is ended. The default time-out value is 3600 seconds.

If any of the above timers expires, the association is aborted (A-ABORT) and the operation in progress is considered to be failed. Any previously received worklist items are discarded.

6.3.1.2.1.2.2 Record Acceptance Policy

The LightSpeed implementation adheres to strict value checking of incoming query responses from the remote AE. Each response received is examined to verify that all Type 1 attributes are present with non-zero length, that all Type 2 attributes are present (possibly with zero length) and that the data for all attributes is consistent with respect to the attributes’ value representation (VR).

Any inconsistencies in the response data, with respect to the categories described above, are considered errors. Upon detecting any such errors in the response data, the Worklist Server AE will issue a C-FIND-CANCEL and, upon receipt of a C-FIND-RSP (or if an applicable timer expires), will abort the association. All previously received worklist items are retained. Note that the absence of requested Type 3 attributes is not considered an error.

Fields considered Type 1 by the Worklist Server include:

- (0010,0010), Patient Name
- (0010,0020), Patient ID
- (0020,000D), Study Instance UID
- (0040,0001), Scheduled Station AE Title
- (0040,0002), Scheduled Procedure Step Start Date ¹
- (0040,0003), Scheduled Procedure Step Start Time ¹
- (0040,0009), Scheduled Procedure Step ID
- (0040,1001), Requested Procedure ID

¹ Start Date must be of the form YYYYMMDD, exactly eight numeric characters, and Start Time must be of the form HHMMSS, exactly six numeric characters.

Fields considered Type 2 by Worklist Server include:

- (0008,0050), Accession Number
- (0008,0060), Modality
- (0008,0090), Referring Physician Name
- (0010,0030), Patient Date of Birth
- (0010,0040), Patient Sex
- (0010,1030), Patient Weight in kg

- (0010,2000), Medical Alerts
- (0010,2110), Contrast Allergies
- (0010,21C0), Pregnancy Status
- (0032,1032), Requesting Physician
- (0032,1070), Requested Contrast Agent
- (0038,0010), Admission ID
- (0038,0050), Special Needs
- (0038,0300), Current Patient Location
- (0038,0500), Patient State
- (0040,0006), Performing Physician
- (0040,0010), Scheduled Station Name
- (0040,0011), Scheduled Procedure Step Location
- (0040,0012), Pre-order Medication
- (0040,1003), Requested Procedure Priority
- (0040,1004), Patient Transport Arrangements
- (0040,3001), Confidentiality Constraint

6.3.1.3 Association Acceptance Policy

The Worklist Server AE does not respond to attempts by a remote AE to open an association.

6.4 Communication Profiles

6.4.1 Supported Communication Stacks (PS 3.8, PS 3.9)

DICOM Upper Layer (PS 3.8) is supported using TCP/IP.

6.4.2 OSI Stack

The OSI Communication Stack is not supported by this implementation.

6.4.3 TCP/IP Stack

The TCP/IP Communication Stack is inherited from the operating system.

6.4.3.1 API

Not applicable to this product.

6.4.3.2 Physical Media Support

Ethernet 802.3 provides the physical network layer for this product.

6.4.4 Point-to-Point Stack

The Point-to-Point Communication Stack is not supported by this implementation.

6.5 Extensions / Specializations / Privatizations

6.5.1 Standard Extended /Specialized/Private SOPs

ModalityWorklist for LightSpeed does not implement any private transfer SOP classes.

6.5.2 Private Transfer Syntaxes

ModalityWorklist for LightSpeed does not implement any private transfer syntaxes.

6.6 CONFIGURATION

The ModalityWorklist feature is configured by GEMS Field Service engineers. The DICOM configuration items below are configurable or re-configurable by a Field Service Engineer and are not accessible by users through the LightSpeed user interface.

6.6.1 AE Title/Presentation Address Mapping

ModalityWorklist allows for the configuration of the following parameters that pertain to the remote AE.

- Remote AE (HIS/RIS) IP address – IP address used to contact the remote AE
- Remote AE (HIS/RIS) IP port – IP port used to contact the remote AE

These parameters define where worklist queries will be directed. Configuration of these parameters is performed by GEMS Field Service engineers using the ModalityWorklist installation facilities.

6.6.2 Configurable Parameters

The following parameters are configurable for the DICOM Worklist Server AE:

- Local (Worklist Server) AE Title (automatically set to host name of scanner)
- Local IP address
- Local IP netmask
- Local IP gateway

The following parameters are configurable by changing their values in the configuration file **/usr/g/config/WLdcm.cfg**. Note that these parameters typically need not be changed. Furthermore, no support is provided for retaining changed settings: the values will require changing again after a system software upgrade.

- Implementation UID
- PDU size
- Association time-out period
- Session time-out period
- C-FIND time-out period

6.7 Support of Extended Character Sets

ModalityWorklist will support only the ISO_IR 100 (ISO 8859-1:1987 Latin alphabet N 1. supplementary set) as extended character sets.

6.8 Modality Worklist Information Model Definition

6.8.1 Introduction

This section specifies the use of the DICOM Modality Worklist Information Model used to organize data and against which a Modality Worklist Query will be performed. The contents of this section are:

6.8.2 - Information Model Description

6.8.3 - Information Model Entity-Relationship Model

6.8.4 - Information Model Module Table

6.8.5 - Information Model Keys

6.8.2 Modality Worklist Information Model Description

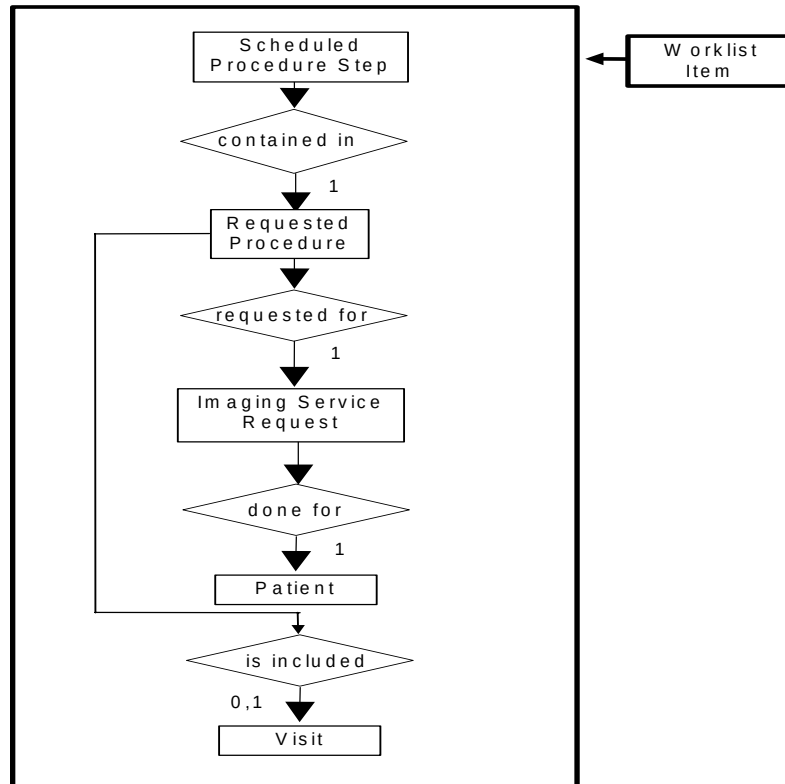
In order to serve as a Service Class Provider (SCP) of the Modality Worklist Service Class, a DICOM Application Entity (AE) possesses information about the attributes of a number of managed worklist items. These items are organized into Modality Worklist Information Modules. In this Service Class, the Information Model plays a role similar to an Information Object Definition of most other DICOM Service Classes.

6.8.3 Modality Worklist Information Model Entity-Relationship Model

The Entity-Relationship diagram for the Modality Worklist Information Model schema is shown in Illustration 3.3-1. It represents the information that composes a Worklist Item. In this figure, the following diagrammatic convention is established to represent the information organization:

- Each entity is represented by a rectangular box
- Each relationship is represented by a diamond shaped box.
- The fact that a relationship exists between two entities is depicted by lines connecting the corresponding entity boxes to the relationship boxes.

Illustration 6.8.3-1 – Modality Worklist Information Model E/R DIAGRAM



6.8.4 Entity Descriptions

Please refer to DICOM Standard PS 3.3. (Information Object Definitions) and PS 3.4 (Service Class Specifications) for a description of each of the Entities contained within the Modality Worklist Information Model.

6.8.4.1 Scheduled Procedure Step

A Scheduled Procedure Step is an arbitrarily defined scheduled unit of service that is specified by the Procedure Plan for a Requested Procedure. It specifies one or more Action Items (events) involving equipment (i.e. imaging modality equipment), human resources, location and time (i.e. start time, stop time, duration).

6.8.4.2 Requested Procedure Entity Description

A Requested Procedure is an instance of a Procedure of a given Procedure Type. An instance of a Requested Procedure includes all of the items of information that are specified by an instance of a Procedure Plan that is selected for the Requested Procedure by the imaging service provider.

6.8.4.3 Imaging Service Request Entity Description

An Imaging Service Request is a set of one or more Requested Procedures selected from a list of Procedure Types. An Imaging Service Request is submitted by one authorized

imaging service requester to one authorized imaging service provider in the context of one Service Episode.

6.8.4.4 Visit Entity Description

A Visit is the context in which the treatment or management of an arbitrary subset of a Patient's medical conditions occurs. A Visit is limited to the description of a Patient's activities at a single facility.

6.8.4.5 Patient Entity Description

A Patient is a person receiving, or registered to receive, healthcare services.

6.8.5 ModalityWorklist Mapping of DICOM Entities

Table 6.8.5-1 —Mapping of DICOM Entities to ModalityWorklist Entities

DICOM	ModalityWorklist Entity
Scheduled Procedure Step	Exam
Requested Procedure	Exam
Imaging Service Request	Exam
Visit	Exam
Patient	Patient

6.8.6 Information Model Module Table

Within an entity of the DICOM v3.0 Modality Worklist Information Model, attributes are grouped together into related set of attributes called modules. A module facilitates the understanding of the semantics concerning the attributes and how the attributes relate to one another. A module grouping does not infer any encoding of information into datasets.

Table 6.8.6-1 identifies the defined modules within the entities that comprise the DICOM v3.0 Modality Worklist Information Model. Modules are identified by Module Name.

See DICOM v3.0 PS 3.3 and PS 3.4 for a complete definition of the entities, modules, and attributes.

Table 6.8.6-1 – Modality Worklist Information Model Modules

Entity Name	Module Name	Reference
Scheduled Procedure Step	SOP Common	6.8.9.1
	Scheduled Procedure Step	6.8.9.2
Requested Procedure	Requested Procedure	6.8.10.1
Imaging Service Request	Imaging Service Request	6.8.11.1
Visit	Visit Identification	6.8.12.1
	Visit Status	6.8.12.2
	Visit Relationship	6.8.12.3
	Visit Admission	6.8.12.4
Patient	Patient Relationship	6.8.13.1
	Patient Identification	6.8.13.2
	Patient Demographic	6.8.13.3
	Patient Medical	6.8.13.4

6.8.7 Information Model Keys

Please refer to DICOM Standard PS 3.3. (Information Object Definitions) and PS 3.4 (Service Class Specifications) for a description of each of the Entities contained within the Modality Worklist Information Model.

The following Module descriptions contain the attributes that are present in a C-FIND request message sent by the Worklist Server AE to a remote AE. It should be noted that they are the same as those defined in the DICOM v3.0 Standard, PS 3.4 (Service Class Specifications) and include:

- Name
- Tag group and element numbers
- Expected Matching Key Type: R-required, O-optional
- Expected Return Key Type:
 - 1 - non-zero value required
 - 1C - conditionally of type 1
 - 2 - required to be present, possibly with zero-length value
 - 3 - optional
- Mapped into The Image - whether this data is mapped into subsequently acquired images
- Notes - clarification of this implementation's use/treatment of this attribute

All data elements in the following Module descriptions are requested by the Worklist Server AE. Values of data elements that are not mapped into images, and are not otherwise dealt with (displayed on the user interface, etc.), are not used and are, thus, discarded upon receipt. See Table B-1 for further information.

Data elements for which values can be sent for matching purposes are described as such. Data elements for which values are not sent are sent with zero length and universal matching will apply. This is the default case if no other description to the contrary is provided.

6.8.8 Supported Matching

The following are the types of matching that can be request by the implementation:

- Single Value matching
- Universal Matching
- Range of date/time

6.8.9 Scheduled Procedure Step Entity

6.8.9.1 SOP Common Module

Table 6.8.9-1 – SOP Common Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Specific Character Set	(0008,0005)	O	1C	No	Matching for this item is supported only for the character set ISO_IR 100. This value is always sent and therefore, must be returned.

6.8.9.2 Scheduled Procedure Step Module

Table 6.8.9-2 – Scheduled Procedure Step Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Scheduled Procedure Step Sequence	(0040,0100)	R	1	No	
Scheduled Station AE Title	(0040,0001)	R	1	No	Matching is supported as follows: either no AE title is supplied (universal matching), or the scanner's Worklist Server AE title is supplied for matching; this is user selectable.
Scheduled Procedure Step Start Date	(0040,0002)	R	1	No	Matching is supported as one of the following; this is user selectable: • all days, • today only, • today and a number of days before today, • today and a number of days after today, • today and a number of days before today and a number of days after today. Number of days before/after is specified by the user. Returned values must be exactly 8 numeric characters in YYYYMMDD format.
Scheduled Procedure Step Start Time	(0040,0003)	R	1	No	This attribute is sent with zero-length. Returned values must be exactly 6 numeric characters in HHMMSS format.
Modality	(0008,0060)	R	1	Yes	Matching is supported as follows: either no Modality is supplied (universal matching), or the scanner's Modality is supplied for matching; this is user selectable.
Scheduled Performing Physician's Name	(0040,0006)	R	2	No	This attribute is sent with zero-length.
Scheduled Procedure Step Description	(0040,0007)	O	1C	Yes	
Scheduled Station Name	(0040,0010)	O	2	No	
Scheduled Procedure Step Location	(0040,0011)	O	2	No	
Scheduled Action Item Code Sequence	(0040,0008)	O	1C	No	
Code Value	(0008,0100)	O	1C	No	
Coding Scheme Designator	(0008,0102)	O	1C	No	
Code Meaning	(0008,0104)	O	3	No	
Pre-Medication	(0040,0012)	O	2C	No	
Scheduled Procedure Step ID	(0040,0009)	O	1	Yes	
Requested Contrast Agent	(0032,1070)	O	2C	No	Displayed on "More Info..." screen.

6.8.10 Requested Procedure Entity

6.8.10.1 Requested Procedure Module

Table 6.8.10.1-1 Requested Procedure Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Requested Procedure ID	(0040,1001)	O	1	Yes	User can enter the value for Requested Procedure Id prior to query. If user has entered the value then that value will be sent as part of the query. The value returned in the response shall be mapped to the image.
Requested Procedure Description	(0032,1060)	O	1C	Yes	Truncated to 22 characters.
Requested Procedure Code Sequence	(0032,1064)	O	1C	No	
Code Value	(0008,0100)	O	1C	No	
Coding Scheme Designator	(0008,0102)	O	1C	No	
Code Meaning	(0008,0104)	O	3	No	
Study Instance UID	(0020,000D)	O	1	Yes (Based on User option)	User (Modality) will be able to configure the following. If the user has set the option to "Use MWL Study UID", then study Instance UID will be copied into the final DICOM image header. If this option is not set then a new Study instance UID is generated locally on the scanner.
Referenced Study Sequence	(0008,1110)	O	2	No	
Referenced SOP Class UID	(0008,1150)	O	1C	No	
Referenced SOP Instance UID	(0008,1155)	O	1C	No	
Requested Procedure Priority	(0040,1003)	O	2	No	
Patient Transport Arrangements	(0040,1004)	O	2	No	
Requested Procedure Location	(0040,1005)	O	3	No	
Confidentiality Code	(0040,1008)	O	3	No	

6.8.11 Imaging Service Request Entity

6.8.11.1 Imaging Service Request Module

Table 6.8.11.1-1 – Imaging Service Request Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Accession Number	(0008,0050)	O	2	Yes	User will be able to enter value for Accession Number prior to query to be sent as part of C-FIND request. Supports maximum of 16 characters.
Requesting Physician	(0032,1032)	O	2	No	
Referring Physician's Name	(0008,0090)	O	2	Yes	Truncated to 32 characters
Requesting Service	(0032,1033)	O	3	No	

6.8.12 Visit Entity

6.8.12.1 Visit Identification

Table 6.8.12.1-1 – Visit Identification Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Admission ID	(0038,0010)	O	2	No	
Institution Name	(0008,0080)	O	3	No	

6.8.12.2 Visit Status

Table 6.8.12.2-1 – Visit Status Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Current Patient Location	(0038,0300)	O	2	No	Displayed on "More Info..." screen.

6.8.12.3 Visit Relationship

Table 6.8.12.3-1 – Relationship Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Referenced Patient Sequence	(0008,1120)	O	2	No	
Referenced SOP Class UID	(0008,1150)	O	2	No	
Referenced SOP Instance UID	(0008,1155)	O	2	No	

6.8.12.4 Visit Admission

No data elements are requested from the Visit Admission Module.

6.8.13 Patient Entity

6.8.13.1 Patient Relationship

No data elements are requested from the Patient Relationship Module.

6.8.13.2 Patient Identification

Table 6.8.13.2-1 – Patient Identification Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Patient's Name	(0010,0010)	R	1	Yes	This attribute is sent with zero-length. Supports up to 32 characters. See also Note1 .
Patient ID	(0010,0020)	R	1	Yes	This attribute is sent with zero-length. Supports up to 16 characters.

Note 1 :

- Modality Worklist server supports 32 characters for patient Name
- Supports DICOM format for patient Name (with “^” as delimiters)
- If patient name in worklist has more than 32 characters then
 - Worklist will be accepted by the server
 - Worklist Browser will display only the first 32 characters
 - “More Info” screen will display the full patient name
 - **Only the FIRST 32** characters is copied into the final DICOM image header

6.8.13.3 Patient Demographic

Table 6.8.13.3-1 – Patient Demographic Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Patient's Birth Date	(0010,0030)	O	2	Yes	This value is also used to calculate the Patient's Age.
Patient's Sex	(0010,0040)	O	2	Yes	
Patient's Weight	(0010,1030)	O	2	Yes	Limited to maximum value of 999 kg.
Confidentiality constraint on patient data	(0040,3001)	O	2	No	
Patient's Size	(0010,1020)	O	3	No	
Patient's Address	(0010,1040)	O	3	No	
Patient's Telephone Numbers	(0010,2154)	O	3	No	

6.8.13.4 Patient Medical

Table 6.8.13.4-1 – Patient Medical Module Attributes

Attribute Name	Tag	Expected Matching Key Type	Expected Returned Key Type	Mapped into the Image	Note
Patient State	(0038,0500)	O	2	No	
Pregnancy Status	(0010,21C0)	O	2	No	Displayed on "More Info..." screen.
Medical Alerts	(0010,2000)	O	2	No	Displayed on "More Info..." screen.
Contrast Allergies	(0010,2110)	O	2	No	Displayed on "More Info..." screen.
Special Needs	(0038,0050)	O	2	No	Displayed on "More Info..." screen.
Additional Patient History	(0010,21B0)	O	3	No	Displayed on "More Info..." screen.

6.9 Private Data Dictionary

The ModalityWorklist implementation does not define any Private Attributes within the Modality Worklist Information Model.

6.10 C-FIND Request Message

This section provides a detailed description of the C-FIND request message data that is provided to the remote AE during a worklist query operation. The dump in Table 6.10-1 below lists, in exact message order, the fields transferred as part of the C-FIND request message for a typical query.

In this particular dump, no values are specified for the Scheduled Procedure Step Start and End Dates (the attributes are sent with zero length). In DICOM this is interpreted as meaning all dates (i.e. universal matching). The Modality is also not specified in this particular dump, meaning all modalities. Note that the user, through the use of the LightSpeed user interface, can submit a worklist query that will cause non-zero values to be sent for these attributes.

Table 6.10-1 – C-FIND Request Message Dump

```
(0008,0000) UL      108          #      4, 1  IdentifyingGroupLength
(0008,0005) CS      [ISO_IR 100] #     12, 1  SpecificCharacterSet
(0008,0050) SH      (no value available) #      0, 0  AccessionNumber
(0008,0080) LO      (no value available) #      0, 0  InstitutionName
(0008,0090) PN      (no value available) #      0, 0  ReferringPhysicianName
(0008,1110) SQ      (Sequence with explicit Length #=1) #     24, 1  ReferencedStudySequence
(fffe,e000) na      (Item with explicit Length #=2) #     16, 1  Item
(0008,1150) UI      (no value available) #      0, 0  ReferencedSOPClassUID
(0008,1155) UI      (no value available) #      0, 0  ReferencedSOPInstanceUID
(fffe,e00d) na      (ItemDelimitationItem for re-encoding) #      0, 1  ItemDelimitationItem
(fffe,e0dd) na      (SequenceDelimitationItem for re-enc.) #      0, 1  SequenceDelimitationItem
(0008,1120) SQ      (Sequence with explicit Length #=1) #     24, 1  ReferencedPatientSequence
(fffe,e000) na      (Item with explicit Length #=2) #     16, 1  Item
(0008,1150) UI      (no value available) #      0, 0  ReferencedSOPClassUID
(0008,1155) UI      (no value available) #      0, 0  ReferencedSOPInstanceUID
(fffe,e00d) na      (ItemDelimitationItem for re-encoding) #      0, 1  ItemDelimitationItem
(fffe,e0dd) na      (SequenceDelimitationItem for re-enc.) #      0, 1  SequenceDelimitationItem
(0010,0000) UL      96          #      4, 1  PatientGroupLength
(0010,0010) PN      (no value available) #      0, 0  PatientName
(0010,0020) LO      (no value available) #      0, 0  PatientID
(0010,0030) DA      (no value available) #      0, 0  PatientBirthDate
(0010,0040) CS      (no value available) #      0, 0  PatientSex
(0010,1000) LO      (no value available) #      0, 0  Other Patient IDs
(0010,1020) DS      (no value available) #      0, 0  PatientSize
(0010,1030) DS      (no value available) #      0, 0  PatientWeight
(0010,1040) LO      (no value available) #      0, 0  PatientAddress
(0010,2000) LO      (no value available) #      0, 0  MedicalAlerts
(0010,2110) LO      (no value available) #      0, 0  ContrastAllergies
(0010,2154) SH      (no value available) #      0, 0  PatientTelephoneNumber
(0010,2160) SH      (no value available) #      0, 0  Ethinc Group
(0010,21b0) LT      (no value available) #      0, 1  AdditionalPatientHistory
(0010,21c0) US      (no value available) #      0, 0  PregnancyStatus
(0010,4000) LT      (no value available) #      0, 0  Patient Comments
(0020,0000) UL      8          #      4, 1  ImageGroupLength
(0020,000d) UI      (no value available) #      0, 0  StudyInstanceUID
(0032,0000) UL      64          #      4, 1  StudyGroupLength
(0032,1032) PN      (no value available) #      0, 0  RequestingPhysician
(0032,1033) LO      (no value available) #      0, 0  RequestingService
(0032,1060) LO      (no value available) #      0, 0  RequestedProcedureDescription
(0032,1064) SQ      (Sequence with explicit Length #=1) #     32, 1  RequestedProcedureCodeSequence
(fffe,e000) na      (Item with explicit Length #=3) #     24, 1  Item
(0008,0100) SH      (no value available) #      0, 0  CodeValue
(0008,0102) SH      (no value available) #      0, 0  CodingSchemeDesignator
(0008,0104) LO      (no value available) #      0, 0  CodeMeaning
(fffe,e00d) na      (ItemDelimitationItem for re-encoding) #      0, 1  ItemDelimitationItem
(fffe,e0dd) na      (SequenceDelimitationItem for re-enc.) #      0, 1  SequenceDelimitationItem
(0038,0000) UL      32          #      4, 1  VisitGroupLength
(0038,0010) LO      (no value available) #      0, 0  AdmissionID
```

```

(0038,0050) LO    (no value available) #    0, 0  SpecialNeeds
(0038,0300) LO    (no value available) #    0, 0  CurrentPatientLocation
(0038,0500) LO    (no value available) #    0, 0  PatientState
(0040,0000) UL    192 #    4, 1  ModalityWorklistGroupLength
(0040,0100) SQ    (Sequence with explicit Length #=1) # 136, 1  ScheduledProcedureStepSequence
(fffe,e000) na    (Item with explicit Length #=12) # 128, 1  Item
  (0008,0060) CS    (no value available) #    0, 0  Modality
  (0032,1070) LO    (no value available) #    0, 0  RequestedContrastAgent
  (0040,0001) AE    (no value available) #    0, 0  ScheduledStationAETitle
  (0040,0002) DA    (no value available) #    0, 0  ScheduledProcedureStepStartDate
  (0040,0003) TM    (no value available) #    0, 0  ScheduledProcedureStepStartTime
  (0040,0006) PN    (no value available) #    0, 0  ScheduledPerformingPhysiciansName
  (0040,0007) LO    (no value available) #    0, 0  ScheduledProcedureStepDescription
  (0040,0008) SQ    (Seq with explicit Length #=1) #   32, 1  ScheduledActionItemCodeSequence
  (fffe,e000) na    (Item with explicit Length #=3) #   24, 1  Item
    (0008,0100) SH    (no value available) #    0, 0  CodeValue
    (0008,0102) SH    (no value available) #    0, 0  CodingsSchemeDesignator
    (0008,0104) LO    (no value available) #    0, 0  CodeMeaning
  (fffe,e00d) na    (ItemDelimitationItem for re-encoding) #    0, 1  ItemDelimitationItem
  (fffe,e0dd) na    (SequenceDelimitationItem for re-enc.) #    0, 1  SequenceDelimitationItem
  (0040,0009) SH    (no value available) #    0, 0  ScheduledProcedureStepID
  (0040,0010) SH    (no value available) #    0, 0  ScheduledStationName
  (0040,0011) SH    (no value available) #    0, 0  ScheduledProcedureStepLocation
  (0040,0012) LO    (no value available) #    0, 0  PreMedication
  (fffe,e00d) na    (ItemDelimitationItem for re-encoding) #    0, 1  ItemDelimitationItem
  (fffe,e0dd) na    (SequenceDelimitationItem for re-enc.) #    0, 1  SequenceDelimitationItem
  (0040,1001) SH    (no value available) #    0, 0  RequestedProcedureID
  (0040,1003) SH    (no value available) #    0, 0  RequestedProcedurePriority
  (0040,1004) LO    (no value available) #    0, 0  PatientTransportArrangements
  (0040,1005) LO    (no value available) #    0, 0  RequestedProcedureLocation
  (0040,1008) LO    (no value available) #    0, 0  ConfidentialityCode
  (0040,1010) PN    (no value available) #    0, 0  Names of Intended Recipients
  (0040,1400) LT    (no value available) #    0, 0  Requested Procedure Comments
  (0040,2400) LT    (no value available) #    0, 0  Imaging Service Request Comm.
  (0040,3001) LO    (no value available) #    0, 0  ConfidentialityConstraintOnPatientData

```

If the query is for a particular date range, the ScheduledProcedureStepStartDate will be filled with a valid date range. If either the start or end date are left blank by the user, they will simply be blank in the query.

Below is an example of a date range for August 30, 1997 through October 12, 1997.

```
(0040,0002) DA    [19970830-19971012] #   18, 1  ScheduledProcedureStepStartDate
```

Below is an example of a date range for August 30, 1997 through the end of time.

```
(0040,0002) DA    [19970830-] #   18, 1  ScheduledProcedureStepStartDate
```

Below is an example of a date range from the beginning of time through August 30, 1997.

```
(0040,0002) DA    [-19970830] #   18, 1  ScheduledProcedureStepStartDate
```

If the query is for records for this modality, the Modality will be filled in as follows:

```
(0008,0060) CS    [CT] #    2, 1  Modality
```

If the query is for records for this Scanner, the Modality will be filled in with CT as above and the Scheduled Station AE Title will be filled in with the value configured for this system. For example, this station was configured as CTRoom1.

```
(0040,0001) AE    [CTRoom1] #    8, 1  ScheduledStationAETitle
```

User will be able to enter the values for "Accession Number" prior to the query. If value is entered then that value will be sent as part of the query. For example, if "1234" is entered then

```
(0008,0050) SH    [1234] #    4, 1  AccessionNumber
```

User will be able to enter the values for “Requested Procedure Id” prior to the query. If value is entered then that value will be sent as part of the query. For example, if “3456” is entered then

(0040,1001) SH [3456] # 4, 1 RequestedProcedureID

6.11 Use of Specific DICOM Data

This section details the use of the DICOM data returned by remote AEs during worklist queries. The LightSpeed user interface fields that display the data, along with the data's mapping into resulting acquired and transferred DICOM images, are presented in following table 6.11-1.

Table 6.11-1 – Specific Data Usage

DICOM Worklist Data Element	Patient Schedule Screen Field	LightSpeed DICOM Image Data Element
Accession Number (0008,0050)	Req Number	Accession Number Supports maximum of 16 characters.
Patient ID (0010,0020)	Patient ID	Patient ID Supports maximum of 16 characters.
Patient Name (0010,0010)	Patient Name	Patient Name Supports maximum of 32 characters.
Patient's Birth Date (0010,0030)	Patient Age (Patient Birth Date user to calculate age)	Patient Birth Date
Patient's Sex (0010,0040)	Sex	Patient's Sex
Patient's Weight (0010,1030)	Weight in Kg	Patient's Weight
Referring Physician's Name (0008,0090)	Referring Physician	Referring Physician's Name
Requested Procedure Description (0032,1060)	Exam Description	Study Description.
Scheduled Procedure Step Start Date (0040,0002)	Date	Not available.
Scheduled Procedure Step Start Time (0040,0003)	Time	Not available.
Study Instance UID (0020,000d)	Study instance UID (only displayed on the "More Info..." screen)	Study Instance UID, if User has set the preference to "Use MWL Study UID" .
Requested Procedure Id (0040,1001)	Requested Proc ID	Requested Procedure Id.
Pregnancy Status (0010,21C0)	Pregnancy Status (only displayed on the "More Info..." screen)	Not available.
Medical Alerts (0010,2000)	Medical Alerts (only displayed on the "More Info..." screen)	Not available.
Contrast Allergies (0010,2110)	Contrast Allergies (only displayed on the "More Info..." screen)	Not available.
Special Needs (0038,0050)	Special Needs (only displayed on the "More Info..." screen)	Not available.
Requested Contrast Agent (0032,1070)	Requested Contrast Agent (only displayed on the "More Info..." screen)	Not available.
Current Patient Location (0038,0300)	Current Patient Location (only displayed on the "More Info..." screen)	Not available.
Additional Patient History (0010,21B0)	Additional Patient History (only displayed on the "More Info..." screen)	Patient History.

Note that the display of a specific data item on the “More Info...” screen is contingent on the item being enabled for display. Depending on the preferences of each specific site, data can either be displayed or not. A GE field service engineer can assist in setting these site preferences.

6.12 Setting User Preferences

6.12.1 Setting “Use MWL Study UID “ Option

Setting this option to “**Yes**” copies the **Study Instance UID** from work-list into the final **DICOM Image header**. If this option is set to “**No**” then a new study instance uid will be generated locally.

1. Click on “Patient Schedule”
2. Click on “Preferences” button
3. Set the option “Use Study Instance UID”, to either “Yes” or “No”

6.12.2 Setting Custom Query Option

This option allows the user to enter values for “Accession Number” and / or “ Requested Procedure Id” , which are used for Custom Query.

1. Click on “Patient Schedule”
2. Click on “Preferences” button
3. Set the option “**Show Update Parameters** ”, to “**Yes**”
4. To do a query click on “Update” button.
A User Interface appears, with provision to enter values for:
 - a) Accession Number
 - b) Requested Procedure Id

7 PERFORMED PROCEDURE STEP CONFORMANCE STATEMENT

7.1 Introduction

This section of the DICOM Conformance Statement specifies the compliance to DICOM conformance requirements for the Performed Procedure Step feature on this GEMS product. Note that the format of this section strictly follows the format defined in DICOM Standard PS 3.2 (Conformance). Please refer to that part of the standard while reading this section. The PPS option for LightSpeed allows a Modality Performed Procedure Step to be communicated to the Hospital/Radiology information system. The PPS feature is providing the DICOM Modality Performed Procedure Step service as a service class user (SCU).

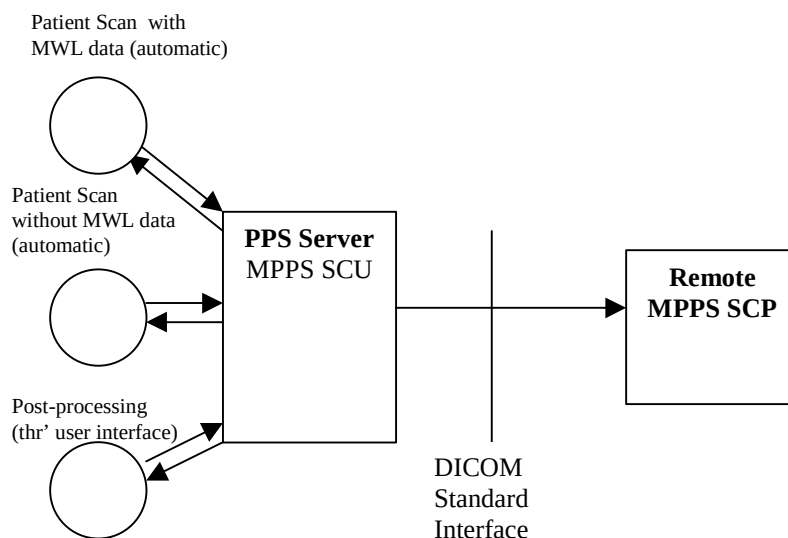
This feature works in conjunction with DICOM Modality Work-list feature, if installed. However the conformance of this feature is independent of Modality work-list feature. For information on conformance of Modality Work-List feature to DICOM standard please refer to the appropriate section or document.

7.2 Implementation Model

The DICOM 'Performed Procedure Step' service is provided by the PPS Server DICOM AE. The PPS Server DICOM AE is commanded to perform Performed Procedure Step services either automatically or through the user interface.

7.2.1 Application Data Flow Diagram

The basic Application models for the feature are shown in the following illustration:



7.2.2 Functional Definition of AEs

The PPS Server AE is implemented as an application process on the scanner host computer. It runs as a daemon serving requests from other applications to send the PPS information to the remote AE and return the results to the requesting application.

The PPS Server AE initiates the following functions.

- *Start PPS*: Initiates a DICOM association in order to create a DICOM Modality Performed Procedure Step SOP instance in the remote AE. If the remote AE accepts a presentation context applicable to Modality performed Procedure Step, the PPS Server AE will issue a request to create the SOP instance in the remote AE via the N-CREATE service.
- *Complete PPS*: Initiates a DICOM association in order to update a DICOM Modality Performed Step instance that is already created with the remote AE. If the remote AE accepts a presentation context applicable to Modality performed Procedure Step, the PPS Server AE will issue a request to update the SOP instance in the remote AE via the N-SET service. The PPS Status is set to 'COMPLETED'.
- *Discontinue PPS*: Initiates a DICOM association in order to update a DICOM Modality Performed Step instance that is already created with the remote AE. If the remote AE accepts a presentation context applicable to Modality performed Procedure Step, the PPS Server AE will issue a request to update the SOP instance in the remote AE via the N-SET service. The PPS Status is set to 'DISCONTINUED'.

7.2.3 Sequencing of Real-World Activities

7.2.3.1 PPS from Acquisition System with MWL data

The system has a Modality Work-list Server AE installed. Work-List information is obtained from HIS/RIS system through the use of Basic Work-list Management Service. Use of the information retrieved in the creation of Image SOP instance is described in the Modality Work-list Conformance statement. Use of the information retrieved in MPPS SOP instances is described later in this document.

- The system initiates a 'Start PPS' before starting a scan, i.e. when the image acquisition is started. The system retrieves necessary information related to the Scheduled Procedure Step from Modality Work-list Server. PPS Server AE initiates a MPPS (Modality Performed Procedure Step) N-CREATE request to the remote AE (MPPS SCP), in-order to create a MPPS SOP instance at the remote AE.
- The MPPS SCP returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user. **The DICOM association is closed.**
- System includes the necessary information related to Scheduled Procedure Steps and the Performed procedure Step in the image instances created.

- At the end of image acquisition, system initiates a 'Complete PPS' or 'Discontinue PPS' based on the choice selected by the user using the user interface provided. The user is also given a choice 'Defer PPS' which is described below. PPS Server AE initiates a MPPS N-SET request to the remote AE, in-order to update the MPPS SOP instance that is already created. **The N-SET is sent over a new DICOM association**
- The remote AE returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.
- At the end of image acquisition, if the user has chosen 'Defer PPS', the user is provided with an interface to 'Complete PPS' or 'Discontinue PPS' at any later time. The user might wish to alter the image set generated through acquisition, before invoking these operations. Note that the user explicitly uses the user interface provided to invoke this operation, as in the case of PPS generated for post-processing, which is described in the following section. **PPS messages N-CREATE (if applicable) and N-SET will be sent over the same DICOM association**
- The remote AE returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.

7.2.3.2 PPS from acquisition system without MWL data

The system either does not have a Modality Work-list Server AE installed or a Modality Work-list Server AE installed but no Work-List information is obtained from HIS/RIS system for the current procedure that is being performed. The information required for performing the procedure is supplied through the user interface of the system. The use of this information in MPPS SOP instances is described later in this document.

- The system initiates a 'Start PPS' before starting a scan, i.e. when the image acquisition is started. PPS Server AE initiates a MPPS (Modality Performed Procedure Step) N-CREATE request to the remote AE (MPPS SCP), in-order to create a MPPS SOP instance at the remote AE.
- The MPPS SCP returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.
- System includes the necessary information related to Scheduled Procedure Steps and the Performed procedure Step in the image instances created.
- At the end of image acquisition, system initiates a 'Complete PPS' or 'Discontinue PPS' based on the choice selected by the user using the user interface provided. The user is also given a choice 'Defer PPS' which is described below. PPS Server AE initiates a MPPS N-SET request to the remote AE, in-order to update the MPPS SOP instance that is already created.

- The remote AE returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.
- At the end of image acquisition, if the user has chosen 'Defer PPS', the user is provided with an interface to 'Complete PPS' or 'Discontinue PPS' at any later time. The user might wish to alter the image set generated through acquisition, before invoking these operations. Note that the user explicitly uses the user interface provided to invoke this operation, as in the case of PPS generated for post-processing, which is described in the following section.
- The remote AE returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.

7.2.3.3 PPS from post-processing system

- The user initiates post-processing on the images generated through acquisition.
- The system creates a Modality performed Procedure Step instance locally in the system. If the source image instance has the Scheduled Procedure Step information, it is copied into the image instances created. Also the system includes the necessary information related to the Modality Performed Procedure Step into the image instance.
- At the end of (one or more) post-processing, the user initiates 'Complete PPS' or 'Discontinue PPS' through the user interface provided. PPS Server AE initiates a MPPS (Modality Performed Procedure Step) N-CREATE request to the remote AE (MPPS SCP), in-order to create a MPPS SOP instance at the remote AE (which is actually a replica of the locally created MPPS SOP instance).
- The remote AE returns response indicating the success/failure of the request execution. If the response indicates success, PPS Server AE initiates a MPPS N-SET request to the remote AE, in-order to update the MPPS SOP instance that is already created, with the additional information.
- The remote AE returns response indicating the success/failure of the request execution. The PPS state information is updated in the system based on the response data, and is presented to the user.

7.3 AE Specification

7.3.1 PPS Server AE Specification

This Application Entity provides Standard Conformance to the following DICOM v3.0 SOP Classes as an SCU:

SOP Class Name	SOP Class UID
Modality Performed Procedure Step	1.2.840.10008.3.1.2.3.3

7.3.1.1 Association Establishment Policies

7.3.1.1.1 General

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
---------------------------------	------------------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU from an association initiated by the PPS Server AE is:

Maximum Length PDU	128 Kbytes
---------------------------	-------------------

The SOP Class Extended negotiation is not supported.

The maximum number of presentation negotiation items that will be proposed is 1.

The user information items sent by this AE are

- Maximum PDU Length
- Implementation UID

7.3.1.1.2 Number of Associations

The PPS Server will initiate only one DICOM association at any time to perform a PPS operation to the remote AE.

7.3.1.1.3 Asynchronous Nature

Asynchronous mode is not supported. All operations are performed synchronously.

7.3.1.1.4 Implementation Identifying information

The implementation UID for this DICOMv3.0 Implementation is:

PPS Feature for LightSpeed Implementation UID	1.2.840.113619.6.55
--	----------------------------

7.3.1.2 Association Initiation Policy

The PPS Server AE initiates a new association for every PPS operation initiated.

7.3.1.2.1 Real-World Activity: Performed Procedure Step creation and update

7.3.1.2.1.1 Associated Real-World Activity

The real-world activities are mentioned in section 7.2.3 *Sequencing of Real-World Activities*. Each of the real world activity results in either creating a new Performed procedure Step SOP instance at the remote SCP or updating an already created Performed Procedure Step SOP instance as per the DICOM v3.0 standard.

7.3.1.2.1.2 Proposed Presentation Context Table

The following table shows the proposed presentation contexts for the PPS Server AE after any of the real-world activity listed in section 7.2.3 *Sequencing of Real-World Activities*, is initiated.

Presentation Context Table – Proposed					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Modality Performed Procedure Step	1.2.840.10008.3.1.2.3.3	Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None

7.3.1.2.1.2.1 SOP Specific DICOM Conformance Statement for MPPS SOP class

If the remote AE does not support the proposed Presentation context, an appropriate error message logged. Only one association is opened at a time.

All the operations used by this SOP class support an association timer, which is configurable. The timer is started when a request (association request, N-CREATE request or N-SET request) is send and stopped when the respective response is received. The default time-out value is 300 seconds.

All the operations used by this SOP class support a “Session Timer”. This timer is started when the association is established and stopped when association is ended. The default time-out value is 3000 seconds.

If any of the above timers expires the association is aborted and the operation in-progress is considered FAILED.

In any case an operation (N-CREATE or N-SET) fails, system updates the state to enable operator to manually invoke the operation at any later time.

7.3.1.3 Association Acceptance Policy

The PPS Server AE does not respond to attempts by remote AE to open an association.

7.4 COMMUNICATION PROFILES

7.4.1 Supported Communication Stacks (PS 3.8)

DICOM Upper Layer (PS 3.8) is supported using TCP/IP.

7.4.2 OSI Stack

The OSI Communication stack is not supported by this implementation.

7.4.3 TCP/IP Stack

The TCP/IP stack is inherited from the operating system

7.4.3.1 API

Not Applicable to this product.

7.4.3.2 Physical Media Support

Ethernet 802.3 provides the physical network layer for this product.

7.4.4 Point-to-Point Stack

The Point-to-Point Stack is not supported by this implementation.

7.5 EXTENSIONS/SPECIALIZATION/PRIVATIZATION

7.5.1 Standard Extended/Specialized/Private SOPs

PPS for LightSpeed does not implement any private SOP classes.

7.5.2 Private Transfer Syntaxes

PPS for LightSpeed does not implement any private Transfer Syntaxes.

7.6 CONFIGURATION

The PPS feature is configured by GEMS Field Service Engineers. The DICOM configuration items below are configurable or re-configurable by the Field Service Engineer and are not accessible by users through the LightSpeed's user interface.

7.6.1 AE Title/Presentation address Mapping

PPS allows for the configuration of the following parameters that pertain to the remote AE.

- Remote MPPS AE IP Address: IP Address used to contact the remote MPPS AE
- Remote AE (HIS/RIS) IP Port: IP Port used to contact the remote MPPS AE
- Remote MPPS AE Title: Application Entity Title used to contact the remote MPPS AE

These parameters define where the MPPS requests will be directed. Configuration of these parameters is performed by GEMS Field Service engineers, using the installation facilities.

7.6.2 Configurable Parameters

The following are configurable for the PPS Server AE:

- Local (PPS Server AE) AE Title (The default is host name of the scanner appended with string “_PPS”)
- Local IP Address
- Local IP net-mask
- Local IP gateway
- The following parameters are configurable, but need change only in case of a system software upgrade:
 - Implementation UID
 - PDU size
 - Association time-out period
 - Session time-out period

7.7 SUPPORT OF EXTENDED CHARACTER SETS

PPS feature for LightSpeed does not support any extended character set.

7.8 N-CREATE & NSET REQUEST MESSAGE

PPS Feature for LightSpeed supports all named attributes listed in Table F.7.2.1 in PS3.4 of DICOM standard. That is, attributes that are not explicitly referenced by name in the table are not supported. (Example is last row in the table reads “All other attributes from Radiation Dose Module and Billing and Material Code Module”. The attributes referenced here are not supported).

For the MPPS associated with an acquisition, the following attributes are copied from the Modality Work-list SCU into the MPPS request Message, if procedure performed corresponds to the SPS information retrieved through the Modality Work-list.

LightSpeed supports the selection of single or multiple SPS for a scan. The following are applicable.

- Single SPS selection results in single PPS message
- Multiple SPS selection results in single PPS message
- Multiple SPS selection is allowed **only if they all correspond to same patient id**
- A maximum of 15 SPS's can be selected
- Referenced Study Sequence - a maximum of ten Item's is supported. *This attribute will be present only if SPS information is available from Modality Work-list SCU.*
- At the end of acquisition the user might choose to 'Defer PPS' and later choose to 'Complete PPS' or 'Discontinue PPS' from the user interface provided in the system. In this case, the date and time when user chooses to 'Complete PPS' or 'Discontinue PPS' is taken as the Performed Procedure Step End Date and Performed Procedure Step End Time respectively (Not the actual end date and end time of acquisition)
- Mapping of SPS data to **MPPS SOP instance** is explained in section 7.10
- Mapping of specific SPS data to **CT DICOM IMAGE HEADER** , for PPS is explained in section 7.11

For the MPPS associated with a post-processing the following restrictions apply on the attributes listed below.

- Referenced Study Sequence – The sequence is not sent in the MPPS message
- Scheduled Step Attribute Sequence – a maximum of ONE item is supported. The attribute will be send only if SPS information is available in the image instance.
- Referenced Patient Sequence – This sequence is not added
- Scheduled action Item Code Sequence – The sequence is not send in the MPPS message
- Performed Procedure Step Start date & Performed Procedure Step start time – The exam date and exam time that is the Start date and Start Time of the associated Study Component (Exam) is used, not the actual time when post-processing started.
- Performed Procedure Step end date & Performed Procedure Step end time - The date and time when user chooses to 'Complete PPS' or 'Discontinue PPS' is taken as the Performed Procedure Step End Date and Performed Procedure Step End Time respectively (Not the actual end date and end time of post-processing).
- Procedure Code Sequence – This sequence is sent with ZERO items in the MPPS message
- Performed Action Item Code Sequence – This sequence is sent with ZERO items in the MPPS message.

- Referenced Standalone SOP Instance Sequence – The sequence is sent with ZERO items in the MPPS message.

7.9 ERROR HANDLING AND RECOVERY

PPS Server AE does not define any extended error codes. The standard error codes are handled. On a response with status 'success' for the N-CREATE or N-SET request, the system updates the state and indicates the same on the user interface. On a response with status other than 'success' the operation is deemed 'Failed' and the system updates the state and indicates the same on the user interface. If the request has failed or response is not received before the association timeout, the operation is deemed 'Failed' and the system updates the state and indicates the same on the user interface.

If the operation is 'Failed', detailed message is logged into system log-file and system provides an alternative mechanism to retry the failed operation through the user interface to ensure that transient failures do not affect the feature performance.

7.10 Use of Specific DICOM Data

The following table gives specific usage of some of attributes in the MPPS SOP instance created, with reference to each of the real-world scenarios mentioned in 7.2.3 of this document.

Attribute Name	Tag	Usage in MPPS Instance		
		Acquisition with MWL data	Acquisition without MWL data	Post-Processing
Specific Character Set	(0008,0005)	Not used, but copied into image header	Not used	Not used, but copied into image header
Scheduled Step Attribute Sequence	(0040,0270)	Only one item		Only one item
>Study Instance UID	(0020,000D)	Copied from SPS, if option to copy is selected or else created at the scanner	Created at the scanner	Copied from source image
>Referenced Study Sequence	(0008,1110)	Copied from SPS, if selected	Not sent as part of image header	Not sent as part of image header
>Accession Number	(0008,0050)	Copied from SPS, if selected	User input on the scanner	Copied from source image
>Placer Order Number/Imaging Service Request	(0040,2016)	Not sent as part of image header	Not sent as part of image header	Not sent as part of image header
>Filler Order Number/Imaging Service Request	(0040,2017)	Not sent as part of image header	Not sent as part of image header	Not sent as part of image header
>Requested Procedure ID	(0040,1001)	Copied from SPS, if selected	User input on the scanner	Copied from source image
>Requested Procedure Description	(0032,1060)	Copied from SPS, if selected	Not used	Not sent as part of image header
>Placer Order Number/Procedure	(0040,1006)	Not sent as part of image header	Not used	Not sent as part of image header
>Filler Order Number/Procedure	(0040,1007)	Not sent as part of image header	Not used	Not sent as part of image header
>Scheduled Procedure ID	(0040,0009)	Copied from SPS, if selected	Not used	Copied from source image
>Scheduled Procedure Step Description	(0040,0007)	Copied from SPS, if selected	Not used	Copied from source image
>Scheduled Action Item Code Sequence	(0040,0008)	Copied from SPS, if selected	Not used	Not sent as part of image header
Patient's Name	(0010,0010)	Copied from SPS, if selected	User input on the scanner	Copied from source image
Patient ID	(0010,0020)	Copied from SPS, if selected	User input on the scanner	Copied from source image
Patient's Birth Date	(0010,0030)	Copied from SPS, if selected	User input on the scanner	Copied from source image
Patient's Sex	(0010,0040)	Copied from SPS, if selected	User input on the scanner	Copied from source image
Referenced Patient Sequence	(0008,1120)	Copied from SPS, if selected	Not used	No item
Performed Procedure Step ID	(0040,0253)	Created at the scanner. Will	Created at the scanner. Will have	New generated, may not be unique

Attribute Name	Tag	Usage in MPPS Instance		
		Acquisition with MWL data	Acquisition without MWL data	Post-Processing
		have the following "PPS_ID_<exam number"	the following "PPS_ID_<exam number"	
Performed Station AE Title	(0040,0241)	Local system host-name	Local system host-name	Local system host-name
Performed Station Name	(0040,0242)	Local system suite id	Local system suite id	Local system host-name
Performed Location	(0040,0243)	Local system suite id	Local system suite id	Not used
Performed Procedure Step Start Date	(0040,0244)	Same as exam start date	Same as exam date	Same as exam date
Performed Procedure Step Start Time	(0040,0245)	Same as exam start time	Same as exam time	Same as exam time
Performed Procedure Step Description	(0040,0254)	Copied from SPS, if selected	Not sent as part of image header	Not sent as part of image header
Performed Procedure Step status	(0040,0252)	See Note 1.	See Note 1.	See Note 1.
Performed Procedure Type Description	(0040,0255)	Not sent as part of image header	Not sent as part of image header	Not sent as part of image header
Procedure Code Sequence	(0008,1032)	Not sent as part of image header. Not sent in MPPS N-Create.	Not sent as part of image header. Not sent in MPPS N-Create.	Not sent as part of image header. Not sent in MPPS N-Create.
Performed Procedure Step End Date	(0040,0250)	Date when all images got installed	Date when all images got installed	The date "Complete PPS" or "Discontinue PPS" is invoked, not the actual end of post-processing
Performed Procedure Step End Time	(0040,0251)	Time when all images got installed	Time when all images got installed	The time "Complete PPS" or "Discontinue PPS" is invoked, not the actual end of post-processing
Modality	(0008,0060)	Value "CT" is stored in image header	Value "CT" is stored in image header	Value "CT" is stored in image header
Study ID	(0020,0010)	Same as exam number	Same as exam number	Copied from source image
Performed Action Item Code Sequence	(0040,0260)	Not used.	Not used	Not used
Performed Series Sequence	(0040,0340)	One item for each series created	One item for each series created	One item for each series created with post-processing
>Performing Physician's Name	(0008,1050)	Copied from SPS, if selected	User input on the scanner	Not sent as part of image header
>Protocol Name	(0018,1030)	The name of the protocol selected on the scanner	The name of the protocol selected on the scanner	Copied from source image
>Operator's Name	(0008,1070)	Copied from SPS if selected and present otherwise	User input on the scanner	Not sent as part of image header

Attribute Name	Tag	Usage in MPPS Instance		
		Acquisition with MWL data	Acquisition without MWL data	Post-Processing
		from user input on the scanner		
>Retrieve AE Title	(0008,0054)	Local system host-name	Local system host-name	host-name of the system
>Referenced Image Sequence	(0008,1140)	One item for each image created within the series	One item for each image created within the series	One item for each image generated by post-processing
>Referenced Standalone SOP Instance Sequence	(0040,0220)	Not sent as part of image header	Not sent as part of image header	Not sent as part of image header
>All other attributes from Performed Series Sequence (which Table F.7.2.1 of DICOM standard PS3.4 does not explicitly list)		Not sent as part of image header	Not sent as part of image header	Not sent as part of image header
All other attributes from Radiation Dose Module and Billing and Material Code Module (which Table F.7.2.1 of DICOM standard PS3.4 does not explicitly list)		Not sent as part of image header	Not sent as part of image header	Not sent as part of image header

- Note 1:**
- When PPS start (N-CREATE) message is sent, this element will have the value "INPROGRESS"
 - When PPS end (N-SET) message is sent, this element will have either "COMPLETE" or "DISCONTINUE" based on user selection

7.11 USE OF SPECIFIC DICOM DATA

7.11.1 Patient Level

Attribute Name	Tag	Usage in CT DICOM Image Header
Patient Name	(0010,0010)	Copied from SPS, if selected
Patient ID	(0010,0020)	Copied from SPS, if selected
Patient Birthdate	(0010,0030)	Copied from SPS, if selected
Patient Sex	(0010,0040)	Copied from SPS, if selected
Referenced Patient Sequence	(0008,1120)	Copied from SPS, if selected
Ref. SOP class uid	(0008,1150)	
Ref. SOP Instance uid	(0008,1155)	

7.11.2 Study Level

Attribute Name	Tag	Usage in CT DICOM Image Header
Study Instance UID	(0020,000D)	Copied from SPS, if selected
Study ID	(0020,0010)	Scanner generated study ID
Referring Physicians name	(0008,0090)	Copied from SPS, if selected
Accession Number	(0008,0050)	If multiple SPS's are selected, then accession number from the first selection (determined by the user) is used
Referenced Study Sequence	(0008,1110)	Copied from SPS, if selected
Ref. SOP class uid	(0008,1150)	
Ref. SOP Instance uid	(0008,1155)	

7.11.3 Series Level

Attribute Name	Tag	Usage in CT DICOM Image Header
Modality	(0008,1060)	Value "CT" is copied
Protocol Name	(0018,1030)	Name of the selected protocol is copied
Operator Name	(0008,1070)	Copied from SPS if selected and present otherwise generated from user input on scanner
Referenced Study Component Sequence	(0008,1111)	Will be present only if SPS , obtained form HIS/RIS is selected for scanning
Ref. SOP class uid	(0008,1150)	Value of MPPS SOP class UID
Ref. SOP Instance uid	(0008,1155)	Scanner generated unique UID
Requested Attribute Sequence	(0040,0275)	Will be present only if SPS , obtained form HIS/RIS is selected for scanning. If multiple SPS's are selected then this will contain multiple items one for each SPS. For all other cases this element will not be present
Requested Procedure Id	(0040,1001)	Copied from SPS, if selected
Scheduled Procedure Step Id	(0040,0009)	Copied from SPS, if selected
Scheduled Procedure Step description	(0040,0007)	Copied from SPS, if selected
Scheduled Action Item Code Sequence	(0040,0008)	Copied from SPS, if selected
Performed Procedure Step Id	(0040,0253)	Created at the scanner. The string " PPS_ID_<examnumber> " is used.
Performed Procedure Step start date	(0040,0244)	Same as exam start date
Performed Procedure Step start time	(0040,0245)	Same as exam start time
Performed Procedure Step description	(0040,0254)	Exam description is used

8 Grayscale Presentation State

8.1 Introduction

This section of the DICOM Conformance Statement specifies the compliance to DICOM conformance requirements for the Grayscale Presentation State (GSPS), referred to as the Exam Split option, feature on this GEMS product. Note that the format of this section strictly follows the format defined in DICOM Standard PS 3.2 (Conformance). Please refer to that part of the standard while reading this section. The Exam Split option for LightSpeed allows a GSPS object to be sent to any remote host that supports the display of images using GSPS. The Exam Split feature provides the DICOM Modality GSPS service as a service class user (SCU).

This feature works in conjunction with DICOM Modality Work-list feature, if installed. However the conformance of this feature is dependent on Modality work-list feature. For information on conformance of Modality Work-List feature to DICOM standard please refer to the appropriate section or document.

8.2 Implementation Model

Exam Split option have the two options i.e. Virtual Exam Split and Hard Exam Split.

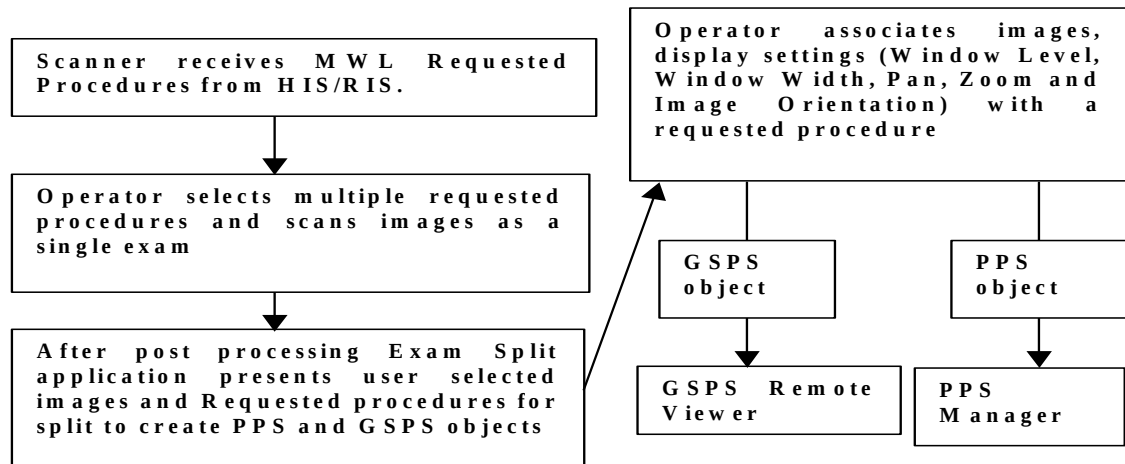
- Virtual Exam Split will create Gray Scale Presentation State and Performed Presentation State.
- In Hard Exam Split, application will send the user selected images from grouped Images that acquired during acquisition. PPS will not sent along with these images.

The DICOM Modality 'Gray Scale Presentation State' service is provided by the Exam Split. The Remote Host DICOM AE is commanded to perform transmission of Gray Scale Presentation State object through the user interface.

The Remote Host DICOM AE is commanded to perform transmission of Hard Split Images through the user interface.

8.2.1 Application Data Flow Diagram

The basic Application models for the feature are shown in the following illustration:



8.2.2 Functional Definition of AEs

8.2.3 Sequencing of Real-World Activities

8.2.3.1 GSPS Acquisition System with MWL data

The system has a Modality Work-list Server AE installed. Work-List information is obtained from HIS/RIS system through the use of Basic Work-list Management Service. Use of the information retrieved in the creation of Image SOP instance is described in the Modality Work-list Conformance statement. Use of the information retrieved in MPPS SOP instances is described later in this document.

- After Post processing Exam Split application presents associated requested Procedure(s) along with selected acquired Images.
- Exam Split application includes the necessary information related to Requested Procedure, scheduled Procedure Steps and the Performed Procedure Step of the images acquired during acquisition.
- After User modifications on Image(s), user has to click on Send button to create Gray Scale Presentation State and Performed Procedure Step objects to transmit to remote host. If there is any Image Orientation, user will be notified about the orientation before sending to Remote Host.
- Exam Split will provide Host Selection user interface to select remote host to transmit of Gray Scale Presentation State object. Performed Procedure Step object will be sent to default Performed Procedure Step host.
- After successful transmission of Gray Scale Presentation State and Performed Procedure Step objects to remote host, user will be notified with success.

8.2.4 GSPS Server AE Specification

This Application Entity provides Standard Conformance to the following DICOM v3.0 SOP Classes as an SCU:

SOP Class Name	SOP Class UID
Grayscale Softcopy Presentation State	1.2.840.10008.5.1.4.1.1.11.1

8.2.4.1 Association Establishment Policies

8.2.4.1.1 General

The DICOM Application Context Name (ACN), which is always proposed, is:

Application Context Name	1.2.840.10008.3.1.1.1
--------------------------	-----------------------

The Maximum Length PDU negotiation is included in all association establishment requests. The maximum length PDU from an association initiated by the PPS Server AE is:

Maximum Length PDU	128 Kbytes
--------------------	------------

The SOP Class Extended negotiation is not supported.

The maximum number of presentation negotiation items that will be proposed is 1.

The user information items sent by this AE are

- Maximum PDU Length
- Implementation UID

8.2.4.1.2 Number of Associations

The GSPS application will initiate only one DICOM association at any time to perform a GSPS operation to the remote AE.

8.2.4.1.3 Asynchronous Nature

Asynchronous mode is not supported. All operations are performed synchronously.

8.2.4.1.4 Implementation Identifying information

The implementation UID for this DICOMv3.0 Implementation is:

PPS Feature for LightSpeed Implementation UID	1.2.840.113619.6.55
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8.2.4.2 Association Initiation Policy

The GSPS AE initiates a new association for every GSPS operation initiated.

8.2.4.2.1 Real-World Activity: GSPS

8.2.4.2.1.1 Associated Real-World Activity

The real-world activities are mentioned in section 8.2.3 *Sequencing of Real-World Activities*. Each of the real world activity results in either creating a new GSPS SOP instance at the remote SCP.

8.2.4.2.1.2 Proposed Presentation Context Table

The following table shows the proposed presentation contexts for the GSPS AE after any of the real-world activity listed in section 7.2.3 *Sequencing of Real-World Activities*, is initiated.

Presentation Context Table – Proposed					
Abstract Syntax		Transfer Syntax		Role	Extended Negotiation
Name	UID	Name List	UID List		
Grayscale Presentation State	1.2.840.10008.5.1.4.1.1.11.1	Explicit VR Little Endian	1.2.840.10008.1.2	SCU	None

8.2.4.2.1.2.1 SOP Specific DICOM Conformance Statement for GSPS SOP class

If the remote AE does not support the proposed Presentation context, an appropriate error message logged. Only one association is opened at a time.

All the operations used by this SOP class support an association timer, which is configurable. The timer is started when a request (association request, N-CREATE request or N-SET request) is send and stopped when the respective response is received. The default time-out value is 300 seconds.

All the operations used by this SOP class support a “Session Timer”. This timer is started when the association is established and stopped when association is ended. The default time-out value is 3000 seconds.

If any of the above timers expires the association is aborted and the operation in-progress is considered FAILED.

8.2.4.3 Association Acceptance Policy

The GSPS AE does not respond to attempts by remote AE to open an association.

8.3 COMMUNICATION PROFILES

8.3.1 Supported Communication Stacks (PS 3.8)

DICOM Upper Layer (PS 3.8) is supported using TCP/IP.

8.3.2 OSI Stack

The OSI Communication stack is not supported by this implementation.

8.3.3 TCP/IP Stack

The TCP/IP stack is inherited from the operating system

8.3.3.1 API

Not Applicable to this product.

8.3.3.2 Physical Media Support

Ethernet 802.3 provides the physical network layer for this product.

8.3.4 Point-to-Point Stack

The Point-to-Point Stack is not supported by this implementation.

8.4 EXTENSIONS/SPECIALIZATION/PRIVATIZATION

8.4.1 Standard Extended/Specialized/Private SOPs

GSPS for LightSpeed does not implement any private SOP classes.

8.4.2 Private Transfer Syntaxes

GSPS for LightSpeed does not implement any private Transfer Syntaxes.

8.5 CONFIGURATION

The Exam Split feature is an option. The DICOM configuration items below are configurable or re-configurable by the Field Service Engineer and are not accessible by users through the LightSpeed's user interface.

8.5.1 AE Title/Presentation address Mapping

Exam Split allows for the configuration of the following parameters that pertain to the remote AE.

- | | |
|-----------------------------|---|
| • Remote GSPS AE IP Address | IP Address used to contact the remote GSPS AE |
| • Remote GSPS IP Port: | IP Port used to contact the remote GSPS AE |
| • Remote GSPS AE Title | Application Entity Title used to contact the remote GSPS AE |
| | |
| • Remote HES AE IP Address | IP Address used to contact the remote HES AE |
| • Remote HES IP Port: | IP Port used to contact the remote HES AE |
| • Remote HES AE Title | Application Entity Title used to contact the remote HES AE |

8.5.2 Configurable Parameters

The Following are configurable for the GSPS.

- Remote Host IP address
- Remote Host port
- Remote AE Title
- VES_ORIENTATION (environment variable for enable/disable Flip/Rotate functionality in VES)
- Changesplit_mode <HES/VES>

8.6 SUPPORT OF EXTENDED CHARACTER SETS

GSPS feature for LightSpeed does not support any extended character set.

8.7 ERROR HANDLING AND RECOVERY

Exam Application has an extended error codes and the standard error codes are also handled. If the sending GSPS/PPS objects request has failed or response is not received before the association timeout, the operation is deemed 'Failed' and the system updates the state and indicates the same on the user interface.

If the operation is 'Failed', detailed message is logged into system log-file and system provides an alternative mechanism to retry the failed operation through the user interface to ensure that transient failures do not affect the feature performance.

8.8 Use of Specific DICOM Data

The following table gives specific usage of some of attributes in the GSPS and splitting PPS SOP instance created, with reference to each of the real-world scenarios mentioned in 8.2.3 of this document.

Attribute Name	Tag	GSPS	Splitting PPS	Comment
Group Length	(0002,0000)	X		Exam Split Application will calculate
File Meta Information Version	(0002,0001)	X		Exam Split Application will calculate
Media Storage SOP Class UID	(0002,0002)	X		Exam Split Application will calculate
Media Storage SOP Instance UID	(0002,0003)	X		Exam Split Application will calculate
Transfer Syntax UID	(0002,0010)	X	X	Exam Split Application will calculate
Implementation Class UID	(0002,0012)	X		Exam Split Application will calculate
Implementation Version Name	(0002,0013)	X		Exam Split Application will calculate
Patient's Name	(0010,0010)	X	X	Copied from acquired image(s)
Patient ID	(0010,0020)	X	X	Copied from acquired image(s)
Patient's Birth Date	(0010,0030)	X	X	Copied from acquired image(s)
Patient's Sex	(0010,0040)	X	X	Copied from acquired image(s)
Instance Creation Date	(0008,0012)	X		Copied from acquired image(s)
Instance Creation Time	(0008,0013)	X		Copied from acquired image(s)
Study Instance UID	(0020,000D)	X		Copied from acquired image(s)
Study Date	(0008,0020)	X		Copied from acquired image(s)
Study Time	(0008,0030)	X		Copied from acquired image(s)
Accession Number	(0008,0050)	X		Copied from acquired image(s)

Referring Physician's Name	(0008,0090)	X		Copied from acquired image(s)
Study ID	(0020,0010)	X		Copied from acquired image(s)
Referenced Results Sequence	(0008,1110)	X		Copied from acquired image(s)
Modality	(0008,0060)	X	X	Copied from acquired image(s)
Study Description	(0008,1030)	X		This will be set as Requested Procedure Code Meaning. If Code Meaning absent, it will be Requested Procedure Description
Procedure Code Sequence	(0008,1032)	X	X	Copied from acquired image(s)
Protocol Name	(0018,1030)	X	X	Set as "Virtual Exam Split"
Series Description	(0008,103E)	X		This will be set as Requested Procedure Code Meaning. If Code Meaning absent, it will be Requested Procedure Description
Series Instance UID	(0020,000E)	X		Exam Split Application will calculate
Series Number	(0020,0011)	X		Copied from acquired image(s)
Laterality	(0020,0060)	X		Set as 0 (zero)
Requested Attribute Sequence	(0040,0275)	X		Copied from source image. Originally from Request Attributes Sequence.
>Accession Number	(0008,0050)			
>Referenced Study Sequence	(0008,1110)			
>Study Instance UID	(0020,000D)			
>Requested Procedure Description	(0032,1060)			
>Requested Procedure Code Sequence	(0032,1064)			
>Scheduled Procedure Step Description	(0040,0007)			
>Scheduled Protocol Code Sequence	(0040,0008)			

>Scheduled Procedure ID	(0040,0009)			
>Requested Procedure ID	(0040,1001)			
Performed Station AE Title	(0040,0241)		X	Exam Application will set as Host name
Performed Station Name	(0040,0242)		X	Exam Application will set as Host name
Performed Location	(0040,0243)		X	Exam Application will set as Host name
Performed Procedure Step Start Date	(0040,0244)	X	X	Exam Split Application will calculate and same time will use in GSPS.
Performed Procedure Step Start Time	(0040,0245)	X	X	Exam Split Application will calculate and same time will use in GSPS.
Performed Procedure Step End Date	(0040,0250)		X	Exam Split Application will calculate
Performed Procedure Step End Time	(0040,0251)		X	Exam Split Application will calculate
Performed Procedure Step Status	(0040,0252)		X	Set as " In Progress"
Performed Procedure Step ID	(0040,0253)	X	X	Will be set as Requested Procedure ID (0040,1001) from (0040,0275)
Performed Procedure Step Description	(0040,0254)	X	X	This will be set as Requested Procedure Code Meaning. If Code Meaning absent, it will be Requested Procedure Description
Performed Procedure Type Description	(0040,0255)		X	Set as "empty"
Performed Protocol Code Sequence	(0040,0260)		X	Set as Empty sequence

Performed Series Sequence	(0040,0340)		X	Exam Split Application will calculate and copy from Image(s) / Study and set these sequence values
Rescale Type	(0028,1054)	X		Exam Split Application will set this value.
Referenced Performed Procedure Step Sequence	(0008,1111)	X		Exam Split Application will calculate and set these sequence values
Manufacturer	(0008,0070)	X		Exam Split Application will set as "GE MEDICAL SYSTEMS"
Instance Number	(0020,0013)	X		Exam Split Application will set as "OFFIS_DCMTK_341"
Presentation label	(0070,0080)	X		This will be set as Requested Procedure Code Meaning or Requested Procedure Description
Presentation Description	(0070,0081)	X		Exam Split Application will set as "Exam Split"
Presentation Create Date	(0070,0082)	X		Exam Split Application will calculate
Presentation Create Time	(0070,0083)	X		Exam Split Application will calculate
Presentation creators name	(0070,0084)	X		Exam Split Application will set this value as "" (null)
Referenced Series Sequence	(0008,1115)	X		Copied from source image.
Displayed area selection sequence	(0070,005A)	X		Exam Split Application will calculate values for Window Width, Window Level, pan and zoom

Softcopy VOI LUT Sequence	(0028,3110)	X		Exam Split Application will calculate and copy the from acquired Image(s)
Image Horizontal flip	(0070,0041)	X		Exam Split Application will calculate from Image Orientation
Image Rotation	(0070,0042)	X		Exam Split Application will calculate from Image Orientation
Presentation LUT Shape	(2050,0020)	X		Exam Split Application will set as "IDENTITY"
SOP Class UID	(0008,0016)	X		Exam Application will set as "1.2.840.10008.5.1.4.1.1.11.1"
SOP Instance UID	(0008,0018)	X	X	Copied from acquired study.
Specific Character Set	(0008,0005)	X	X	Copied from acquired study
Referenced Patient Sequence	(0008,1120)		X	
Scheduled Step Attribute Sequence	(0040,0270)		X	Copied from source image. Originally from SPS sequence.
>Accession Number	(0008,0050)			
>Referenced Study Sequence	(0008,1110)			
>Study Instance UID	(0020,000D)			
>Requested Procedure Description	(0032,1060)			
>Scheduled Procedure Step Description	(0040,0007)			
>Scheduled Action Item Code Sequence	(0040,0008)			
>Scheduled Procedure ID	(0040,0009)			
>Requested Procedure ID	(0040,1001)			

8.9 Image Header Changes Supporting GSPS

8.9.1 Request Attributes Sequence

The (0040, 0275) Request Attributes Sequence has been expanded to include dicom tags that the VES application requires to complete the VES GSPS and PPS. This sequences now contains the following elements:

- (0040, 0275) Request Attributes Sequence
 - >(0008, 0050) Accession Number
 - >(0008, 1110) Reference Study Sequence
 - >>(0008, 1150) Referenced SOP Class UID
 - >>(0008, 1155) Referenced SOP Instance UID
 - >(0020, 000D) Study Instance UID
 - >(0032, 1060) Requested Procedure Description
 - >(0032, 1064) Requested Procedure Code Sequence
 - >>(0008, 0100) Code Value
 - >>(0008, 0102) Coding Scheme Designator
 - >>(0008, 0104) Code Meaning
 - >(0040, 0007) Scheduled Procedure Description
 - >(0040, 0008) Scheduled Action Item Code Sequence
 - >>(0008, 0100) Code Value
 - >>(0008, 0102) Coding Scheme Designator
 - >>(0008, 0104) Code Meaning
 - >(0040, 0009) Scheduled Procedure Step ID
 - >(0040, 1001) Requested Procedure ID

The Requested Attributes Sequence may contain a maximum of 15 sequence items.
This sequence will only appear in the image header if the VES/HES option is installed or PPS is enabled.

8.9.2 Accession Number

If scheduled procedures are grouped, the accession number for each procedure is stored in the Request Attributes Sequence (0040, 0275) in the image header.

If two or more of the accession numbers in the grouped case are different, the top level Accession Number (0008, 0050) shall be present but empty.

APPENDIX A: OPTIONAL/CONDITIONAL MODULES/ATTRIBUTES (IOD DEFINITIONS)

Enclosed is a listing of only the Optional/Conditional modules/attributes used by this implementation for CT/MR/SC and Standalone Overlay IOD's.

Note: All other modules that are mandatory and attributes that are type 1 & 2 in nature per the DICOM standard are supported by this implementation but are not explicitly listed here.

A.1 CT Specific IOD Definition

A.1.1 CT Image IOD Modules

Entity Name	Module Name	Référence	Usage
Patient	Patient	A.5.1	M
Study	General Study	A.5.2	M
	Patient Study	A.5.3	U
Series	General Series	A.5.4	M
Frame of Reference	Frame of Reference	A.5.12	M
Equipment	General Equipment	A.5.5	M
Image	General Image	A.5.6	M
	Image Plane	A.5.7	M
	Image Pixel	Only the type 1 elements are sent.	M
	Contrast/Bolus	A.5.8	C
	CT Image	A.1.2	M
	VOI LUT	A.5.11	U
	SOP Common	A.5.9	M

A.1.2 CT Image Module

Attribute Name	Tag	Type	Notes
KV	(0018,0060)	2	Value always sent.
Acquisition Number	(0020,0012)	2	Value always sent.
Scan Options	(0018,0022)	3	Value always sent.
Data Collection Diameter	(0018,0090)	3	Value always sent.
Reconstruction Diameter	(0018,1100)	3	Value sent for all images except scouts.
Distance Source to Detector	(0018,1110)	3	Value always sent.
Distance Source to Patient	(0018,1111)	3	Value always sent.
Gantry / Detector Tilt	(0018,1120)	3	Value always sent.

Attribute Name	Tag	Type	Notes
Table Height	(0018,1130)	3	Value always sent.
Rotation Direction	(0018,1140)	3	Not sent for scout or axial.
Exposure Time	(0018,1150)	3	Value always sent.
X-Ray Tube Current	(0018,1151)	3	Value always sent.
Exposure	(0018,1152)	3	Value always sent.
Filter Type	(0018,1160)	3	Sent. Defined terms: <u>BODY FILTER</u> <u>BOWTIE FLAT FILT</u> <u>FLAT FILTER</u> <u>LARGE BOWTIE FIL</u>
Focal Spot	(0018,1190)	3	Sent. Fixed value of 0.7 or 1.2.
Convolution Kernel	(0018,1210)	3	Not sent for scouts. Defined terms: <u>SMOOTH</u> <u>SOFT</u> <u>STANDARD</u> <u>DETAIL</u> <u>BONE</u> <u>EDGE</u> <u>SHARP</u> <u>LUNG</u>
Revolution Time	(0018, 9305)	3	Sent for spiral scan only
Single Collimation Width	(0018, 9306)	3	Sent for spiral scan only
Total Collimation Width	(0018, 9307)	3	Sent for spiral scan only
Table Speed	(0018, 9309)	3	Sent for spiral scan only
Table Feed per Rotation	(0018, 9310)	3	Sent for spiral scan only
CT Pitch Factor	(0018, 9311)	3	Sent for spiral scan only

CAUTION

It is possible for the operator of CT LightSpeed QX/i system to change the table height while scanning a series of images. Therefore, implementations must use the Frame of Reference UID (0020,0052) in conjunction with the Table Height (0018,1130) to determine if two images are spatially related.

A.1.2.1 Implementation Specific details

- For all CT Images created by LightSpeed patient birth date element (0010,0030) is sent, if the operator has entered the details. If the operator has not entered the details it will be sent as a “Zero length element”
- The length of Accession number element (0008,0050) is a maximum of 16 characters
- Window Center (0028,1050) and Window Width (0028,1051) are sent for all CT Images created by LightSpeed
- Based on whether contrast was used or not the following applies for CT Images created by LightSpeed

Mode	(0018,0010)	(0018,1040)
No Contrast	Not sent	Not sent
Oral Contrast	Oral Contrast Agent name	The string “Oral”
IV Contrast	IV Contrast agent name	The String “IV”
Oral and IV contrast	Oral contrast agent “ & “ IV Contrast agent	The string “Oral & IV”

- Supported field length for DICOM fields

DICOM Tag	Field Name	Field Length
(0008, 0050)	Accession Number	16 characters
(0010, 0010)	Patient Name	32 characters
(0010, 0020)	Patient Id	16 characters

A.2 SC Specific IOD Definition

A.2.1 SC Image IOD Modules

Entity Name	Module Name	Reference	Usage
Patient	Patient	A.5.1	M
Study	General Study	A.5.2	M
	Patient Study	A.5.3	U
Series	General Series	A.5.4	M
Equipment	General Equipment	A.5.5	U
	SC Equipment	A.5.13	M
Image	General Image	A.5.6	M
	Image Pixel	Only the type 1 elements are	M

		sent.	
	SC Image	Not sent (consists entirely of type 3 element).	M
	Modality LUT	Only the type 1C elements are sent.	U
	VOI LUT	A.5.11	U
	SOP Common	A.5.9	M

A.3 Standalone Overlay Specific IOD Definition

A.3.1 Standalone Overlay Image IOD Modules

Entity Name	Module Name	Reference	Usage
Patient	Patient	A.5.1	M
Study	General Study	A.5.2	M
	Patient Study	A.5.3	U
Series	General Series	A.5.4	M
Equipment	General Equipment	A.5.5	M
Overlay	Overlay Identification	A.5.14	M
	Overlay Plane	A.5.10	M
	SOP Common	A.5.9	M

A.4 CT/MR/SC/Standalone Overlay IOD Common Module Definitions

A.4.1 Patient Module

Attribute Name	Tag	Type	Notes
Patient's Name	(0010,0010)	2	As entered at user interface or from worklist. Supports 5 different components delimited by "^". Supports a maximum length of 32 characters including the delimiter .
Patient ID	(0010,0020)	2	As entered at user interface or from worklist.
Patient's Birth Date	(0010,0030)	2	Always sent zero length.
Patient's Sex	(0010,0040)	2	Always sent.

A.4.2 General Study Module

Attribute Name	Tag	Type	Notes
Study Instance UID	(0020,000D)	1	Generated uniquely for each exam. Value from worklist, if present, is not filled in here.
Study Date	(0008,0020)	2	Generated for each exam and always sent.
Study Time	(0008,0030)	2	Generated for each exam and always sent.
Accession Number	(0008,0050)	2	Value from user interface or worklist sent.
Referring Physician's Name	(0008,0090)	2	Value from user interface sent.

Attribute Name	Tag	Type	Notes
Study ID	(0020,0010)	2	Value from user interface or worklist sent.
Study Description	(0008,1030)	3	Value from user interface sent.
Name of Physician(s) Reading Study	(0008,1060)	3	Sent if entered at the user interface.

A.4.3 Patient Study Module

Attribute Name	Tag	Type	Notes
Patient's Age	(0010,1010)	3	Calculated from Date of Birth entered at user interface.
Patient's Weight	(0010,1030)	3	Value from user interface.
Additional Patient's History	(0010,21b0)	3	Value from user interface.

A.4.4 General Series Module

Attribute Name	Tag	Type	Notes
Series Number	(0020,0011)	2	Generated sequentially, always sent.
Laterality	(0020,0060)	2C	Always sent zero-length.
Series Date	(0008,0021)	3	Generated for each series and always sent.
Series Time	(0008,0031)	3	Generated for each series and always sent.
Performing Physician's Name	(0008,1050)	3	
Protocol Name	(0008,1030)	3	Sent if entered at user interface.
Series Description	(0008,103E)	3	Value from user interface is sent.
Operators Name	(0008,1070)	3	Value from user interface is sent.
Patient Position	(0018,5100)	2C	Sent. As selected by operator when patient is positioned. Defined terms are: HFP = Head-First Prone HFS = Head-First Supine HFDR = Head-First Decubitus Right HFDL = Head-First Decubitus Left FFDR = Feet-First Decubitus Right FFDL = Feet-First Decubitus Left FFP = Feet-First Prone FFS = Feet-First Supine

A.4.5 General Equipment Module

Attribute Name	Tag	Type	Notes
Manufacturer	(0008,0070)	2	Always sent as "GE MEDICAL SYSTEMS"
Institution Name	(0008,0080)	3	Sent. Value is configurable.
Station Name	(0008,1010)	3	Sent. Value is configurable.
Manufacturers Model Name	(0008,1090)	3	
Device Serial Number	(0018,1000)	3	Sent if present in image.
Software Versions	(0018,1020)	3	Always sent as "05" for this version - does not distinguish individual software releases.
Spatial Resolution	(0018,1050)	3	Sent only for MR images.
Pixel Padding Value	(0028,0120)	3	Sent.

A.4.6 General Image Module

Attribute Name	Tag	Type	Notes
Image Number	(0020,0013)	2	Generated sequentially, always sent.
Patient Orientation	(0020,0020)	2C	Always sent zero length.
Image Date	(0008,0023)	2C	Generated for each image, always sent.
Image Time	(0008,0033)	2C	Generated for each image, always sent.
Image Type	(0008,0008)	3	Always sent.
Acquisition Number	(0020,0012)	3	Generated for each acquisition, always sent.
Acquisition Date	(0008,0022)	3	Generated for each acquisition, always sent.
Acquisition Time	(0008,0032)	3	Generated for each acquisition, always sent.

A.4.7 Image Plane Module

Attribute Name	Tag	Type	Notes
Slice Thickness	(0018,0050)	2	Value always sent.
Image Slice Location	(0020,1041)	3	Value always sent.

A.4.8 Contrast Bolus Module

Attribute Name	Tag	Type	Notes
Contrast/Bolus Agent	(0018,0010)	2	Sent if contrast exam, as entered in user interface.
Contrast/Bolus Route	(0018,1040)	3	Sent if contrast exam, as entered in user interface.

A.4.9 SOP Common Module

Attribute Name	Tag	Type	Notes
Specific Character Set	(0008,0005)	1C	ISO_IR 100

A.4.10 Overlay Plane Module

Attribute Name	Tag	Type	Notes
Overlay Data	(60xx,3000)	1C	Only for SC & Standalone Overlay IOD Definitions

A.4.11 VOI LUT Module

Attribute Name	Tag	Type	Notes
Window Center	(0028,1050)	3	Window Center for display. Always sent.
Window Width	(0028,1051)	1C	Window Width for display. Always sent.

A.4.12 Frame of Reference Module

Attribute Name	Tag	Type	Notes
Frame of Reference UID	(0020,0052)	1	
Position Reference Indicator	(0020,1040)	2	Value as entered at the user interface.

CAUTION

It is possible for the operator of CT LightSpeed QX/i system to change the table height while scanning a series of images. Therefore, implementations must use the Frame of Reference UID (0020,0052) in conjunction with the Table Height (0018,1130) to determine if two images are spatially related.

A.4.13 SC Equipment Module

Attribute Name	Tag	Type	Notes
Modality	(0008,0060)	3	Modality of original image (MR or CT).

A.4.14 Overlay Identification Module

Attribute Name	Tag	Type	Notes
Overlay Number	(0020,0022)	2	The image number of the image this overlay is for.
Overlay Date	(0008,0024)	3	Date of overlay.
Overlay Time	(0008,0034)	3	Time of overlay.

APPENDIX B: PRIVATE DATA ELEMENTS

Enclosed is a listing of private data elements used in this implementation for CT/MR Image IOD definition.

B.1 CT Image IOD Private Data Elements Definition

B.1.1 Private Creator Identification (GEMS_IDEN_01)

Attribute Name	Tag	VR	VM
Full fidelity	(0009,1001)	LO	1
Suite id	(0009,1002)	SH	1
Product id	(0009,1004)	SH	1
Image actual date	(0009,1027)	SL	1
Equipment UID	(0009,10E3)	UI	1

Note: For all images created by LightSpeed system (0009,xx01) element will have the value "CT_LIGHTSPEED".

B.1.2 Private Creator Identification (GEMS_ACQU_01)

Attribute Name	Tag	VR	VM
Number of cells I in Detector	(0019,1002)	SL	1
Cell number at Theta	(0019,1003)	DS	1
Cell spacing	(0019,1004)	DS	1
Horiz. Frame of ref.	(0019,100F)	DS	1
Series contrast	(0019,1011)	SS	1
First scan ras	(0019,1018)	LO	1
Last scan ras	(0019,101A)	LO	1
Table speed	(0019,1023)	DS	1
Mid scan time	(0019,1024)	DS	1
Mid scan flag	(0019,1025)	SS	1
Degrees of azimuth	(0019,1026)	SL	1
Gantry period	(0019,1027)	DS	1
Number of triggers	(0019,102C)	SL	1
Angle of first view	(0019,102E)	DS	1
Trigger frequency	(0019,102F)	DS	1
Scan FOV type	(0019,1039)	SS	1
Segment number	(0019,1042)	SS	1
Total segments requested	(0019,1043)	SS	1
View compression factor	(0019,1047)	SS	1
Recon post proc. Flag	(0019,1052)	SS	1
Dependent on #views processed	(0019,106A)	SS	1

B.1.3 Private Creator Identification (GEMS_RELA_01)

Attribute Name	Tag	VR	VM
Series from which Prescribed	(0021,1003)	SS	1
Series Prescribed From	(0021,1035)	SS	1
Image Prescribed From	(0021,1036)	SS	1
Biopsy position	(0021,1091)	SS	1
Biopsy T location	(0021,1092)	FL	1
Biopsy ref location	(0021,1093)	FL	1

B.1.4 Private Creator Identification (GEMS_STDY_01)

Attribute Name	Tag	VR	VM
Start time(secs) in first axial	(0023,1070)	FD	1

B.1.5 Private Creator Identification (GEMS_IMAG_01)

Attribute Name	Tag	VR	VM
Scout Type	(0027,1010)	SS	1
Vma mamp	(0027,101C)	SL	1
Vma mod	(0027,101E)	SL	1
Vma clip	(0027,101F)	SL	1
Smart scan ON/OFF flag	(0027,1020)	SS	1
Plane Type	(0027,1035)	SS	1
Center R coord of plane image	(0027,1042)	FL	1
Center A coord of plane image	(0027,1043)	FL	1
Center S coord of plane image	(0027,1044)	FL	1
Normal R coord	(0027,1045)	FL	1
Normal A coord	(0027,1046)	FL	1
Normal S coord	(0027,1047)	FL	1
Table start location	(0027,1050)	FL	1
Table end location	(0027,1051)	FL	1

B.1.6 Private Creator Identification (GEMS_PARM_01)

Attribute Name	Tag	VR	VM
Window value	(0043,1010)	US	1
X-ray chain	(0043,1012)	SS	3
Number of overranges	(0043,1016)	SS	5
Delta start time	(0043,101E)	DS	1
Max overranges in a view	(0043,101F)	SL	1
Corrected after glow terms	(0043,1021)	SS	1
Reference channels	(0043,1025)	SS	6
No views ref chans blocked	(0043,1026)	US	6
Scan pitch ratio	(0043,1027)	SH	1
Unique image iden	(0043,1028)	OB	1
Private Scan Options	(0043,102B)	SS	4
RA cord of target recon center	(0043,1031)	DS	2

Attribute Name	Tag	VR	VM
Trigger on position	(0043,1040)	FL	4
Degree of rotation	(0043,1041)	FL	4
DAS trigger source	(0043,1042)	SL	4
DAS fpa gain	(0043,1043)	SL	4
DAS output source	(0043,1044)	SL	4
DAS ad input	(0043,1045)	SL	4
DAS cal mode	(0043,1046)	SL	4
Start scan to X-ray on delay	(0043,104D)	FL	4
Duration of X-ray on	(0043,104E)	FL	4
Recon filter	(0043, 1064)	LO	1

B.1.7 Private Creator Identification(GEMS_HELIOS_01)

Note: Dicom elements (0045, 1030-1033) and (0045, 1036-1039) and (0045, 103B) are present only if the appropriate cardiac option is installed on the scanner.

Attribute Name	Tag	VR	VM
Number of Macro Rows in Detector	(0045, 1001)	SS	1
Macro width at ISO Center	(0045, 1002)	FL	1
DAS type	(0045, 1003)	SS	1
DAS gain	(0045, 1004)	SS	1
DAS Temperature	(0045, 1005)	SS	1
Table Direction	(0045, 1006)	CS	1
Z smoothing Factor	(0045, 1007)	FL	1
View Weighting Mode	(0045, 1008)	SS	1
Sigma Row number	(0045, 1009)	SS	1
Minimum DAS value	(0045, 100A)	FL	1
Maximum Offset Value	(0045, 100B)	FL	1
Number of Views shifted	(0045, 100C)	SS	1
Z tracking Flag	(0045, 100D)	SS	1
Mean Z error	(0045, 100E)	FL	1
Z tracking Error	(0045, 100F)	FL	1
Start View 2A	(0045, 1010)	SS	1
Number of Views 2A	(0045, 1011)	SS	1
Start View 1A	(0045, 1012)	SS	1
Sigma Mode	(0045, 1013)	SS	1
Number of Views 1A	(0045, 1014)	SS	1
Start View 2B	(0045, 1015)	SS	1
Number Views 2B	(0045, 1016)	SS	1
Start View 1B	(0045, 1017)	SS	1
Number of Views 1B	(0045, 1018)	SS	1
Itebone Flag	(0045, 1021)	SS	1
Peristaltic Flag	(0045, 1022)	SS	1
CardiacReconAlgorithm	(0045, 1030)	CS	1
AvgHeartRateForImage	(0045, 1031)	CS	1
TemporalResolution	(0045, 1032)	FL	1
PctRpeakDelay	(0045, 1033)	CS	1
EkgFullMaStartPhase	(0045, 1036)	CS	1

Attribute Name	Tag	VR	VM
EkgFullMaEndPhase	(0045, 1037)	CS	1
EkgModulationMaxMa	(0045, 1038)	CS	1
EkgModulationMinMa	(0045, 1039)	CS	1
NoiseReductionImageFilterDesc	(0045, 103B)	LO	1

B.1.8 Private Creator Identification (GEMS_CT_CARDIAC_001)

Note: Private Group 49 is present only if the appropriate cardiac option is installed on the scanner.

Attribute Name	Tag	VR	VM
CT Cardiac Sequence	(0049, 1001)	SQ	1
HeartRateAtConfirm	(0049, 1002)	CS	1
AvgHeartRatePriorToConfirm	(0049, 1003)	FL	1
MinHeartRatePriorToConfirm	(0049, 1004)	CS	1
MaxHeartRatePriorToConfirm	(0049, 1005)	CS	1
StdDevHeartRatePriorToConfirm	(0049, 1006)	FL	1
NumHeartRateSamplesPriorToConfirm	(0049, 1007)	US	1
AutoHeartRateDetectPredict	(0049, 1008)	CS	1
SystemOptimizedHeartRate	(0049, 1009)	CS	1
EkgMonitorType	(0049, 100A)	ST	1
NumReconSectors	(0049, 100B)	CS	1
RpeakTimeStamps	(0049, 100C)	FL	256

APPENDIX C: DICOMDIR DIRECTORY INFORMATION

Enclosed here is a listing of only the optional (conditional) modules and optional attributes used by this implementation in the DICOMDIR definition. All standard attributes as defined in Part 3 Addendum (Basic Directory Information Object) are supported by this implementation but not listed here.

C.1 Basic Directory IOD Definition

Module	Reference	Usage	Notes
Directory Information	B.X.3.2.1 (DICOM PS3.10)	U	

C.2 Directory Information Module

Attribute Name	Tag	Type	Notes
Offset of the Next Directory Record	(0004,1400)	1C	
Record In-use Flag	(0004,1410)	1C	
Offset of Referenced Lower-Level Directory Entity	(0004,1420)	1C	
Directory Record Type	(0004,1430)	1C	PATIENT, STUDY, SERIES and IMAGE
Referenced File ID	(0004,1500)	1C	Present only in IMAGE Directory Record
Referenced SOP Class UID in file	(0004,1510)	1C	Present only in IMAGE Directory Record
Referenced SOP Instance UID in File	(0004,1511)	1C	Present only in IMAGE Directory Record
Referenced Transfer Syntax UID in File	(0004,1512)	1C	Present only in IMAGE Directory Record

C.3 Directory Record Selection Keys

As indicated in table C.1.2, the PRI-CTMR-MOD12 application profile will have only the Patient, Study, Series and Image directory record types. Given below are the list of attributes supported under each of these directories.

C.3.1 Patient Keys

Attribute Name	Tag	Type	Notes
Specific Character Set	(0008,0005)	1C	ISO_IR 100
Patient's Name	((0010,0010)	2	
Patient ID	(0010,0020)	1	

C.3.2 Study Keys

Attribute Name	Tag	Type	Notes
Specific Character Set	(0008,0005)	1C	ISO_IR 100
Study Date	(0008,0020)	1	
Study Time	(0008,0030)	1	
Accession Number	(0008,0050)	2	
Study Description	(0008,1030)	2	
Study Instance UID	(0020,000D)	1C	
Study ID	(0020,0010)	1	

C.3.3 Series Keys

Attribute Name	Tag	Type	Notes
Specific Character Set	(0008,0005)	1C	ISO_IR 100
Image Type	(0008,0008)	3	
Modality	(0008,0060)	1	
Manufacturer's Id	(0008,0070)	3	
Series Description	(0008,103E)	3	
Manuf. Model Name	(0008,1090)	3	
Series Instance UID	(0020,000E)	1	
Series Number	(0020,0011)	1	

C.3.4 Image Keys

Attribute Name	Tag	Type	Notes
Specific Character Set	(0008,0005)	1C	ISO_IR 100
SOP Instance UID	(0008,0018)	3	Redundant to (0004,1511)
Referenced Image Sequence	(0008,1140)	1C	Not included.
Sequence Variant	(0018,0021)	3	
Slice Thickness	(0018,0050)	3	
Repetition Time	(0018,0080)	3	
Echo Time	(0018,0081)	3	
Inversion Time	(0018,0082)	3	
Number of Averages	(0018,0083)	3	
Echo Number	(0018,0086)	3	MRI profile only.
Spacing Between Slices	(0018,0088)	3	
Data Collection Diameter	(0018,0090)	3	
Contrast/Bolus Route	(0018,1040)	3	
Trigger Time	(0018,1060)	3	
Reconstruction Diameter	(0018,1100)	3	
Gantry/Detector Tilt	(0018,1120)	3	
Convolution Kernel	(0018,1210)	3	
Flip Angle	(0018,1314)	3	
Image Number	(0020,0013)	1	
Image Position (Patient)	(0020,0032)	1C	
Image Orientation (Patient)	(0020,0037)	1C	
Frame of Reference UID	(0020,0052)	1C	Absent from PRI-CTMR-MOD12
Slice Location	(0020,1041)	3	
Samples Per Pixel	(0028,0002)	R	
Photometric Interpretation	(0028,0004)	R	Absent from all profiles.
Rows	(0028,0010)	1	
Columns	(0028,0011)	1	
Pixel Spacing	(0028,0030)	1C	
Bits Allocated	(0028,0100)	R	
Bits Stored	(0028,0101)	R	MRI – Absent from profiles.
High Bit	(0028,0102)	R	MRI – Absent from profiles.

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